

Customer Prioritisation under Supply Constraints:

A Decision Framework for Semiconductor Suppliers in the
Global Automotive Industry with Application to Nexperia

Master's Thesis

for the award of the academic degree

Master of Business Administration (MBA)

at

Technische Hochschule Nürnberg Georg Simon Ohm

OHM Professional School

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Date:	01.07.2026

Table of contents

List of figures	iii
List of tables	iv
List of abbreviations	v
Executive summary	vii
Introduction	viii
1. Current industry situation	1
1.1. Industry background and automotive semiconductor dependence	1
1.2. Structural supply constraints, concentration and slow capacity expansion	4
1.3. Structural allocation pressure and uneven customer conditions	7
1.4. Government intervention, regionalisation and Nexperia's regulatory exposure	10
2. Emerging trends shaping the automotive semiconductor industry	14
2.1. Electrification-driven demand shifts	15
2.2. Software-defined vehicles and centralised computing	18
2.3. Mature-node dependence and the paradox of supply in a growing market	22
2.4. Geopolitical reconfiguration of semiconductor supply, regionalisation, and industrial policy	25
2.5. Relationship restructuring and governance implications for constrained allocation	28
3. Analysis of challenges, risks, and opportunities	30
3.1. Structural challenges in supplier decision-making	31
3.2. Major risks generated by prioritisation errors	35
3.3. Strategic opportunities arising from constrained supply	41
3.4. Application to Nexperia	44
4. Development of a customer prioritisation framework	49
4.1. Basic assumptions and decision logic	49
4.2. Evaluation criteria for prioritisation	58
4.3. Balancing profitability, strategic relevance, and risk	62
4.4. Governance and implementation aspects	65
4.5. Application to Nexperia	69

4.6.	Outlook for Nexperia	75
4.7.	Management implications for Nexperia	77
5.	Conclusion and next steps	81
5.1.	Answering the research questions	81
5.2.	Contribution of the developed framework.....	85
5.3.	Limitations of the framework.....	88
5.4.	Future research and future framework development	90
	References.....	93

List of figures

Figure 1: Indicators of rising automotive semiconductor dependence	1
Figure 2: Divergence between chip volume and revenue (2026, projected) and between revenue and physical wafer-shipment growth (2025)	5
Figure 3: Diverging projected growth of automotive component segments, 2025-2035.	8
Figure 4: Current market size and future growth of automotive component segments: The automotive component allocation conflict	16
Figure 5: Shift to software-defined vehicles is raising semiconductor and software demand across the automotive industry.....	21
Figure 6: Global semiconductor capex allocation and automotive supply vulnerability by technology segment.....	24
Figure 7: Global automotive production concentration and degree of regulatory restrictions imposed by U.S. and EU semiconductor policies on manufacturing expansion in those regions	27
Figure 8: Structural challenges affecting Nexperia's allocation decision quality: Decision visibility vs. Impact on allocation quality	34
Figure 9: Risk landscape of unstructured customer prioritisation at Nexperia: Likelihood vs. business impact	40
Figure 10: Nexperia product portfolio: Allocation risk exposure vs. strategic opportunity from constrained supply	46
Figure 11: Visualisation of criteria weight shifts across allocation horizons in the Nexperia customer prioritisation framework	57
Figure 12: Composite programme scores and priority band evolution across allocation horizons	72

List of tables

Table 1: Strategic opportunities for Nexperia arising from semiconductor supply constraints	43
Table 2: Nexperia customer prioritisation framework: Stage 1 Gates and Stage 2 criteria weights by allocation horizon	57
Table 3: Customer prioritisation scoring matrix	60
Table 4: Priority bands and allocation rules for constrained supply	63
Table 5: Functional responsibilities and ownerships within the semiconductor allocation decision framework at Nexperia	67
Table 6: Governance KPIs for the allocation decision framework at Nexperia	68
Table 7: Programme-level scoring of Programmes A-D on criteria C1-C6.....	70
Table 8: Composite programme scores and priority bands by allocation horizon.....	72
Table 9: Summary of managerial implications derived from the developed customer prioritisation framework.....	79

List of abbreviations

ACEA	European Automobile Manufacturers' Association
ADAS	Advanced Driver Assistance Systems
AEC-Q100	Automotive Electronics Council Qualification Standard for Integrated Circuits
AEC-Q101	Automotive Electronics Council Qualification Standard for Discrete Semiconductors
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
ASML	ASML Holding N.V.
ATP	Assembly, Testing and Packaging
BCG	Boston Consulting Group
BIS	Bureau of Industry and Security
BMFTR	Bundesministerium für Forschung, Technologie und Raumfahrt
BMW	Bayerische Motoren Werke
CAGR	Compound Annual Growth Rate
CEO	Chief Executive Officer
DC	Direct Current
DCM	Demand and Capacity Management
DESY	Deutsches Elektronen-Synchrotron
E/E	Electrical and Electronic Architecture
ECFR	European Council on Foreign Relations
ECU	Electronic Control Unit
EPS	Electronic Power Steering
ESD	Electrostatic Discharge
EU	European Union
EV	Electric Vehicle
FDI	Foreign Direct Investment
FET	Field-Effect Transistor
FY	Fiscal Year
GaN	Gallium Nitride
IC	Integrated Circuit
ICE	Internal Combustion Engine
IEA	International Energy Agency
KPI	Key Performance Indicator
KPMG	Klynveld Peat Marwick Goerdeler
LTSA	Long-Term Supply Agreement

MAPE	Mean Absolute Percentage Error
MEMS	Micro-Electro-Mechanical Systems
MERICS	Mercator Institute for China Studies
MOSFET	Metal–Oxide–Semiconductor Field-Effect Transistor
nm	Nanometre
OECD	Organisation for Economic Co-operation and Development
OEM	Original Equipment Manufacturer
R&D	Research and Development
RFQ	Request for Quotation
RQ	Research Question
SDV	Software-Defined Vehicle
SiC	Silicon Carbide
SIA	Semiconductor Industry Association
SKU	Stock Keeping Unit
SOP	Start of Production
S&P	Standard & Poor's
TVS	Transient Voltage Suppression
UK	United Kingdom
US	United States
USD	United States Dollar
VDA	Verband der Automobilindustrie e.V.

Executive summary

Semiconductors have become a critical component for automotive production because vehicles are becoming more electrified, software-driven and electronically complex (Parliamentary Office of Science and Technology, 2024). With electric car sales of 17 million units in 2024 (20% of global sales), which increased to 20 million units (25% of global sales) in 2025 (International Energy Agency, 2025a, 2026a), indicating almost 18% growth, shows increasing trend in electric vehicle demand. Moreover, Level 2 Advanced Driver Assistance Systems (ADAS) may account for 52% of vehicle sales by 2030 (McKinsey & Company, 2026b). This indicates rising semiconductor demand. Because semiconductor supply chains are complex, capital-intensive and slow to expand (Boston Consulting Group and Semiconductor Industry Association, 2024), suppliers should prioritise customers in case of limited available capacity. This is because the impact can be critical: in 2021, semiconductor shortages cost the industry more than USD 200 billion (Parliamentary Office of Science and Technology, 2024).

The customer prioritisation framework is developed using Nexperia as the central case for customer prioritisation under supply constraints (Nexperia B.V., 2025c). With USD 2.06 billion in FY2024 revenue (Nexperia B.V., 2025d), over 50% revenue generated from automotive customers, Europe, U.S. and China being the largest markets (Rosina, Boulay and Yang, 2025), a Shanghai-listed parent company Wingtech on the U.S. Entity List (Bureau of Industry and Security, 2024a), headquartered in the Netherlands and significant back-end operations in China (Poitiers and Schenk, 2026), Nexperia operates simultaneously under Dutch, EU, U.S. and Chinese regulations (European Commission, 2023; Bureau of Industry and Security, 2024a; Brattli, Meldahl and Hermann, 2025; Government of the Netherlands, 2025a; Nexperia B.V., 2025f). This combination of regulatory, ownership and supply complexity makes Nexperia a uniquely relevant case.

The developed framework consists of two stages. Stage 1 applies a regulatory and export-control eligibility gate (G1). Stage 2 evaluates eligible programmes using six weighted criteria, namely contractual severity, qualification lock-in, strategic importance, geopolitical and operational risk, gross margin, and demand stability. The weights are adjusted across immediate-response, recovery and future-growth horizons.

Customer prioritisation remains the strategic objective of this thesis. Because each OEM has multiple programmes with different contractual obligations, qualification lock-in, strategic relevance, profitability, demand stability and risk exposure, this thesis operationalises customer prioritisation using programme-level allocation. This also supports long-term customer relationships under constrained semiconductor supply.

Introduction

During September-October 2025, a series of events related to a single automotive semiconductor supplier brought a part of the global automotive supply chain almost to a standstill. The invocation of the 1952 Goods Availability Act and the temporary takeover of governance at Nexperia by the Dutch government on 30 September 2025 (Government of the Netherlands, 2025a), subsequently blocking the exports of Dongguan-produced Nexperia semiconductors by China on 4th October 2025 (Nexperia B.V., 2025f), appear to have contributed to global disruptions in the automotive supply chain during September-October 2025. Within weeks, its impact became evident through subsequent events. Warnings from European and Japanese automotive companies about the shortage of semiconductors (García-Herrero, 2025), temporary workforce suspensions of 300 to 400 workers at Bosch Salzgitter and approximately 650 at Bosch Ansbach locations (TrendForce, 2025c), 21% downward revision of Honda's operating profit, including approximately 150 billion yen attributed to the Nexperia disruption (WardsAuto, 2025) and the production cuts at Nissan, Kyushu (TrendForce, 2025c) suggest that operational disruptions at Nexperia may create customer prioritisation challenges across leading OEMs and Tier-1 operations.

Moreover, an increase in global electric car sales to 17 million (20% of total new car sales) in 2024 (International Energy Agency, 2025a), and the report by McKinsey predicting an increase in Level 2 driver-assistance systems, such as adaptive cruise control, lane-keeping assistance, and traffic jam assist, to be included in about 52% of vehicles by 2030 (McKinsey & Company, 2026b), accelerating the transition toward a software-defined and technology-centric automotive model by top carmaker BMW Group (BMW Group, 2026), suggest that the demand for automotive semiconductors is likely to increase in the coming years because each new function requires chips for power conversion, sensing, and control. Furthermore, as the automotive semiconductors such as AEC-Q100, AEC-Q101 have long qualification cycles (Automotive Electronics Council, 2021, 2023), once the semiconductor is qualified for a vehicle programme, it remains in the programme for the entire lifecycle of the programme, lasting up to 7-8 years (Center for Automotive Research, 2023). Due to long qualification cycles, changing the semiconductor supplier may not be easy, limiting the available response options during shortages, and during a semiconductor shortage Nexperia may not ask customers to wait. Moreover, an estimated removal of more than 9.5 million vehicles from production, causing the global automotive industry to lose over USD 200 billion in revenue in 2021 due to semiconductor shortages (Parliamentary Office of Science and

Technology, 2024) suggests that the automotive semiconductor shortages can create disruptions significant enough to affect global automotive production.

As discussed above, increasing dependency of the automotive industry on semiconductors, and its ability to disrupt the global automotive supply chain, suggests the need for a customer prioritisation framework under supply constraints.

Furthermore, with USD 2.06 billion in revenue in FY2024 (Nexperia B.V., 2025c), shipments of almost 110 billion components per year (Nexperia B.V., 2025c), operating under Dutch, EU, US and Chinese rules (European Commission, 2023; Bureau of Industry and Security, 2024b, 2024a; Government of the Netherlands, 2025a, 2025b; Sterling, 2025b; van Klinken, 2026), owned by Shanghai-listed firm Wingtech, headquartered in the Netherlands, relies on China for back-end activities such as cutting, packaging and testing before supplying semiconductors to customers in Europe (Poitiers and Schenk, 2026), makes Nexperia a unique case that represents a particularly relevant semiconductor supply constraint problem in a particularly pronounced form. For this reason, Nexperia is selected as the central case for the thesis to develop a customer prioritisation framework under constrained supply.

This thesis answers three connected research questions:

1. How can semiconductor suppliers (in case of Nexperia) prioritise automotive OEM customers under conditions of limited supply?
2. Which key factors (e.g. profitability, strategic importance, and risk) influence customer prioritisation decisions in such situations?
3. How can these factors be structured into a practical and applicable decision-making framework for suppliers?

This thesis also examines the level at which customer prioritisation is implemented and treats prioritisation at the level of individual vehicle programmes as the operational form of customer prioritisation rather than as a separate objective. Developed using publicly available information, this thesis is structured as the logical evolution of the concepts in five chapters. The thesis starts with understanding of the automotive semiconductor industry situation in Chapter 1. Chapter 2 discusses current and future trends in the industry. Further, the thesis analyses challenges, risks and opportunities in Chapter 3. After discussing the background, Chapter 4 develops and applies the framework for customer prioritisation. Once the framework is developed and tested, Chapter 5 answers the research questions, the contribution of the framework, its limitations and discusses potential future research and framework development in the same direction.

1. Current industry situation

1.1. Industry background and automotive semiconductor dependence

As reported by the UK Parliamentary Office of Science and Technology, a modern car contains more than 3,000 chips controlling functions from engine management to airbags (Parliamentary Office of Science and Technology, 2024). Therefore, semiconductors do not seem to be an optional requirement but appear to be an integrated part of modern vehicles. Moreover, the recent announcement by one of the biggest global automotive OEM BMW about moving towards a software-defined and technology-centric automotive model (BMW Group, 2026) may further set a trend for the future path for the semiconductor-centric vehicle architecture.

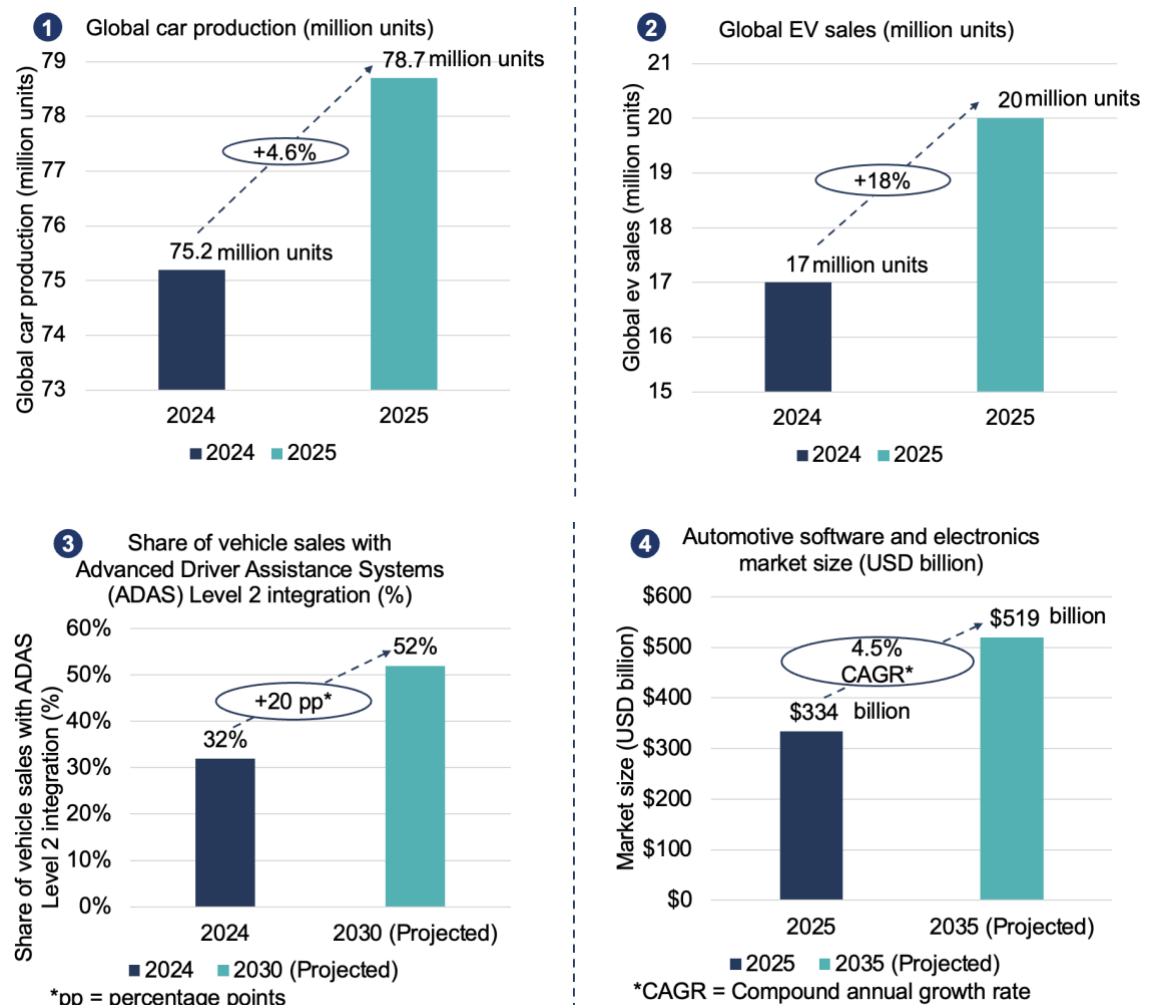


Figure 1: Indicators of rising automotive semiconductor dependence
Source: Own illustration based on International Energy Agency (2025a, 2026a), European Automobile Manufacturers' Association (2026), McKinsey & Company (2026b)

Figure 1 shows the rising dependence through one combined chart consisting of four panels: (1) global car production rising from 75.2 to 78.7 million units (+4.6%) in 2024-25, (2) global EV sales rising from 17 to 20 million units (+15%), (3) ADAS Level 2 penetration rising from 32% to 52% by 2030 (+20 pp), and (4) the automotive software and electronics market growing from USD 334 to 519 billion (4.5% CAGR). All four graphs generally move upward together, suggesting that automotive semiconductor demand is rising structurally.

Three distinct factors appear to push the use of semiconductors in the automotive sector: First, electrification, as shown in Figure 1 (Panel 2) with more than 20% of total cars sold in 2024, amounting to more than 17 million units (International Energy Agency, 2025a), being electric, with sales expected to reach 20 million units in 2025 (International Energy Agency, 2025a), show that the demand for electric cars appears to be rising at a rapid rate. As electric vehicles rely heavily on power electronics, including components such as onboard chargers, DC/DC converters and high-voltage inverters as compared to internal combustion engines (McKinsey & Company, 2026b), rising electric car demand also increases semiconductor demand in automotive applications. The second reason could be regulation. Requirements imposed by the European Union that every new car must carry safety and driver-assistance functions such as automated braking, lane-keeping and event data recording since July 2026 (European Commission, 2024), further make the use of semiconductors in vehicles legally mandatory, requiring the OEMs to demand more from automotive semiconductor suppliers such as Nexperia. The third reason is more structural in nature. Automotive semiconductor applications still depend heavily on mature-node categories which are receiving comparatively less investment as compared to AI and advanced computing *segments* (Kilian et al., 2026). Therefore, the rising market demand for electric vehicles and regulatory push by governments seem to increase semiconductor demand significantly. The rising automotive software and electronics market (Figure 1, Panel 4) also reflects the shift toward software-defined and electrified architectures, which, in turn, increases demand for the power, logic and protection devices that Nexperia supplies (Nexperia B.V., 2025a).

This increasing dependence of the automotive industry on semiconductors may become an allocation problem when the supply of semiconductors is also constrained. Nexperia supplies MOSFETs, logic devices, ESD protection and TVS diodes for ADAS, body, lighting, chassis, safety, conventional powertrain, and electrified powertrain applications (Nexperia B.V., 2017). Because these components supplied by Nexperia are embedded in many important vehicle functions, during supply shortages they may contribute to the stoppage of the production line. Due to stringent automotive qualification AEC-Q100/Q101 requirements and time required for such qualification processes (Automotive

Electronics Council, 2021, 2023), switching to another supplier if Nexperia does not supply these parts appears to be extremely difficult for customers and therefore non-supply by Nexperia may disrupt multiple OEM programmes simultaneously. However, depending on the criticality of the shortage component, not all programmes may be affected equally, therefore a programme-level consideration for supply prioritisation becomes necessary. As a result, the allocation challenge is not only operational but also strategic, because different programmes contribute differently to profitability, future design-in opportunities and customer relationships.

Inclusion of Wingtech, which is the parent company of Nexperia, on BIS Entity List in December 2024 (Bureau of Industry and Security, 2024a, 2024b), invocation of the Goods Availability Act in September 2025 by the Dutch government citing concerns related to strategic technology (Government of the Netherlands, 2025a), export controls restrictions imposed by China on Nexperia (Nexperia B.V., 2025f), uncertainty regarding the extension of restrictions imposed on Wingtech to Nexperia under the Affiliates Rule in September 2025 and the subsequent clarification (Brattli, Meldahl and Hermann, 2025) make supply route related issues more intense for Nexperia. Combined with constraints related to the limited availability of automotive mature-node semiconductors due to lower investment and increasing demand (Kilian *et al.*, 2026), appears to increase the complexity of allocation decisions for Nexperia. Even though Nexperia remained profitable and reported USD 300.3 million revenue in Q1 2026 (Nexperia B.V., 2026c), the value is significantly lower compared to USD 2.06 billion revenue reported for FY2024 (Nexperia B.V., 2025d). Therefore, serving low-availability semiconductor categories to more vehicle programmes may increase the importance of allocation decisions for each programme. This may create a trade-off between protecting current production volumes and supporting programmes with higher future strategic value. This, in turn, increases the importance of addressing the allocation issue at the programme-level where specific qualified semiconductor devices are allocated to specific vehicle programmes.

Supply of a wide variety of semiconductors by Nexperia to automotive customers includes safety-related components as well, making some components potentially more important than others. The Nexperia BUK9M airbag MOSFET illustrates this challenge. This component is specifically qualified for airbag deployment systems satisfying AEC-Q101 requirements (Nexperia B.V., 2022). Strict qualification requirements and the long duration of 12 to 24 months involved for such qualification and testing processes make the requalification of a new supplier for the same component practically difficult (International Automotive Task Force, 2016; Automotive Electronics Council, 2021, 2023). This also suggests that if Nexperia is unable to supply these components,

customers may not be able to obtain them from other suppliers as well. Because shortages of such low-volume but safety-related components may create comparatively more critical effects on vehicle programmes, such low-volume components may sometimes justify higher prioritisation than high-volume components that may have alternative suppliers available with comparatively lesser resistance. At the same time, Nexperia's SiC partnership with Semikron Danfoss (Nexperia B.V., 2026b) and GaN collaboration into advanced electrified powertrain and software-defined battery architectures with IAV (Nexperia B.V., 2026a) further reflects its expansion into higher-value automotive applications. As replacing such components once embedded in vehicle programmes may require significant qualification and validation efforts (International Automotive Task Force, 2016; Automotive Electronics Council, 2021, 2023), the optimised customer prioritisation decisions for constrained automotive semiconductor supply may become even more necessary because allocation decisions today may influence future design-in opportunities and eventually Nexperia's long-term strategic position.

1.2. Structural supply constraints, concentration and slow capacity expansion

While the previous Section 1.1 described why demand for automotive semiconductors is rising and highlighted why shifting to other suppliers is difficult for customers, this chapter analyses why the supply of semiconductors required for automotive applications cannot expand at the same pace.

With design, wafer fabrication, and assembly, testing and packaging (ATP) activities distributed across multiple countries and tightly interconnected segments of a semiconductor supply chain as described by the Organisation for Economic Co-operation and Development (OECD) (Organisation for Economic Co-operation and Development (OECD), 2025), the semiconductor supply chain appears to be highly complex and interconnected. As a result, any disruption at one point in the network may affect the broader semiconductor supply chain due to its high complexity and interconnectivity. Despite the globally distributed supply chain network, approximately 87% of global wafer fabrication capacity is concentrated in five economies, namely China, Chinese Taipei, Korea, Japan and the United States as reported by OECD (Organisation for Economic Co-operation and Development (OECD), 2025). Such high dependency on a concentrated geographical area may increase the vulnerability of the semiconductor supply chain as any disturbance, geopolitical or otherwise, in this area can affect the global semiconductor supply chain network.

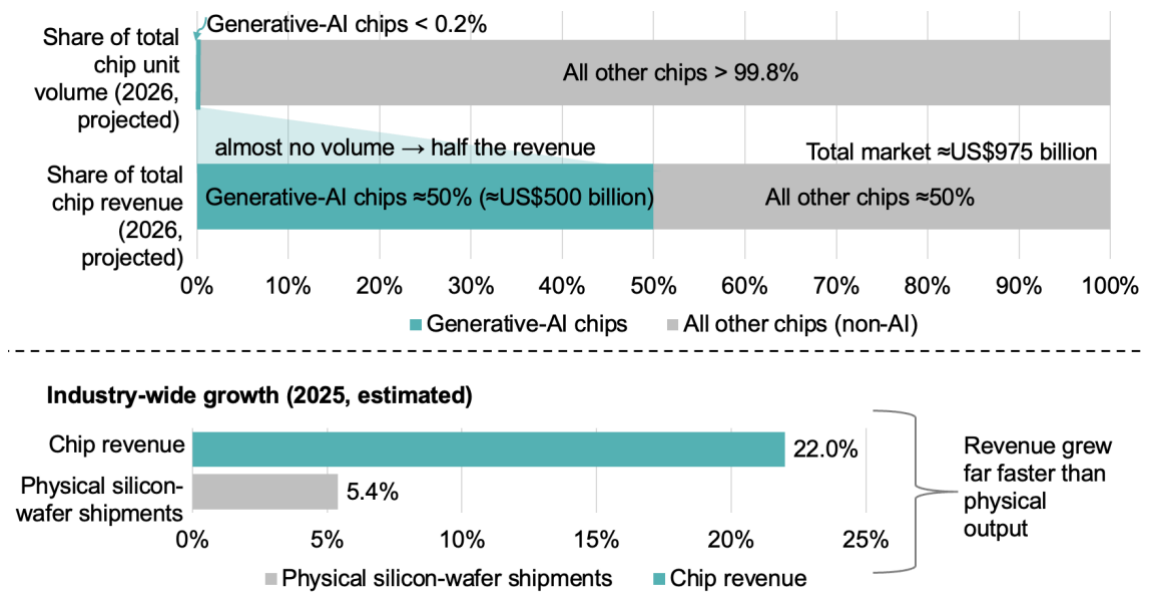


Figure 2: Divergence between chip volume and revenue (2026, projected) and between revenue and physical wafer-shipment growth (2025)

Source: Own illustration based on 2026 Global Semiconductor Industry Outlook published by Deloitte (Kusters et al., 2026)

Moreover, as shown in Figure 2, with Deloitte projecting the semiconductor industry to reach USD 975 billion in 2026, less than 20 million chips, representing roughly 0.2% of the total chips by volume, generative-AI chips are expected to contribute to about 50% of revenue (Kusters et al., 2026). The criticality is further shown by the lower part of figure 2, where it is shown that, as per estimation done by Deloitte, chip revenue grew by 22% in 2025, but the actual silicon-wafer shipments are only estimated to be increased by 5.4% in 2025. This shows that the revenue increase is not due to increase in chip sales, but it is mainly due to increased value per chip. Such high revenue generating chips for AI applications seem to be attractive and this may be the reason for the increased competition for the automotive industry to receive the capacity for automotive applications (S&P Global Mobility, 2025b; Astute Group, 2026).

Furthermore, Roland Berger reports that automotive applications compete with other industries such as consumer electronics, cryptocurrency mining and renewable energy applications for capacity due to high SKU complexity, stringent quality and reliability requirements associated with automotive semiconductors (Meissner and Kirschstein, 2021). The same report by Roland Berger also found that during 2020-22 semiconductor demand grew by 17%, but the capacity grew only by 6% overall and much lower in 40 nm logic, MEMS and legacy analog categories, which are widely used in automotive applications. This argument is further strengthened by a combined report published by Boston Consulting Group (BCG) and Semiconductor Industry Association (SIA) projecting the semiconductor investment to reach approximately USD 2.3 trillion between

2024 and 2032. However, 70% of future capital expenditure is expected to be directed toward sub-10 nm technologies which are used for AI, data-centre, networking and advanced computing applications, as well as selected automotive ADAS processors (Boston Consulting Group and Semiconductor Industry Association, 2024).

Analysing the above findings together does not show the investment issue in semiconductor industry overall, but suggests the profit-driven direction in which the investments are happening in the semiconductor industry. Because of the high revenue potential that generative AI chips generate and the complexity associated with automotive semiconductors, suggests that the problem that Nexperia has related to automotive semiconductor may not be temporary but structural. The mature-node scarcity may not be only a supply chain issue but may also reflect how investment decisions are motivated by revenue opportunities within the semiconductor industry. This observation also suggests a change of the perspective of allocation problem from a day-to-day crisis management task to a long-term structural challenge. This is because suppliers like Nexperia may not be able to fix this problem by waiting and expecting that more investment will flow in mature-node category because current revenue incentives favour generative AI chips. Therefore, Nexperia may need to decide which automotive programmes to protect when allocating limited mature-node capacity. This also makes customer prioritisation appear to be almost a long-term challenge and core operating requirement at Nexperia.

For Nexperia the situation appears to be even more complex because with USD 2.06 billion in revenue in FY2024 (Nexperia B.V., 2025c) and over 50% being contributed by the automotive sector alone (Rosina, Boulay and Yang, 2025), also Nexperia's core automotive products such as small-signal devices, logic ICs, and power discretes are manufactured on non-leading-edge silicon processes or as planar power structures that lack the revenue density of AI chips (Nexperia B.V., 2025a). As above reports show that mature nodes being the area where the semiconductors have to compete with other industries including AI requiring semiconductors due to profit attractiveness and also mature-node category receiving less investment, may create a strategic challenge for Nexperia because Nexperia does not deal with AI related semiconductors, its revenue opportunities may be comparatively lower than those available in AI-related segments. This may limit the speed at which additional mature-node automotive capacity becomes economically attractive for the broader semiconductor industry. As a result, even if the overall investment in the semiconductor industry continues to increase, automotive-relevant mature-node categories may still face constraints due to revenue factor favouring generative-AI related chips. Even though Nexperia's Malaysia and Philippines back-end expansion and partnership with Polar Semiconductor show Nexperia's effort

to expand the capacity, qualifying and ramping automotive-grade capacity takes time (van Gerven, 2025; Nexperia B.V., 2026d).

1.3. Structural allocation pressure and uneven customer conditions

Section 1.2 analysed that mature-node capacity cannot expand at the same pace at which the demand is growing due to investments tied to higher revenue attractiveness of other industries such as generative AI. The investment does not seem to be dependent on the needs of the automotive sector, creating supply and demand imbalance for mature-node semiconductors required for automotive industry. However, this imbalance further seems not to affect all the customers equally but appear to be partly rooted in customer's current focus and strategic goals. This unevenness of the supply and demand imbalance across customers is an important reason for turning this automotive semiconductor shortage into customer prioritisation at Nexperia in this thesis. Figure 1, panel 1, shows that the global vehicle production reached 78.7 million units in 2025, but this growth remained regionally differentiated due to tariffs and energy prices (European Automobile Manufacturers' Association, 2026). Furthermore, the supplier base is being divided between suppliers serving high-growth areas such as semiconductors, batteries and software, and those focused on traditional automotive components. This results in a widening performance gap between stronger and weaker segments of suppliers (Kilian *et al.*, 2026). Moreover, despite the strong growth of the semiconductor industry, strategic concerns of the industry due to significant operational and geopolitical risks such as tariffs and trade policies (KPMG, 2026) may push Nexperia to examine all the demand signals received from customers to ensure that customers are not inflating the demand signal expecting supply constraints in the near future due to the bullwhip effect (Lee, Padmanabhan and Whang, 1997).

Instead of analysing the above observation separately, observing how they interact with each other may make the observation more meaningful. Supplier-segment divergence due to the types of programmes they serve indicates an important factor that can influence the future strategic market position of the supplier, therefore, this might play an important role in customer prioritisation for scarce semiconductor capacity at Nexperia. On the other hand, possible demand inflation, causing a supplier to receive an inflated demand signal may not be as impactful a factor as the first one due to low trust in the demand signal. This may make demand information relatively a weaker basis for decisions precisely when supply is most constrained and the cost of error is highest. However, the demand signal still has to be considered in the customer prioritisation, since an allocation decision ultimately requires some estimate of real need. Consistency

of a customer's past demand provides a verifiable basis for testing whether a current signal is credible or inflated. Therefore, despite the low trust in the demand signal, its role is therefore to support the decision, but without being the main basis for the customer prioritisation under constrained supply.

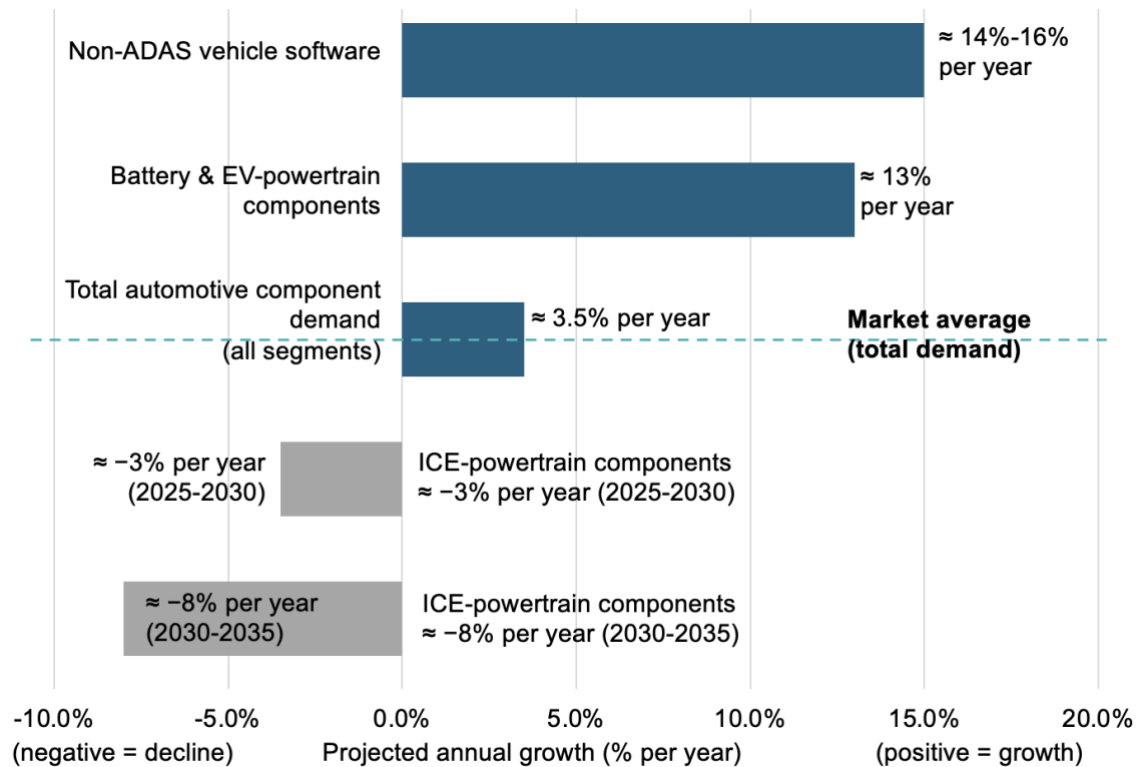


Figure 3: Diverging projected growth of automotive component segments, 2025-2035
Source: Own illustration based on BCG 2026 Global automotive supplier study report (Kilian et al., 2026)

Further on this discussion, Figure 3 shows the overall demand growth projection of automotive component of about 3.5% per year until 2035, demand growth projection of battery and EV-powertrain components of about 13% per year, demand growth projection of non-ADAS vehicle software of 14-16% per year, and at the same time a decline in demand of internal-combustion-engine (ICE) powertrain components projected at roughly -3% per year until 2030 and a decline of -8% until 2035, shows that the unevenness in growth at programme level is even more intense and sharp. This observation is also supported by International Energy Agency (IEA), which finds that electric-vehicle-specific components make up about 10% of total supplier market but have grown nearly seven times since 2019. On the other hand, ICE-specific components make approximately 20% of the supplier market and are shrinking (International Energy Agency, 2026b). Even though both BCG and IEA indicate the same structural shift independently, neither suggests what this structural shift means for Nexperia's allocation and customer prioritisation decisions.

An important analytical point that the above market studies do not make is that a customer can have a declining ICE programme as well as EV related programmes, that are expected to show growth in the coming years, running simultaneously. If this divergence exists inside a particular customer account, the account-level consideration hides the details that are present only at programme-level within the same customer account. An account-level customer analysis may prioritise the customer account, where the semiconductor components are being used in an ICE programme. ICE programmes may leave Nexperia tied to declining programmes and Nexperia may risk losing future EV design-in opportunities due to design-in requirements and replacement barriers associated with such programmes (Automotive Electronics Council, 2021, 2023). Because Nexperia's components reach around a hundred Tier-1 suppliers across powertrain, ADAS, infotainment and body domains (Rosina, Boulay and Yang, 2025), a single allocation decision done at Nexperia may touch many vehicle programmes of different values. Therefore, the allocation challenge appears to be more appropriately assessed at programme-level rather than account-level.

Customer prioritisation remains the strategic objective of this thesis. However, automotive OEMs do not represent a single, uniform unit for allocation decisions. This is because each customer consists of multiple vehicle programmes with different contractual obligations, qualification lock-in, strategic importance, profitability, demand stability and associated operational risk. Evaluating the customer only at account level may therefore not preserve these programme-specific characteristics that a particular customer has, that determine how scarce semiconductor capacity should be allocated.

For this reason, this thesis operationalises customer prioritisation through programme-level prioritisation. Programme-level prioritisation is used as the operational mechanism through which customer prioritisation is implemented. This does not treat programme-level prioritisation as an alternative objective but as the operational implementation of customer prioritisation framework at Nexperia.

Furthermore, customer prioritisation through programme-level prioritisation enables the relevant commercial, technical and strategic characteristics to be evaluated more accurately. Since every vehicle programme belongs to a customer, these individual programme-level allocation decisions may collectively support the broader long-term customer relationship objectives. Therefore, prioritising customers based on programme-level details may strengthen the customer relationship more effectively as compared to the scenario when the customer relationship is maintained without the individual details of customers' important programmes. Consequently, programme-level analysis appears more customer-centric than account-level analysis for customer

prioritisation because it preserves the programme-specific characteristics that collectively shape the customer relationship.

Overall, the above discussion suggests that automotive customers and their programmes do not expose Nexperia to the same values and risks under constrained semiconductor supply. Customer prioritisation decisions under scarce semiconductors appear to be heavily influenced by the strategic value of the programme, profitability and demand-signal reliability.

1.4. Government intervention, regionalisation and Nexperia's regulatory exposure

Sections 1.2 and 1.3 discussed how much can be produced and which demand needs attention. This section discusses which supplies are legally permitted. Nexperia remained profitable in Q1 2026 despite regulatory and operational disruptions affecting its China operations (Nexperia B.V., 2026c). Regulatory eligibility appears to be a separate screening compared to other parameters discussed in Sections 1.1, 1.2 and 1.3.

Regulatory eligibility appears to be different from business criteria: It is binary (permitted or not), prior (settled before commercial ranking), exogenous (cannot be priced or negotiated), dynamic (shifts within days of a regulatory act), route-dependent (status follows the assembly path), entity-dependent (European- and Chinese-origin parts differ), non-compensable (capacity or margin cannot offset ineligibility), and non-linear (one act blocks a whole flow, not just lowers its score). These characteristics make eligibility a gate, and not a parameter that can be negotiated and adjusted.

A technically qualified and commercially attractive semiconductor component may become ineligible for supply simply due to the geographical route involved in its supply chain because of regulatory and export controls through that route, which can be imposed by multiple countries at the same time. Historically, more than 70% of Nexperia's chips were routed through China for assembly, testing or packaging. Due to front-end fabrication activities in Europe and major back-end operations in China (Poitiers and Schenk, 2026; Tan, 2026), Nexperia remains heavily dependent on the Europe-China route. This front-end-Europe/back-end-China split appears to keep Nexperia exposed to regulatory risk and Bruegel characterises it as one of Europe's core vulnerabilities. Even though Deloitte expects a growing share of semiconductor back-end assembly and test activities to shift towards Southeast Asia and India (Kusters *et al.*, 2026), each route in the supply chain may attract its own regulatory constraints and may expose the supply flow to regulatory restrictions. Even though demand is there, products are technically qualified, any node in the supply chain can make the supply

illegal by exposing it to regulations and controls. This suggests that the same vehicle programme can be permissible if it follows one route and may become ineligible if it follows another route. Therefore, different routes involved in the supply chain act as a gate rather than a score and must be cleared before any commercial criterion is considered.

Even though Nexperia is ramping assembly and test capacity in Malaysia and the Philippines to reduce its dependence on Chinese operations (van Gerven, 2025) and Nexperia shows signs that its European and Chinese operations are becoming more separate from each other (TrendForce, 2026), China continues to appear to be a major node in the Nexperia supply chain. Farrell and Newman (2019) describe such points as chokepoints, making supply eligibility not only a supply chain issue but also a strategic issue for Nexperia.

Not only the supply chain route can make a supply ineligible to serve for Nexperia, but also the ownership can be a reason for supply restrictions. Appearance of Nexperia's parent company Wingtech on the U.S. Entity List and subsequent extension of the U.S. Entity List restriction of the parent company to the majority-owned subsidiaries by the US BIS Affiliates Rule automatically transferred restrictions on Wingtech to Nexperia as well (Bureau of Industry and Security, 2024a). Even though this restriction imposed by the US BIS Affiliates Rule is temporarily suspended until 9th November 2026 (Brattli, Meldahl and Hermann, 2025), Nexperia remains under risk as the future still holds uncertainty until November 2026. Morgan Lewis records that U.S. officials told The Hague that Nexperia would be considered controlled by Wingtech unless the CEO changes and corporate separation of Nexperia from Wingtech (Renner *et al.*, 2025). Therefore, a customer programme can still become ineligible to supply based on its ownership and on which country's law is applicable to Nexperia. In such cases, despite profit, or the available capacity, supply still gets fully blocked.

As most automotive semiconductor components have strict qualification criteria and procedures, which are also time consuming, switching the supplier may become very difficult for the OEMs across entire vehicle programmes (Automotive Electronics Council, 2021, 2023). Switching suppliers like Nexperia may not always be a question due to the semiconductor itself, it may emerge from the supply route, ownership, destination country etc. A semiconductor requirement cannot also be resolved by paying more if exposed to regulatory restrictions. Although Yangzhou Yangjie Electronic Technology Co., Ltd. was initially considered as an alternative for reduction of dependence on Nexperia's restricted supply chain by many automakers and many automakers also shifted to Yangjie, sanctions were imposed on Yangjie by the European Union in April 2026, and later a temporary nine-month exemption period until early 2027 was proposed

(Mariani, 2026). This suggests that shifting the supplier from Nexperia to another may not remove the regulatory risks. Such an alternate source may remove the regulatory exposure temporarily, but it may not be able to resolve the issue completely. Therefore, customer programmes may remain exposed to regulatory risks and the issue may not be removed by switching suppliers.

Even though Nexperia China reports that its MOSFET and logic-IC supply chains now operate largely as independent local operations (TrendForce, 2026; van Gerven, 2026) and Nexperia's Chinese and European operations appear to be moving towards a permanent operational separation (van Gerven, 2026), this may give rise to another problem. Parts from Nexperia Europe with back-end outside China might be eligible for supply but the parts from Nexperia China may not be eligible due to regulatory issues, so the part number would not be able to decide the eligibility of a component alone and its route and associated countries must be examined to confirm whether the component is really eligible for supply. This also means that two customers ordering the same automotive semiconductor component from Nexperia may face different regulatory exposure depending on its China or Europe connection. Furthermore, temporary suspension of the U.S. BIS Affiliates Rule until 9th November 2026 only protects Nexperia Europe until 9th Nov 2026 (Brattli, Meldahl and Hermann, 2025), but if imposed again, Nexperia may also be affected due to the presence of Wingtech on the U.S. Entity List. Therefore, restructuring can reduce one regulatory exposure but might create another problem. A supply chain independent from China can avoid regulations imposed on Nexperia due to China and also can avoid the China export control but Nexperia may still remain vulnerable to the Affiliates Rule as discussed above. On the other hand, a supply chain independent from Europe can avoid Dutch intervention but still carries China's Entity-List risk. As a result, for Nexperia, each solution appears to shift the exposure without eliminating it completely.

These discussions suggest that a programme may be eligible today but it can become ineligible overnight. The reverse also appears structurally asymmetric as restoring the same eligibility may require years, because making any programme eligible again for supply may take years of diplomatic negotiations, management adjustments, as seen in the case of Nexperia (Section 1.4). A supplier may be profitable, but still remain operationally vulnerable. A qualified device can be delivered through one route but may be blocked through another route in the supply chain. Therefore, the constraint may not only be the semiconductor itself but also can be different transactions in the supply chain. As demand for automotive semiconductors is rising through increasing electrification, regulation, and software-defined architectures (Section 1.1) and they are hard to replace with other suppliers due to constraints associated with changing the supplier once

qualified, the problem appears to be structural in nature. Semiconductors required for automotive applications also appear to be investment-deprived due to competition from other industries requiring semiconductors (Section 1.2). The same OEMs simultaneously may run declining ICE programmes with high current volume and growing EV programmes with high strategic value for Nexperia (Section 1.3). These forces appear to act simultaneously at Nexperia. While capacity scarcity raises the cost of any misallocation, programme divergence (one declining ICE, one growing EV) raises the risk of unknowingly supplying the wrong programme. One potential cost of error is therefore not only the loss of one customer but also the bigger loss of design-ins and eventually loss of market position of Nexperia along with the compliance cost of an attempted illegal delivery. These factors suggest that Nexperia should balance multiple factors simultaneously for customer prioritisation of automotive semiconductors.

2. Emerging trends shaping the automotive semiconductor industry

While the previous Chapter 1 analysed the current industry situation and established why the automotive semiconductor demand is increasing, why the supply and demand imbalance is structural and not a temporary issue, this chapter analyses current forward-looking trends that change which customers and programmes will matter for Nexperia in the future. This analysis and the differentiation of customers and programmes are important because the customer prioritisation decision built only on today's volume by ignoring such emerging trends could have a strong impact on the future competitive position of Nexperia.

Emerging trends discussed in this chapter have a significant impact not only on semiconductor demand, but also on the strategic value of Nexperia as well as automotive OEMs to which Nexperia supplies. This is due to the intense use of electronic components, software-defined vehicle architectures, connected-vehicle ecosystems, and advanced safety systems, which are increasing the importance of semiconductor-intensive vehicle platforms (BMW Group, 2026; McKinsey & Company, 2026b). Moreover, as discussed in Chapter 1, heavy dependence of the automotive industry on mature-node semiconductors, limited investment, lower profit margins compared to other industries and rising semiconductor demand from other industries combined with the increasing geopolitical tensions, regional industrial policies and restrictions appear to make the semiconductor allocation decision strategically significant for the future of Nexperia and automotive OEMs. This combined effect appears to significantly increase the complexity of the allocation decision for scarce capacity across automotive OEM programmes.

Furthermore, the complexity of the situation increases as the automotive sector depends on a wide mix of semiconductor products, ranging from power discretes, logic devices, analog ICs and microcontrollers to sensors, processors and semiconductors used in ADAS related systems (Nexperia B.V., 2017). As discussed in Chapter 1, even though the semiconductor industry is growing (KPMG, 2026), most investments are being done in the categories related to advanced computing and AI applications (Boston Consulting Group and Semiconductor Industry Association, 2024). Mature-node categories, on which automotive semiconductor applications still heavily depend, are receiving comparatively less investment as compared to AI and advanced computing segments (Kilian *et al.*, 2026). This investment direction within the semiconductor industry is critical for the thesis because customer prioritisation under constrained supply may not be separated from the technological direction of the market itself.

2.1. Electrification-driven demand shifts

Around 17 million unit sales of electric car in 2024, contributing to more than 20% of the total car sales, and a year-on-year growth of approximately 18%, exceeding 20 million units sales in 2025 globally, representing 25% of total units sold (International Energy Agency, 2025a), suggests that demand for electric vehicle is increasing at a rapid rate. As electrification increases the importance of semiconductor-enabled systems within the vehicle, including power electronics, energy-management functions, software-driven vehicle functions, advanced driver-assistance systems, sensors, computing capabilities, and increasingly centralised electrical and electronic (E/E) architectures (McKinsey & Company, 2025), suggests that the significance of this shift towards EV reflects the fact that an electric vehicle requires a different electronic architecture than a conventional combustion-engine vehicle. Consequently, as discussed in Chapter 1, automotive semiconductor demand appears to rise faster as compared to the overall automotive market growth. For Nexperia, this may change the perspective of “an important customer”. In the past, the concept of “an important customer” might often be linked primarily to production scale and purchasing volume, but today, due to the changing landscape of electrification as discussed above, future strategic relevance of a customer appears to be more dependent on how strongly and strategically the customer is positioned in electrified vehicle architecture platforms. This is gradually shifting allocation logic from being purely volume-based toward a broader assessment of future technological relevance and the long-term semiconductor demand potential of a customer.

The geography of vehicle production and sales further strengthens the above discussed point. With China accounting for more than 70% of global EV production in 2024 and nearly 75% in 2025 and sales of EV exceeding 11 million in 2024 and nearly 55% in 2025 (International Energy Agency, 2025a, 2026a), China appears to be leading the EV market growth and is likely to play a significant role in driving the automotive semiconductor growth globally. Moreover, the increase in EV imports from China, outside Europe and the United States, from 5% five years back to 55% in 2025 (International Energy Agency, 2026a), indicates that the demand is significantly growing at other geographical locations as well. Increasing dominance of China in EV production and international market presence, growing investments in semiconductor-intensive technologies such as SiC power electronics, software-defined vehicle architectures, and advanced battery-management systems (Nexperia B.V., 2026a, 2026b), suggest that the future semiconductor demand growth may be strongest in the regions driving faster electrification.

One of the other main destinations of this EV expansion is expected to be in Europe (International Energy Agency, 2025a). While China currently dominates EV production, Europe may remain strategically important due to its significant role in shaping regulatory standards, vehicle electrification pathways, and the implementation of advanced automotive technologies (European Commission, 2023, 2024). The implication is that the electrification trend may not be only a technological trend, but it also relocates the gravity of the future global automotive industry. Therefore, automotive OEMs in regions with faster EV scale-up are likely to generate higher future semiconductor demand and design-in opportunities, whereas OEMs in slower-transition regions may remain important for current profitability but may offer weaker growth. The implication is that the strategic relevance of electrification also carries a regional dimension within itself. A global and volume-only customer prioritisation might treat customers from these regions as interchangeable, but in reality the same purchasing volume from these regions may point to very different future demand depending on where the EV transition is moving fastest. Consequently, electrification appears to be reshaping not only what types of semiconductors are required, but also where future semiconductor demand is likely to emerge most strongly.

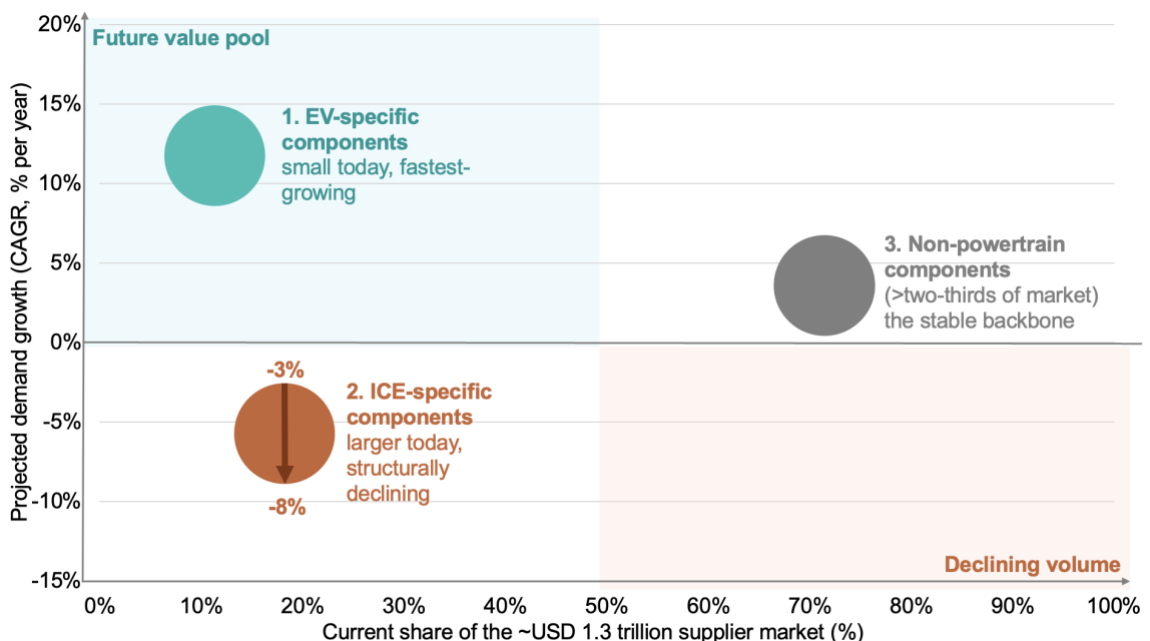


Figure 4: Current market size and future growth of automotive component segments:
The automotive component allocation conflict
Source: Own illustration based on International Energy Agency (2026a), Kilian et al., (2026)

Figure 4 maps current automotive supplier market size in percentage of approximately USD 1.3 trillion on X-axis and the projected compound annual demand growth rate (CAGR) of supplier market on Y-axis. Although the two axes of Figure 4 come from two

independent bodies, IEA and BCG, measuring different things, but both point to the same structural shift from ICE-related components toward EV-related components in the automotive industry. Analysing both findings together highlights an allocation conflict that neither IEA nor BCG answers independently. The IEA finds that EV-specific components are still only about 10% of the roughly USD 1.3 trillion supplier market (represented by Bubble 1 in Figure 4), but this market share has grown almost seven times since 2019. On the other hand, ICE-specific components make approximately 20% of the market share (represented by Bubble 2 in Figure 4), which represents double the market share of EV-specific components (International Energy Agency, 2026a). Non-powertrain components (Bubble 3 in Figure 4) make up more than two-thirds of the roughly USD 1.3 trillion supplier market and grow at a moderate, and almost stable rate, forming the stable base against which EV- and ICE-specific components move in opposite directions (International Energy Agency, 2026a; Kilian *et al.*, 2026). Despite the double market size as compared to EV specific components, the market share of ICE-specific components is shrinking. BCG, on the other hand, measures the projected growth rate, and finds EV-powertrain and battery content is rising at around 13% per year, while ICE-powertrain content is falling to roughly 3% per year until 2030 and then 8% per year until 2035 (Kilian *et al.*, 2026). Figure 4 combines the measurements from IEA and BCG to analyse their combined effect. The IEA identifies the conflict between current market size and future direction, but BCG quantifies the magnitude of that divergence and reinforces its strategic significance. The ICE-segment that makes the larger market share currently, is precisely the one in structural decline. Whereas the fastest-growing EV-segment is still small today but is showing growing tendency in future. By highlighting the broader automotive landscape, the combined result could support Nexperia in identifying which customer programmes may deserve higher strategic weight under scarce semiconductor capacity. If allocation decisions are driven mainly by today's volume, scarce capacity may continue to flow toward larger ICE-related demand, even though this demand is declining, while growing electrification demand may receive less attention over time. For Nexperia, current market share alone may not fully reflect future relevance. Allocation decisions should therefore consider a programme's strategic value and not only its present volume.

This uneven redistribution is critical for Nexperia because its portfolio covers both rising EV-related and declining ICE-related applications, while its newer SiC and GaN line is directly related to electrification (Nexperia B.V., 2017, 2019, 2024a). Therefore, electrification does not appear to lift Nexperia's demand uniformly across its product segment, but it tends to shift the demand toward specific power-electronics and wide-bandgap families and away from products associated with ICE applications. Extending

this argument, another implication of this is that if we consider the demand signal at account-level, we may supply the scarce capacity to the programmes associated with the declining ICE applications and the supply may not be allocated to growing EV programmes (Kilian *et al.*, 2026). Even though Nexperia can trace each part to its programme, a customer prioritisation ranking rule that operates on the account aggregate still collapses these opposite trajectories into a single priority ranking while making allocation decision, this is why account-based rule may misallocate the scarce semiconductor capacity at Nexperia.

2.2. Software-defined vehicles and centralised computing

Electrification is not the only trend increasing semiconductor intensity in a vehicle. The next major trend in the automotive industry increasing the intensity of semiconductors is through the vehicle architecture itself. Although Nexperia does not develop vehicle software (Nexperia B.V., 2017), software-defined vehicle (SDV) architectures depend on advanced semiconductor hardware to support centralized computing, connectivity, and over-the-air update capabilities (McKinsey & Company, 2026a). As a result, the transition toward SDVs may increase the strategic importance of semiconductor suppliers such as Nexperia within future vehicle architectures.

The automotive industry appears to be moving away from distributed electronic control systems toward more centralised and software-defined vehicles (McKinsey & Company, 2026b). Also, software-driven functionality is becoming more connected, customizable, and can be updated, while also becoming a source of competitive advantage, recurring post-sale revenue through subscriptions, on-demand feature activations. As also confirmed by BMW (BMW Group, 2026), demand for software-defined vehicles is increasing. OEMs increasingly develop vehicles around software platforms that allow updates, cybersecurity functions, and ongoing improvements. OEMs may become more competitive when they can use vehicle data to improve software and bring new features, fixes, or updates to customers. Therefore, the speed at which OEMs can improve and deploy software updates may influence their competitive position in the software-defined vehicle market (S&P Global Mobility, 2025d).

Given these circumstances discussed above, for automotive semiconductor suppliers, the OEMs who are moving towards centralised software-defined vehicle architectures may become strategically important as compared to those still tied to ICE-oriented designs or slower transition software-driven platforms. As recurring revenue is related to the initial installation of the software, the competition to secure initial platform installation may therefore become strategically important. This may shift the attention of Nexperia to software-defined vehicle programmes. As computing, connectivity and software become

central to vehicle functionality, future semiconductor demand becomes increasingly tied to the architecture OEMs choose. This argument is further supported by McKinsey's semiconductor analysis that estimates that around 30% of global vehicle production could feature zonal architectures by 2032. Furthermore, another McKinsey analysis also projects that around one-third of vehicle production could feature zonal architectures by 2035. The same study also estimates that zonal architectures can reduce material and engineering costs and will accelerate time-to-market by a significant margin (McKinsey & Company, 2026a). Moreover, BCG estimates that software-defined vehicles could create more than USD 650 billion in value for the automotive industry by 2030 (Koster *et al.*, 2023).

However, the deeper strategic insight for Nexperia may be understood as shifting the value of the allocation decision to the qualification stage and this could happen well before the actual volume materialises. This is because an OEM that decides to implement and win the software-defined architectures may not buy more semiconductors, but the platform Nexperia qualifies today may lock Nexperia in for the future business with that vehicle architecture. For Nexperia, this means a customer prioritisation rule based on current purchasing volume might unknowingly undervalue the OEMs whose zonal platforms are still in the design-in stage, even though these programmes may carry higher lifetime value per scarce capacity unit. For customer prioritisation, competitive position in the SDV market may serve as an indicator of future strategic relevance, since programmes associated with successful software-defined architectures are likely to generate sustained semiconductor demand once the architecture enters production.

Due to these significant factors, under constrained supply conditions, allocating semiconductors to such customers may support future design-in opportunities and long-term positioning in faster-growing architecture generations. At the same time, this shift to software-defined architectures may increase the importance of advanced processors while also increasing demand for a broader range of semiconductor components across the vehicle architecture (McKinsey & Company, 2026b). Due to this transition, more connected and software-intensive vehicles may require additional power management, protection, logic, interface and sensing components across the vehicle architecture. This may be particularly relevant for Nexperia as most of these surrounding semiconductor categories are already represented within the company's automotive portfolio including automotive MOSFETs, analog ICs and logic devices. As a result of this emerging trend, future customer prioritisation done by suppliers may depend not only on current semiconductor purchasing volume by a customer, but also on whether an OEM's future

vehicle strategy is strongly aligned with software-defined architectures that may generate sustained long-term semiconductor demand and related revenue.

This trend appears to be strategically very important for Nexperia as beyond this technological transformation alone, the regulatory bodies are also accelerating this transition toward software-defined vehicle architectures. For instance, from July 2024 onwards, for the new vehicles sold within the European Union, several software-, sensing- and safety-related functions such as ADAS capability, cyber-security measures, lane-keeping support and automated braking have been made mandatory (European Commission, 2024). In practice, this means that the software-driven architecture and electronics are no longer optional premium features but are being integrated into the regulatory baseline of modern vehicles.

However, even though this trend is emerging at a faster pace, vehicles already in the market will continue to exist along with the investments already done in production sites that still produce the traditional vehicle electronic architectures. These vehicle electronic architectures operate with many separate ECUs, older distributed electronic systems, conventional electronic layouts, and older vehicle platform designs. This will lead to a situation where these old vehicle architectures will continue to exist for many years alongside newer software-defined systems. This will ultimately lead to a prolonged hybrid transition period where semiconductor suppliers must balance allocations and business relationships with customers related to fast-growing software-defined platforms and customers still generating stable demand through traditional vehicle architectures.

Under constrained supply conditions, this hybrid period may force Nexperia to run two allocation logics at the same time that can protect the stable, high-volume demand of legacy distributed architectures, and can also secure the low-volume, design-in-stage capacity that future software-defined platforms will need. Because these two priorities can rank customers in opposite orders, an allocation decision considering only one factor could potentially ignore others in the decision. Therefore, a multidimensional decision framework for customer prioritisation seems to be a better fit in order to balance these factors while making the allocation decision of scarce capacity.

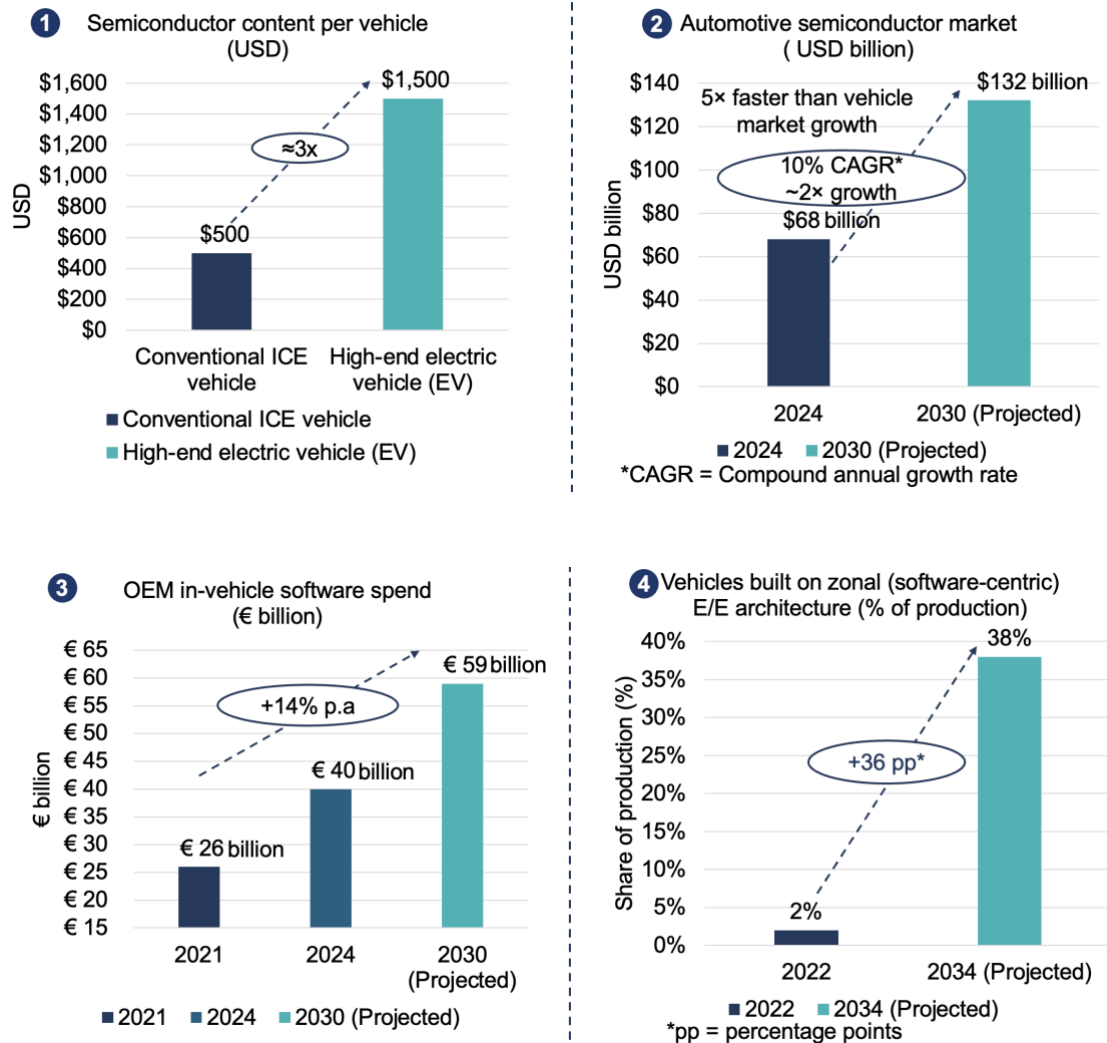


Figure 5: Shift to software-defined vehicles is raising semiconductor and software demand across the automotive industry

Source: Own illustration based on Dixon (2023), Robeco (2023), Shirokinskiy (2025), Yole Group (2025b)

Figure 5 combines four panels namely 1,2,3 and 4 to show how electrification and software-defined vehicle architectures are increasing semiconductor demand and show where future strategic value is emerging in the automotive industry. Panel 1 shows the intensity of semiconductor use in conventional ICE vehicles and inside high-end EVs and reflects how electrification structurally increases per-vehicle chip demand by almost three times (~USD 500 to ~ USD 1,500), (Robeco, 2023). Panel 2 extends this estimate to the market level, and reflects how the automotive semiconductor market is estimated to nearly double to \$132bn by 2030. This growth is five times faster than vehicle market growth itself (Yole Group, 2025b). This indicates that semiconductor demand in the automotive industry is increasing significantly. However, as analysed in Section 1.2, the semiconductor supply is constrained by structural factors and it is not a temporary issue, these structural constraints may prevent semiconductor production from expanding at

the same pace as demand. Therefore, allocating the limited capacity effectively becomes more critical in the coming years. The immediate managerial challenge is therefore no longer expanding production but deciding which customers or programmes receive limited capacity. That is exactly the problem this thesis tries to solve. Panels 3 and 4 take a deeper look and locate where that demand is concentrated.

Panel 3 and 4 of Figure 5 show OEM software spend and the shift to zonal, software-defined architectures are both expected to rise sharply (Shirokinskiy, 2025). As discussed in Chapter 1, Nexperia's portfolio consists of power discretes, MOSFETs, analog and logic devices, as well as newer SiC and GaN technologies (Nexperia B.V., 2017, 2024a), the greatest long-term value may be created when these components are designed into emerging vehicle platforms. Although Nexperia does not develop vehicle software (Nexperia B.V., 2017, 2024a), this trend makes Nexperia's portfolio relevant to the architecture transition as software-defined vehicle (SDV) architectures depend on advanced semiconductor hardware to support centralized computing, connectivity, and over-the-air update capabilities (McKinsey & Company, 2026a).

2.3. Mature-node dependence and the paradox of supply in a growing market

As discussed in Section 1.2, requirements for semiconductors in automotive applications are specific, and general growth and investment in the semiconductor industry shall not be understood as growth in the segment needed for automotive applications. One of the most consequential analytical errors a semiconductor supplier can make in the current market environment is to understand overall semiconductor industry growth as a signal that allocation pressures are reducing. When examined at the product-family and node-level, the evidence shows that the issue is further intensifying for mature-node semiconductors, which are mainly used for automotive applications and are relevant for Nexperia.

Planned capital investment is often read as a strong signal that an industry will grow and supply constraints will ease. According to Boston Consulting Group and the Semiconductor Industry Association (2024), a report entitled "Emerging Resilience in the Semiconductor Supply Chain," private investments in wafer fabrication are projected at about USD 2.3 trillion between 2024 and 2032 (Boston Consulting Group and Semiconductor Industry Association, 2024). At first glance, these numbers look very promising and suggest that the supply issue may get resolved to a significant extent. However, the potential managerial error lies in what these figures do not capture. The above industry report identifies supply chain vulnerabilities and issues in advanced packaging, and key materials as areas where challenges remain and finds that the

supply-demand issue could become more evident, despite the broader overall investment increase.

This point is further reinforced by a report issued by Roland Berger, which provides evidence of the problem through analytical clarity. The findings are critical for Nexperia and for the automotive industry as well, because conventional vehicles remain heavily dependent on mature and legacy semiconductor nodes. At the same time, the overall semiconductor demand increased by about 17% between 2020 and 2022, and with only a 6% capacity increase, the impact of this shortage may further intensify for the automotive industry due to its dependency on the legacy and mature node semiconductors. This is because semiconductor capacity expansions are primarily focused on leading-edge technologies (Meissner and Kirschstein, 2021). Therefore, despite increasing investments in semiconductors as a whole, the one required for automotive applications still appears to be investment-deprived. This could lead to a situation where the balance between supply and demand further deteriorates.

This loss is large enough that the effects can be measured significantly through real economic figures. AlixPartners reported that due to this shortage of semiconductors in 2021 alone, the automotive industry lost approximately USD 210 billion in revenue on account of the chip shortages. In 2024, this information was also quoted by the UK Parliamentary Office of Science and Technology (AlixPartners, 2021; Parliamentary Office of Science and Technology, 2024). This scale of financial loss due to scarcity demonstrates that even temporary supply interruptions in any of the mature-node components can rapidly translate into significant financial losses for the overall automotive industry. Due to this, for Nexperia, this asymmetry is strategically very important because the negative economic impact experienced by automotive OEMs in the case of a supply shortfall is significantly larger as compared to the lost revenue of the respective supplier in any single transaction. This large gap between customer-side consequences and supplier-side lost revenue may shift the bargaining power during shortages to semiconductor suppliers, and wrong allocation decisions may also damage customer trust and relationships in future business relationships.

Nexperia supplies to a broader range of end markets including automotive, industrial & power, AI and data centres, telecommunications, medical, building & home, mobile & wearables, and computing & consumer applications (Nexperia B.V., 2025a). As discussed in Section 1.2, semiconductor industry investments are most likely to follow higher-revenue opportunities despite the higher demand from the automotive industry (Kusters *et al.*, 2026). Therefore, even though the automotive segment contributes around 50% of Nexperia's revenue (Rosina, Boulay and Yang, 2025), future investment and capacity-expansion decisions across the semiconductor industry may increasingly

favour Nexperia's higher-growth and higher-margin end markets, potentially limiting the expansion of mature-node capacity widely used in automotive applications.

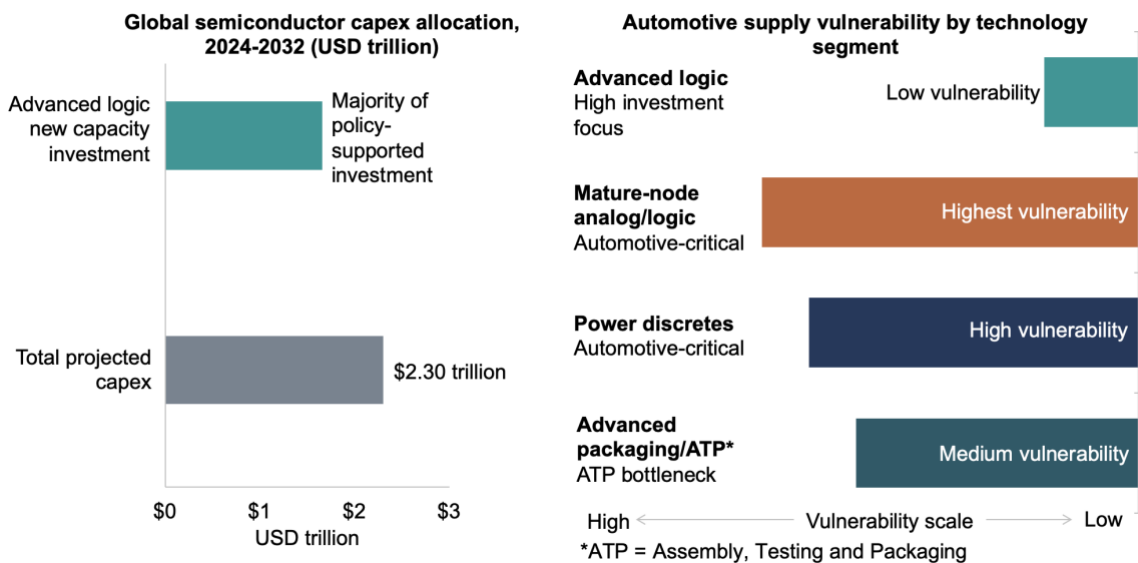


Figure 6: Global semiconductor capex allocation and automotive supply vulnerability by technology segment

Source: Own illustration based on AlixPartners (2021), Meissner and Kirschstein (2021), Boston Consulting Group and Semiconductor Industry Association (2024)

Figure 6 shows that the current semiconductor industry is not growing evenly. Advanced logic refers to leading-edge semiconductor technologies that are currently receiving a major share of new wafer fabrication investment, with global private-sector investment estimated at approximately USD 2.3 trillion between 2024 and 2032 (Boston Consulting Group and Semiconductor Industry Association, 2024). As a result, this segment is expected to experience lower supply vulnerability. In contrast, as discussed in Section 1.2, the automotive industry remains heavily dependent on mature and legacy semiconductors, while industry investment continues to focus primarily on leading-edge technologies rather than the legacy nodes on which conventional vehicles rely due to attractive commercials associated with it (Meissner and Kirschstein, 2021). Therefore, Figure 6 identifies mature-node analog and logic semiconductors as the segment with the highest supply vulnerability. Figure 6 also shows Power discretes as a highly vulnerable segment because capacity expansions receive only limited investments for discrete devices compared to leading-edge technologies (Meissner and Kirschstein, 2021). This is particularly relevant for Nexperia, as power discretes such as MOSFETs form a core part of its automotive portfolio, making this high-vulnerability segment directly tied to Nexperia's supply exposure (Nexperia B.V., 2017, 2025a). Advanced packaging and ATP (assembly, test and packaging) are shown with medium vulnerability because they remain an important area of supply-chain concentration and resilience concern

despite ongoing investment and geographic diversification efforts (Boston Consulting Group and Semiconductor Industry Association, 2024). Overall, the Figure 6 illustrates that the issue is not a lack of investment in the semiconductor industry, but a mismatch between where investment is concentrated and where automotive demand remains structurally dependent.

2.4. Geopolitical reconfiguration of semiconductor supply, regionalisation, and industrial policy

The above structural supply constraints are further intensified by regional politics, geopolitical tensions and frequent government interventions in semiconductor related areas. The decision regarding semiconductor manufacturing was taken earlier by the companies based on the supply chain optimisation and commercial factors. But after 2020, governments appear to be started treating it as a matter of national security. Regional governments seem to believe that matters related to semiconductors are strategically critical for national security and technological independence, and due to this, these matters cannot be fully left for private firms to decide (Shivakumar, Wessner and Howell, 2024). For Nexperia, this means allocation may no longer be a purely commercial decision but must also account for regulatory and geopolitical constraints outside the firm's control.

As a consequence, regional governments are making policies and regulations that strongly influence how produced semiconductors can be distributed among global customers. If we consider the three major global economies, we find that all these regions have some regional restrictions on the distribution by law. The European Chips Act gives the government the power to intervene in the supply and force companies to prioritise customers in Europe (European Commission, 2023). In the United States, the semiconductor subsidies given to companies also puts restrictions on expanding the manufacturing facilities in politically sensitive areas, especially in China (U.S. Department of Commerce, 2023). Due to all these interventions and laws, Nexperia may no longer be able to decide the semiconductor supply allocation independently based purely on commercial factors as regional laws and political restrictions override the Nexperia's own business decisions. As discussed in Section 1.4, Nexperia appear to be most affected by this recent trend being observed in semiconductor industry and this makes Nexperia an interesting case to study.

Nexperia operates both in Europe and in Asia (Nexperia B.V., 2026e) and as each region has its own regulations and as Nexperia operates there, it must also fulfil these regional requirements while fulfilling customer's supply commitments. Therefore, any customer prioritisation framework that considers only commercial factors and ignores regional

governmental regulations might fail during the allocation decisions as law and regulation is always bigger than Nexperia's own supply commitments despite commercial attractiveness of the vehicle programmes.

Even though these regulations and restrictions imposed by governments aim to achieve regional technological independence, developing a semiconductor facility is extremely costly, investment-intensive and slow (Boston Consulting Group and Semiconductor Industry Association, 2024; Organisation for Economic Co-operation and Development (OECD), 2025). The European Chips Act aims to achieve 20% of global semiconductor production by 2030 (European Commission, 2023) but the European Court of Auditors warns that with the current level of investments, it will be extremely difficult to achieve the target by 2030 and at the time the target was established, the EU did not possess manufacturing capabilities below 22 nanometres (nm) (European Court of Auditors, 2025). Nevertheless, rules, regulations and policy pressure appear to be already creating restrictions on semiconductor supply but the benefits of increased production capacity and improved supply conditions anticipated from these policies and initiatives have not yet fully materialised (European Court of Auditors, 2025).

Another layer of complexity is also observed due to geographical distribution of the regions where automotive production is concentrated and where the governments are restricting the semiconductor companies from selling or producing. Apart from the factors analysed in Section 2.1 regarding China, China also accounts for approximately 35% of global light vehicle production and Asia-Pacific as a region accounts for around 60% of global production in 2024. China also leads in EV production and building domestic semiconductor facilities at scale. Europe accounts for approximately 18% of light vehicle production (German Association of the Automotive Industry, 2025). The U.S. CHIPS Act restricts the semiconductor suppliers to open production facilities in regions of concern for ten years (U.S. Department of Commerce, 2023).

Above discussion shows that a clear pattern is emerging, Asia-Pacific is the biggest market for automotive semiconductor suppliers such as Nexperia, and the regulations imposed by European governments, United States and China (European Commission, 2023; Bureau of Industry and Security, 2024a; Tan, 2026) might be restricting Nexperia to capture this growing market. This may become a business dilemma for any automotive semiconductor supplier. Suppliers may try to fulfil the political expectations very carefully by reducing dependencies on China or executing supply decisions that satisfy these laws, but at the same time, Asia-Pacific region, especially China, is growing at a higher pace and if a supplier does not capture the market, then there might be chance of losing market share, future growth potential and long-term strategic importance in the region. Another issue arising from this is that it will give rise to more

competition globally. This may also invite more competition. Other companies could enter to fill the demand Nexperia is restricted from serving, eventually weakening Nexperia's global position relative to Europe and the United States. While making allocation decisions, balancing multiple conflicting dimensions is the biggest challenge for Nexperia considering the long-term future growth potential.

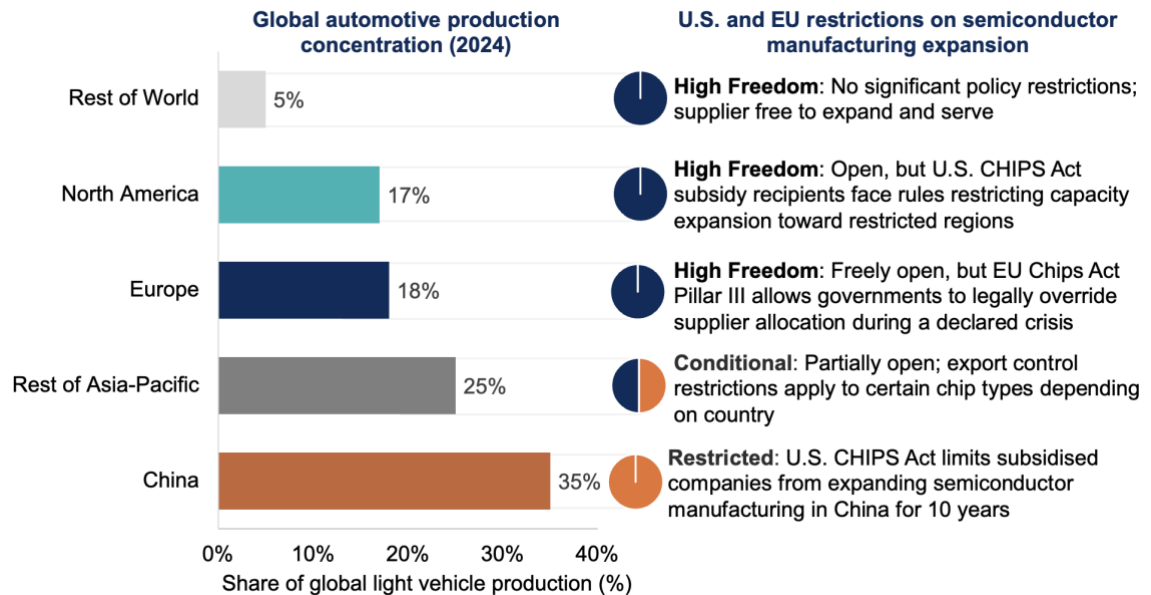


Figure 7: Global automotive production concentration and degree of regulatory restrictions imposed by U.S. and EU semiconductor policies on manufacturing expansion in those regions

Source: Own illustration based on European Commission (2023), U.S. Department of Commerce (2023), European Court of Auditors (2025), German Association of the Automotive Industry (2025)

Figure 7 shows that the Asia-Pacific region including China has around 60% of the vehicle production and this is also the region where semiconductor suppliers face the strongest political and regulatory restrictions from the United States and Europe. Governments and semiconductor policies are creating restrictions for a semiconductor supplier to expand facilities in these areas, especially in China. The figure also shows a regulatory gradient: suppliers remain free to expand in most regions, but they face conditional export controls in the rest of Asia-Pacific including China, which is precisely the highest-production region. This is a significant long-term strategic risk for Nexperia. This suggests that any allocation decision made by Nexperia should fulfil both regulatory restrictions and commercial interests. An allocation framework that treats both separately will be either legally exposed or commercially suboptimal.

2.5. Relationship restructuring and governance implications for constrained allocation

The current market trends are not limited to the supply and demand problems only, but the relationship landscape between the OEMs and semiconductor suppliers is also changing. Earlier OEMs mainly focused on Tier-1 suppliers and communication with semiconductor suppliers was established through Tier-1 suppliers (Arvidsson, Garcia and Govik, 2026).

However, after 2021, OEMs increasingly tried to achieve greater visibility and security across semiconductor supply chains. Research conducted by Capgemini Research Institute (2023) revealing 50% of OEMs did not consider their semiconductor supply sufficiently secure, and 56% reported investing in a secure semiconductor supply capacity apart from the traditional Tier-1 and Tier-2 structure, and the declining automotive procurement from offshore locations by 22% between 2021 and 2023, indicate that the improving transparency and supply reliability were the main drivers for this pattern (Capgemini Research Institute, 2023). This report further highlights that OEMs have longer planning horizons, but information communicated to suppliers through Tier-1s for shorter periods, highlighting the same core problem from both sides. This is because OEMs realised that the lack of direct relationships and transparency with semiconductor suppliers was affecting their production visibility and delivery stability while semiconductor suppliers also observed that they were not getting correct and sufficient inputs to plan their allocation and operational inputs. This also indicates that OEMs and semiconductor suppliers have now identified this and started to solve this structural communication and transparency gap by establishing direct communication and cooperation (Arvidsson, Garcia and Govik, 2026).

These findings suggest that the semiconductor shortage may not only be a capacity-related problem but also appear to be a transparency problem related to information sharing. The traditional automotive sourcing may have created a system in which semiconductor suppliers such as Nexperia made allocation and capacity decisions based on indirect demand signals communicated through Tier-1 suppliers. As a result, both sides operated with incomplete information, increasing the risk of non-optimum allocations and planning differences between what OEMs needed and what suppliers optimised their allocation decisions for. The growth of direct OEM and semiconductor supplier relationships therefore represents not only a change in communication structure but also a structural attempt to reduce information imbalance across the automotive semiconductor supply chain. For suppliers such as Nexperia, this implies that customer prioritisation increasingly depends not only on commercial attractiveness or strategic

importance, but also on the credibility and reliability of the demand information provided by customers, as inaccurate demand signals can lead to inefficient allocation of constrained semiconductor capacity.

To address these transparency and information-sharing issues, industry initiatives such as Catena-X are being developed. By supporting standardized multi-tier data exchange across the automotive supply chain, Catena-X tries to improve visibility and reduce information gaps between OEMs and suppliers (Catena-X Automotive Network e.V., 2024). As a result, allocation decisions are expected to be based on greater visibility into demand and supply information across the automotive supply chain rather than forecasts based on historical purchase patterns. This may eventually lead to verified and trusted demand signals and OEM programmes may receive more optimum supply allocations, thereby reducing uncertainty in the real-time information exchange and related risks.

Insufficient information exchange and transparency may lead to suboptimal allocation, which can force denial of orders for important OEM programmes when capacity is insufficient to satisfy demand. Ramani, Ghosh and Sodhi (2022) show that semiconductor shortages and supply constraints can lead to line-stop events and operational delays. Such disruptions may also influence how suppliers are evaluated by OEMs and may create risks for customer relationships. This reflects that a denial is not only the immediate loss of revenue for the supplier but may also create longer-term commercial consequences. To protect future interests, OEMs may reduce their reliance on the same supplier and trust may decline. The potential cost of a denial therefore may not only be considered the margin on the denied order itself, but may appear to affect a broader range as the risk-adjusted value of the entire customer relationship placed at stake.

The trends discussed in Chapter 2 demonstrate a complex emerging situation and show why allocation decisions require a multidimensional decision framework at Nexperia. Ongoing geopolitical developments may further increase the complexity of allocation decisions at Nexperia. On one side, the emerging trends such as electrification, software-defined vehicle architectures, mature-node dependency may strengthen the bargaining position of semiconductor suppliers to such an extent that OEMs are more closely involved in managing direct relationships with semiconductor suppliers, but at the same time, denial of orders related to OEM programmes may weaken supplier evaluations and create long-term customer relationship risks. Therefore, the next chapter examines the major risks, challenges, and strategic opportunities that further intensify this allocation dilemma under constrained supply conditions.

3. Analysis of challenges, risks, and opportunities

This chapter analyses challenges, risks, and opportunities affecting Nexperia's customer prioritisation and converts them into managerial implications for customer prioritisation. First, this chapter discusses the challenges affecting allocation decisions. Second, this chapter analyses the risks that could arise if the allocation decisions fail to reflect the strategic direction of Nexperia. Lastly, this chapter also examines how a good allocation decision can create opportunities for Nexperia if implemented correctly. With USD 2.06 billion in revenue in FY2024 (Nexperia B.V., 2025d) and a dominant share being contributed by the automotive sector (Rosina, Boulay and Yang, 2025), decisions made for automotive semiconductor allocation may not only change Nexperia's current revenue, but also its future access to automotive programmes and Nexperia's future market position.

Although Nexperia's headquarters are situated in the Netherlands, the company is owned by Wingtech Technology, which is a Shanghai-listed Chinese firm, and has its largest back-end facility in Dongguan, China (S&P Global Mobility, 2025a), this unique structural condition may further intensify Nexperia's allocation challenges by placing Nexperia at the intersection of Dutch, Chinese, and U.S. regulatory jurisdictions as discussed in Section 1.4 (U.S. Department of Commerce, 2023; Bloomberg, 2025; Government of the Netherlands, 2025a; S&P Global Mobility, 2025a). Poitiers and Schenk (2026) use Nexperia's geographical distribution with the front-end in Europe and the back-end in China, as this is also a common feature of the semiconductor industry across different geographical regions, to show that this cross-regional interdependence must be managed instead of being eliminated. Because of these parallel factors, allocation decisions at Nexperia can be influenced by both commercial objectives and regulatory requirements imposed by multiple countries. Herman Quarles van Ufford (2025) further argues that the Dutch intervention in Nexperia became part of the wider US-China technology conflict and may create a risk of disrupting automotive supply chains instead of protecting Dutch strategic interests.

Additionally, as discussed in Sections 1.3 and 2.3, the capacity expansion in mature-node technologies appears relatively slow compared to the increasing demand for automotive semiconductors (International Energy Agency, 2025b; Semiconductor Equipment and Materials International, 2025). Because of this supply and demand imbalance, the allocation pressure at Nexperia may become increasingly complex and critical.

3.1. Structural challenges in supplier decision-making

Demand signal distortion: When there is an anticipation of supply constraints, OEM and Tier-1 customers may increase order quantities and reduce lead times to secure available material, as observed during the 2025 crisis when Nexperia's controlled order entry process was triggered by such behaviour (TrendForce, 2025a). Lee, Padmanabhan and Whang (1997) define this phenomenon as the bullwhip effect, where the inflated projections and demand signals given by customers become further amplified while moving upward through the supply chain. As Nexperia's semiconductor components are being used in multiple automotive parts including body electronics, powertrain, chassis, and also for overall electrification purposes to nearly a hundred Tier-1 suppliers (Nexperia B.V., 2017, 2025a; Rosina, Boulay and Yang, 2025), there is a possibility that distorted demand signals further compound across multiple product lines simultaneously due to the bullwhip effect discussed above. The automotive industry's long qualification cycles and the limited number of suppliers available for qualified parts may intensify the situation even further. As a result, a Nexperia customer with high future demand potential may not receive sufficient quantity today, but customers exerting greater bargaining pressure may receive inflated quantities, which may not represent actual demand. Also, a current high-volume customer may not be the future high-volume customer and vice versa. Consequently, if Nexperia relies heavily on projected customer demand and is influenced by the bargaining pressure of the customer, then its allocation decision can become suboptimal, diverting constrained capacity, for example from its Hamburg-fabricated wafers (Nexperia B.V., 2024a) to inflated orders rather than verified demand. For Nexperia, a large portfolio of AEC-Q100/Q101 automotive components (Nexperia B.V., 2025e), which are being used across body electronics, powertrain, chassis, and electrification applications, may become directly affected by distorted customer demand signals. Even though the individual value of these components is relatively low, due to the large-scale use of these components across automotive applications, any inflated allocation to non-verified demand can result in lost opportunities elsewhere as manufacturing capacity remains constrained at Nexperia. Nexperia therefore may require an allocation framework that can differentiate between actual programme demand and the inflated demand from customers.

Multi-tier programme opacity: As Nexperia's automotive supply chain is highly multilayered, almost all major OEMs contain Nexperia components in their vehicles, and they may not even be aware of it (Rosina, Boulay and Yang, 2025). This indicates that Nexperia's components may be flowing through Tier-1 and other lower-tier suppliers before reaching OEMs. As is the case in hub-centric supply chain structures, lower-tier supply chain visibility remains limited for the suppliers (Garcia, Arvidsson and Jonsson,

2024), Nexperia's supply chain visibility across the broader downstream automotive supply chain may also remain limited. Therefore, during semiconductor allocation, these lower-tier suppliers or geographical regions in which the supplied components are used may remain unclear to Nexperia because the supplied components can move to other lower-tier suppliers depending on the overall vehicle programme structure. Furthermore, as the origin of semiconductor components cannot always be clearly identified within complex automotive supply chains (Bureau of Industry and Security, 2024b), this creates a possibility that semiconductor products from Nexperia might unknowingly reach China through the supply chain. Given Nexperia's scale of operations and product portfolio (Nexperia B.V., 2017, 2025a, 2026e), this may create a direct allocation problem for Nexperia. A customer account may appear commercially and regulatorily safe at the immediate level, but may carry risks that are not visible at the time of Nexperia's allocation decision. With approximately 80% of Nexperia's back-end capacity being centred in mainland China (S&P Global Mobility, 2025a), regulatory interventions can significantly affect supply availability and allocation outcomes for Nexperia.

Therefore, it is important that the decision framework also includes transparency across lower tiers. Otherwise, suppliers may allocate the scarce capacity to programmes that may lead to regulatory, geopolitical and strategic risks at later stages of the programme. As discussed in Section 1.4, with back-end operations in Dongguan, China, and customers distributed across China, Europe, North America, and Asia (Nexperia B.V., 2026e; Poitiers and Schenk, 2026; Tan, 2026), any lack of transparency issue in multi-tier customer visibility could potentially create regulatory and geopolitical exposure for Nexperia. As discussed in Sections 1.4 and 2.4, regulatory restrictions from multiple countries can emerge suddenly for Nexperia. Such unpredictable situations related to multi-tier transparency may unknowingly expose Nexperia to regulatory restrictions and government controls. This highlights the importance of a dedicated regulatory compliance screening layer within Nexperia's allocation framework so that the uncertainty can be minimised and documented.

Dependence on major customers and power asymmetry: With over 50% of Nexperia's USD 2.06 billion revenue in 2024 derived from the automotive industry and nearly a hundred Tier-1s distributed across powertrain, ADAS, infotainment, and body domains (Rosina, Boulay and Yang, 2025), any supply shortage could create allocation conflict between competing automotive programmes. This, in turn, can influence a broader set of downstream automotive programmes. Strongly dominant sales to automotive customers (Nexperia B.V., 2025d; Rosina, Boulay and Yang, 2025) make Nexperia's financial performance and future participation of Nexperia in automotive programmes highly dependent on maintaining relationships with automotive customers.

Because of this, semiconductor allocation decisions become strategically very significant for Nexperia's future programme participation, customer relationships, and long-term strategic positioning. Jiang *et al.* (2023) show that a high dependency on a few customers reduces the flexibility of a supplier because during periods of limited availability or disruption, suppliers tend to prioritise these high-revenue customers (Jiang *et al.*, 2023). This may be driven by fear of losing major customers along with potential negative effects on quarterly revenue performance. This may create a strategic imbalance in Nexperia's allocation decisions. Nexperia's 4% market share in silicon MOSFET discretes (Yole Group, 2025a) suggests that Nexperia's future market position depends heavily on retaining strategically important customer relationships and future programme participation. Under such conditions, any allocation decision that negatively affects these relationships may weaken Nexperia's competitive position in comparison to Nexperia's larger competitors such as Infineon, onsemi, and STMicroelectronics.

On the contrary, however, such allocation decisions based purely on volume and power asymmetry may deprioritise strategically important future Nexperia customers. Once deprioritised during shortage periods by giving more priority to a limited number of high-volume customers, these deprioritised customers may be less likely to return in the future due to loss of trust. This dependence may influence the future strategic position of Nexperia. Allocating constrained volume only to current major revenue-generating customers may limit Nexperia's access to future strategic programmes. The dominance of 91.9% of the SiC power device market by five suppliers, namely Infineon, STMicroelectronics, onsemi, Wolfspeed, and Rohm (TrendForce, 2024), reflects Nexperia's relatively limited position in this market. Nexperia's USD 200 million Hamburg investment (Nexperia B.V., 2024a) represents a significant step towards strengthening Nexperia's market position. Under such conditions, any allocation decision that weakens future design-in opportunities may not only reduce the strategic value of Nexperia, but may also reduce the strategic use of Nexperia's Hamburg investment.

Qualification lock-in and design-in lifecycle economics: An automotive semiconductor supplier can participate in a vehicle programme only after qualifying for certain industry standards related to the product. For automotive semiconductors, these qualifications include AEC-Q100 standards for integrated circuits and AEC-Q101 standards for discrete semiconductors as standardised by the Automotive Electronics Council (Automotive Electronics Council, 2023). Once qualified, the components can be integrated into a vehicle programme. As these qualification cycles are time-consuming and technically demanding, they require effort from both OEMs and suppliers, changing suppliers after qualification is time-consuming and costly (Center for Automotive Research, 2023) and it may impact customer operation, therefore, customers are less

likely to switch suppliers after the qualification process. This also means that once Nexperia has qualified for a programme, the business for the entire 7-8 year lifecycle (Center for Automotive Research, 2023) of the vehicle programme may become relatively secured for Nexperia. This may significantly reduce the business uncertainty for the qualified products from competitive pressure and may also strengthen future business opportunities. Thus, the qualification process itself becomes very significant for the future business relationships with customers. As discussed above, customers may not wish to invest time and money again in repeating the qualification cycle. Therefore, the business may become relatively secure for Nexperia once the initial design-in has been approved. This, in turn, creates a structural allocation challenge because the future demand potential, future strategic opportunities, and related potential future revenue are not reflected in the current order volumes. For Nexperia's SiC MOSFET portfolio, this may become one of the most challenging issues. The announcement by Nexperia to invest USD 200 million in Hamburg to expand SiC and GaN manufacturing capacity (Nexperia B.V., 2024a) may provide opportunities for Nexperia's future business, but at the same time, commercial use of this investment is more likely to be realised if future electrification programmes are identified today and the customer relationships are developed today to secure future design-ins. As discussed above, once an alternative supplier is selected for a vehicle programme, that may remain locked for the entire programme lifecycle, and this can represent a strategically significant loss of potential revenue for Nexperia.

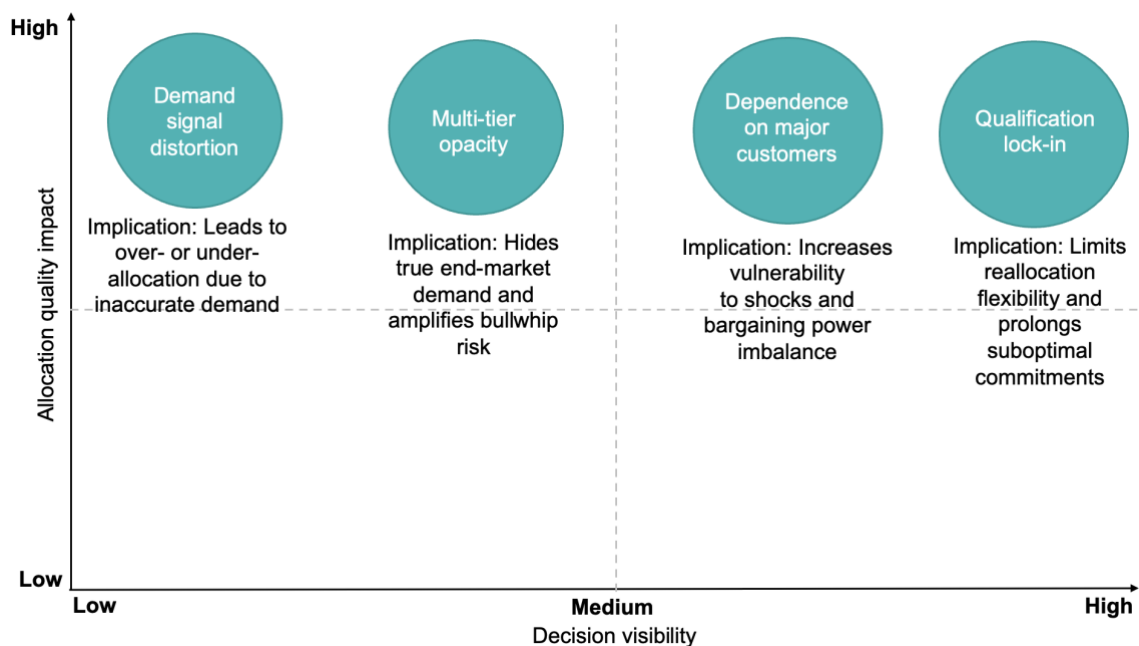


Figure 8: Structural challenges affecting Nexperia's allocation decision quality: Decision visibility vs. Impact on allocation quality

Source: Own illustration

Figure 8 maps each of the challenges discussed above with respect to decision visibility and their degree of impact on the allocation quality. Due to inflated demand from customers, strong customers' bargaining power and fear of losing customers may contribute to the low visibility of demand signal distortion and due to this practice, the volume impact could also be significant. Due to this, this challenge is positioned at the top-left corner of the diagram. A similar volume impact is also noticeable with "Qualification lock-in". This challenge appears to have significantly higher visibility due to RFQs and other development information from customers about vehicle programmes already available to Nexperia. However, the challenge of "Multi-tier opacity" seems to have low visibility, but as discussed in Section 2.5, with the help of governance tools such as Catena-X DCM (Catena-X Automotive Network e.V., 2024), this can be improved to some extent. Therefore, this sits near the medium-to-low visibility area. But as the impact is related to regulations and policies introduced by governments, the volume impact remains higher. Even though "Dependence on major customers" seems fully visible, it appears to be resistant to change due to current revenue dependence on these major customers. The volume dependence is high as this deals only with major customers of Nexperia. Thus, for Nexperia, an effective customer prioritisation framework seems to reduce distortion-driven misallocation in the high-volume portfolio while structurally protecting SiC and GaN qualification-stage capacity.

3.2. Major risks generated by prioritisation errors

Economic misallocation: During disruptions, a supplier's flexibility is highly reduced by dependence on high-volume customers (Jiang *et al.*, 2023). 50% of the USD 2.06 billion revenue makes Nexperia highly dependent on automotive customers (Rosina, Boulay and Yang, 2025). This high dependence of Nexperia also reflects that under constrained supply conditions, high-volume automotive accounts may gain bargaining power and may potentially reduce the economic value generated from Nexperia's constrained capacity. As discussed in Section 3.1, this might be driven by fear of losing major customers and to maintain the trust gained from these customers over time. This may also weaken Nexperia's bargaining power because the volume purchased by such customers constitutes a major share of Nexperia's sales volume. Due to this high-volume dependency, Nexperia may need to offer the available semiconductors at comparatively lower prices to these automotive customers. Additionally, to maintain the volume and customer relationships, Nexperia may also offer better delivery terms to such customers along with other favourable commercial terms that may be economically unfavourable to Nexperia. As a result, Nexperia may allocate constrained Hamburg and Dongguan capacity to customers that contribute to Nexperia's current revenue while unintentionally overlooking future strategic opportunities by not supplying the constrained capacity to

strategically important customers. With R&D spending of USD 106 million in 2024 (Nexperia B.V., 2025c), capacity allocation favouring current high-volume customers may reduce the long-term commercial value generated from these innovation investments such as in Hamburg, and may limit Nexperia's access to future growth opportunities due to factors discussed in Section 3.1.

The current high-volume customers might not remain strategically important in the near future, but as trust has already been lost during the shortage allocation, they may be less likely to return to that supplier and the supplier will end up losing a strategically important future customer. This also clarifies that the customer allocation framework should incorporate risk-adjusted profitability and future value of any programme. Otherwise, scarce capacity may be allocated to customers that preserve current revenue and volume but weaken long-term economic growth and revenue potential. Increase of Nexperia's market share from 8.9% in 2023 to 9.7% in FY2024 in Nexperia's defined market (Nexperia B.V., 2025d) could suggest Nexperia's participation in design-in activities and involvement in new vehicle programmes. Even though once qualified during the design-in phase, Nexperia may be locked-in for the entire lifecycle of the vehicle programme, poor allocation can force customers to start evaluating alternative suppliers for the same component. Once replaced, Nexperia may not be considered again during the lifecycle of the programme. The extensive qualification requirements of AEC-Q100 and AEC-Q101 could make supplier replacement difficult and require a full qualification and evaluation cycle again (Automotive Electronics Council, 2021, 2023). On the other hand, this also means that failure to secure design-in qualification for a vehicle programme may result not only in current revenue loss, but, more importantly, in lost revenue throughout the lifecycle of the vehicle programme and eventually also a weaker market position. This may further impact Nexperia's market position and competitiveness negatively to secure future business.

Strategic position loss in future value pools: Strong demand for electric vehicles has increased the semiconductor intensity per vehicle as shown in Figure 5, Panel 1 and discussed in Section 2.2. This may create additional allocation pressure for Nexperia and shifts allocation importance from the overall vehicle programme lifecycle towards the design-in phase, because once locked-in, a qualified component is most likely to remain in the vehicle programme for the entire life cycle of the programme as discussed earlier. This increases the importance of getting the semiconductor qualified for the vehicle programme during the design-in phase; otherwise, the semiconductor will not be used in the programme. Design-in qualifications for a programme may increase the chances of securing the long-term business and reduce competitive pressure. If Nexperia continues to prioritise capacity allocation for ICE-oriented vehicle programmes, it may reduce its

ability to support strategic initiatives such as the KOSTAL partnership for SiC MOSFETs in EV onboard chargers (Nexperia B.V., 2025b). Over time, the reduced ability to support such strategic initiatives could eventually weaken Nexperia's position in future SiC and GaN design-in opportunities and strengthen the competitive position of alternative suppliers. Under such conditions, if design-in opportunities are lost repeatedly for multiple programmes, Nexperia might eventually reconsider its participation in that product portfolio. These situations, in turn, create a strategic contradiction between current revenue and future growth potential. Nexperia's USD 200 million Hamburg investment in SiC/GaN capacity reflects Nexperia's commitment to SiC/GaN technologies (Nexperia B.V., 2024a), but any non-strategic allocation of the scarce capacity can turn this investment into a cost rather than a source of future growth. If design-in opportunities are secured by competitors such as Infineon, onsemi, or STMicroelectronics, while Nexperia is still prioritising high-volume customers, then instead of strategic returns, this investment may fail to generate the expected commercial returns as allocation decisions need to be made well before the actual vehicle programme design-in phase to strengthen the relationships with customers.

Operational fragility: Following Nexperia's force majeure declaration in 2025, as observed when European Automobile Manufacturers' Association (ACEA) warned that existing Nexperia stocks would last only several weeks, this resulted in a supply-chain uncertainty (Ozsevim and McLoughlin, 2025). Under such operating conditions, Nexperia may face increasing pressure to make allocation decisions under limited time and information. Due to these firefighting conditions, the scope for long-term strategic planning may be significantly reduced. This may cause uneven, partial-load, and non-optimised deliveries to customers, which may further reduce customer confidence in Nexperia and customers may start projecting their requirements as even more critical and urgent in front of Nexperia to secure the scarce capacity. Under the constant pressure of escalation emails and impulsive reactions from customers, the overall allocation process may become disrupted. Such delivery uncertainties trigger over-ordering through the bullwhip effect (Lee, Padmanabhan and Whang, 1997).

Temporary workforce suspensions of 300 to 400 workers at Bosch Salzgitter and approximately 650 at Bosch Ansbach locations (TrendForce, 2025c), a 21% downward revision of Honda's operating profit, including approximately 150 billion yen attributed to the Nexperia disruption (WardsAuto, 2025), and the production cuts at Nissan, Kyushu (TrendForce, 2025c) illustrate how operational disruptions at Nexperia can have significant operational effects on leading OEMs and Tier-1 operations and can eventually spread across the broader automotive supply chain. Furthermore, apart from the disturbance to the broader automotive supply chain, increasing net loss of Nexperia's

parent company Wingtech to 8.7 billion yuan (approximately USD 1.3 billion) in 2025 from 2.8 billion yuan in 2024 (Reuters, 2026b), suggests that this also affected the company as a whole. Following this disruption, the sanctioning of Yangzhou Yangjie Electronic Technology by the European Union in April 2026 created a second supply-chain disruption as this was an alternative supplier sourced by automakers to fill the gap created by Nexperia and automakers warned of semiconductor inventories being exhausted (van Klinken, 2026). The fact that this decision was taken after the EU concluded that Yangzhou Yangjie has supplied semiconductor technologies to Russia, further shows the volatility of the semiconductor supply chain and thus supports the view that without a pre-established prioritisation structure, such cascading disruptions may accelerate beyond the supplier's control.

Geopolitical and regulatory exposure: Nexperia operates with a highly interconnected automotive supply network including OEMs, Tier-1 suppliers, and other downstream automotive manufacturing participants (Bureau of Industry and Security, 2024b; Rosina, Boulay and Yang, 2025). This complex network may limit downstream supply-chain visibility. Furthermore, invocation of the Goods Availability Act in September 2025 by the Dutch government (Government of the Netherlands, 2025a) also highlights that regulatory intervention can become a significant constraint on Nexperia's operational and strategic decision-making, citing concerns related to strategic technology (Government of the Netherlands, 2025a). Considering the complexity attached to Nexperia, these risks appear even more significant due to export control restrictions imposed by China, ownership-related issues, and Dutch government intervention. As detailed in Sections 1.4 and 2.4, the combination of Wingtech's inclusion on U.S. Entity List, the Affiliates Rule uncertainty, has left Nexperia's cross-country operational flexibility extremely restricted (Nexperia B.V., 2024b; Brattli, Meldahl and Hermann, 2025).

Furthermore, the Dutch court order to investigate suspected management problems at Nexperia in February 2026 (Reuters, 2026a), initiation of an international legal case against the Netherlands by Wingtech demanding USD 8 billion from the Dutch state (Yang, 2026), and simultaneous initiation of a separate legal case by Wingtech in China against Nexperia and three of its executives, on the basis of China's Anti-Foreign Sanctions Law demanding 8 billion yuan in damages, arguing that the actions taken by Nexperia and its management had harmed Wingtech's interests (Dahlgren, 2026), further escalated the legal challenges at Nexperia. Due to these recent legal events, allocation decisions at Nexperia may increasingly need to account for these parallel legal proceedings in three jurisdictions, which may affect Nexperia's operational and business interests.

Relational erosion: During development of a vehicle programme, Nexperia and its customers may be required to collaborate at multiple stages including design-in, qualification, and volume-production stages in order to develop a vehicle programme. In the process, the continuous exchange of information and design standards may strengthen trust between Nexperia and its customers. However, Nexperia sits under competitive pressure because alternative suppliers, including Infineon, NXP, ROHM, and STMicroelectronics, compete in almost the same automotive semiconductor categories. Because Nexperia's position in strategic growth remains smaller compared to leading competitors, any impact on relationships may further strengthen the position of competitors to secure design-in, further weakening the position of Nexperia (Nexperia B.V., 2024a; TrendForce, 2024; Yole Group, 2025a).

Moreover, this relationship might become imbalanced due to increased competition and design-in potential related to future orders. Jiang et al. (2023) show that concentrated and power-imbalanced relationships react adversarially during periods of supply limitation and disruption. This also means that trust in the supplier may decline during supply constraints and, as a result, customers may reduce the forecast transparency related to future vehicle programmes. This is strategically significant as it is directly related to the future business opportunities for which Nexperia might have already built long-term relationships over time. This may become even more critical considering the supplier's past investments in these vehicle programmes as customers reduce order quantities and start exploring the possibilities of alternative suppliers. Temporary production-related workforce suspensions of 300 to 400 workers at Bosch Salzgitter and approximately 650 at Bosch Ansbach locations (TrendForce, 2025c) illustrate how operational disruptions at Nexperia can have significant operational effects on leading Tier-1 operations and can eventually spread across the broader automotive supply chain. Even though the design-in phase is over and the requalification process is time-consuming and costly, customers considering alternative suppliers may not only weaken the current position of Nexperia, but the customer may also not consider Nexperia for future vehicle programmes including future design-in phase consideration. The fact that technical products are also offered by the competitors such as Infineon, onsemi, ROHM, and STMicroelectronics (S&P Global Mobility, 2025a), suggests that sustained business based purely on technical grounds may not be sufficient. For Nexperia, relationship building and perceived supply reliability despite the practical issues may result in consideration for future design-in programmes and continuation of existing business.

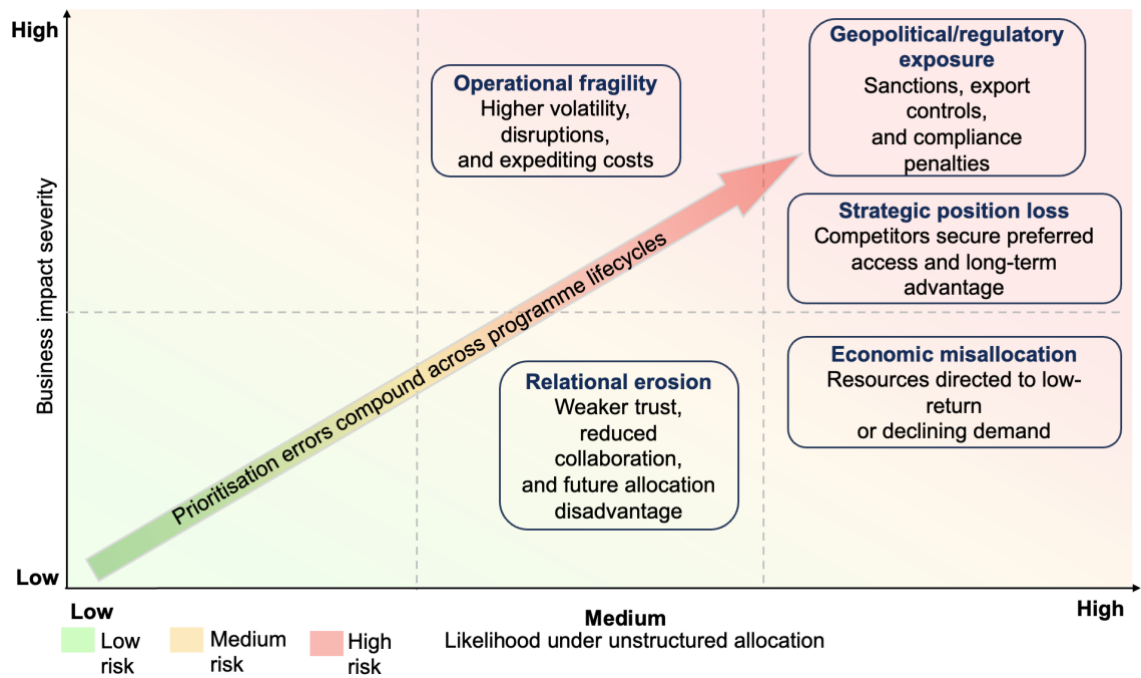


Figure 9: Risk landscape of unstructured customer prioritisation at Nexperia:
Likelihood vs. business impact

Source: Own illustration based on Lee, Padmanabhan and Whang (1997), Jiang et al., (2023), Bureau of Industry and Security (2024b)

Figure 9 maps different risks discussed in Section 3.2 based on their likelihood of occurrence and their business impact severity under an unstructured allocation. “Geopolitical and regulatory exposure” is placed in high probability and the high business impact zone as Nexperia already operates under significant geopolitical and regulatory pressure, including export controls, ownership-related restrictions, and government intervention. Moreover, due to the fact that regulatory actions can override commercial decisions entirely, the potential business impact is also severe for future business. “Strategic position loss” also sits in the high-risk zone as customers affected by supply disruptions may increasingly explore alternative suppliers and introduce dual-sourcing strategies, which could reduce their dependence on Nexperia. This, in turn, may become highly critical for Nexperia’s business. “Economic misallocation” is placed in the high-risk area because the likelihood of prioritising high-volume customers is higher, considering the fact that big customers have higher bargaining powers as discussed earlier in this Section. As consequences resulting from this are often not immediately visible, the resulting impact on business remains moderate. However, if done repeatedly, the design-in opportunities may be missed for future strategic customers. “Operational fragility” may not occur regularly and which is why it is placed in the moderate-risk zone. However, export restrictions, supply constraints, geopolitical events, or inventory

shortages can affect Nexperia's operations severely. "Relational erosion" sits in the moderate-risk zone as it does not occur after every allocation decision. However, repeated shortages faced by customers, delayed deliveries, or perceived unfair allocation can eventually weaken customer trust in Nexperia and cooperation for future programmes. The business impact also tends to be limited because customers do not take immediate action and terminate the supplier, but if repeated over time, the impact may become higher. The upward direction of the diagonal line shows that the prioritisation errors may compound over programme lifecycles and do not vanish or reduce if not addressed at the very beginning of the programme.

3.3. Strategic opportunities arising from constrained supply

Design-in opportunities in future growth segments: During normal supply conditions, Nexperia delivers semiconductors to customers through its portfolio of more than 15,000 products (Nexperia B.V., 2025c) and during unconstrained supply conditions the differentiation between current-revenue accounts and future-growth programmes and customers may not appear clear. The scarcity, however, if analysed carefully, creates an opportunity to identify different vehicle programmes of multiple customers and understand which programmes and OEMs are prioritising electrification, software-defined vehicle architectures, and power electronics as these programmes increasingly require semiconductors. International Energy Agency (IEA) (2025a) shows that the EV market is growing, which suggests that future vehicle OEM programmes may eventually shift towards ADAS, electrification, software-defined vehicle architectures, and power electronics. The impact of this may be strategically significant and may shift the automotive industry towards EV-enabled features. New vehicle programmes not concentrating on electrification may eventually lose market share. A supplier analysing this opportunity may invest in getting the design-in approved for a vehicle programme, which can potentially reduce future business competition and strengthen future business opportunities. Nexperia's partnership with KOSTAL, a Tier-1 supplier whose components are present in one in every two vehicles globally (Nexperia B.V., 2025b), to supply and qualify SiC MOSFETs for EV onboard charger programmes appears to be a highly strategic decision to harness the design-in opportunity and may secure long-term revenue. This precise logic may be captured by the decision-making framework.

Demand governance formalisation: Supply scarcity may create a situation that requires Nexperia to filter inflated demand noise, which can be projected by customers due to impulsive ordering behaviour in anticipation of potential supply constraints. This may lead to excess supply to one customer and less supply to another, irrespective of the actual requirement. This may create risks for long-term customer relationships, as

the customer with stronger bargaining power may receive the supply but the customer with actual demand and greater future strategic value might not receive the supply. In order to avoid such situations, suppliers may introduce forecast-verification systems and collaborative planning structures. This may reduce defensive and reactive over-ordering behaviour of customers and may shift short-term firefighting towards more structured and strategically coordinated planning practices. Due to the potential benefits of such demand planning practices, suppliers may continue these practices even after the supply conditions stabilise and may reduce overall inventory levels, operational costs, and uneven supply situations. “Controlled order entry and fulfilment process” imposed by Nexperia during 2025 disruption and related workarounds that involved direct wafer purchases from Nexperia’s Hamburg facility for separate shipment and packaging illustrate the challenges that Nexperia faced when the semiconductor-related supply chain faced intense disruption (TrendForce, 2025b, 2025a). This case highlights the potential value of a formal demand-process mechanism that can improve transparency, coordination with customers, inflated demand signal filtration, and a more accurate prioritisation during periods of constrained supply. Subsequently, direct wafer sales to customers as an operational workaround (Nexperia B.V., 2025f) also suggest the importance of the demand governance formalisation strategy. Both the above responses by Nexperia during the crisis showcased a reactive nature and may have lacked pre-agreed customer commitments, and therefore made allocation decisions more difficult to justify commercially, but these decisions appear to have been imposed and may not be optimal. A pre-established demand processing framework with trusted predefined allocation rules and with a system for forecast verification could have converted the same supply situation during the supply disruption of 2025 into a structured, defensible process instead of a reputational and relational crisis for Nexperia.

Regional resilience positioning: Even though USD 200 million investment in Hamburg in SiC MOSFET and GaN production by Nexperia (Nexperia B.V., 2024a) strengthens its European manufacturing footprint, Nexperia’s exposure to geopolitical and regulatory risks associated mainly with its Chinese ownership structure may weaken perceptions of regulatory stability and market access among some customers (Bureau of Industry and Security, 2024a). This regional focus by a supplier may lead to improved long-term strategic advantages as compared to scattered short-term revenue. As this also signals resilience during geopolitical conflicts, this model has an additional advantage over purely price-based allocation models. A supplier may also be able to secure subsidy access and policy support in the region. In the long run, the position of the regional supplier may strengthen and might receive additional support in government-supported ecosystem. Moving in the same direction, the decision of Nexperia to invest USD 200

million in Hamburg to expand SiC MOSFET and GaN FET manufacturing capacity (Nexperia B.V., 2024a) not only strengthens the position of Nexperia as a semiconductor supplier but also projects Nexperia as a trusted regional partner. This is due to increasing geopolitical tensions and the need to secure Europe's automotive industry in case such tensions further increase. This decision alone can position Nexperia as a strategically relevant supplier within Europe, which is also a major centre for global automotive OEMs.

	Design-in consolidation	Demand governance	Regional resilience positioning
Reason	Increase semiconductor intensity in vehicle due to EV and SDV growth	Actual demand visibility distorted by defensive over-ordering by customers	Increasing geopolitical issues create pressure for regional supplier
Required supplier action	OEMs with stronger EV and SDV plans are prioritised	Implementing forecast-verification and collaborative planning with customers helps govern demand better	Investing in regions sensitive to law and with high OEM concentration
Outcome	High probability of securing design-in and future business lock-in by suppliers	Results in better forecast accuracy and filter inflated demand projections	A preferred supplier status and regional governments support

Table 1: Strategic opportunities for Nexperia arising from semiconductor supply constraints

Source: Own illustration based on the analysis presented in Chapters 1, 2 and 3

Table 1 is developed by consolidating opportunities that semiconductor supply constraints may bring. Firstly, Increasing EV and SDV adoption also brings higher semiconductor intensity per vehicle. If allocation of scarce material is planned properly, it helps to identify the vehicle programmes that are more EV and SDV focused, even though the current volume is not the highest. If allocated properly, design-in opportunities may be secured before competitors, increasing the likelihood of long-term programme participation. Secondly, semiconductor scarcity also provides an opportunity to verify the demand projected by customers. This can be done by implementing a forecast verification system and developing collaborative planning with customers. The result can be used even after the scarcity period is over to increase the efficiency and accuracy

of the supplier and reduce inventory costs. Lastly, increasing geopolitical tension also creates demand for suppliers that can fulfil regional semiconductor needs. Using this opportunity to gain the trust of regional customers and governments may provide long-term protection to the business and reduce the global competition and price war.

3.4. Application to Nexperia

With USD 2.06 billion in revenue in 2024, a market share of 9.7% (up from 8.9% in 2023), R&D investment of USD 106 million (Nexperia B.V., 2025d), and a broad portfolio of automotive semiconductor products, automotive represents Nexperia's largest end-market segment (Nexperia B.V., 2025e). This also means that allocation decisions in the automotive sector are directly connected to the current growth and future business outcomes and can significantly affect the market position of Nexperia and may influence overall financial performance. Its product range, covering both mature semiconductor categories and emerging SiC and GaN technologies (Nexperia B.V., 2025a), creates a trade-off where established revenue streams and new design-in qualifications become equally important.

Nexperia appears to have unique ownership, manufacturing, and regulatory complexities that further intensify allocation complexity. The fact that approximately 70% of semiconductors that are being manufactured at its European front-end facilities are being sent to its Chinese facilities for back-end assembly and then are re-exported to European customers (James, 2026a) exposes Nexperia to multiple regional laws and regulations (S&P Global Mobility, 2025a). Furthermore, Nexperia represents approximately 57% of its parent company Wingtech's total assets, which, due to verification issues, resulted in a write-down to approximately USD 3.43 billion, which, in turn, resulted in a delisting-risk warning from the Shanghai Stock Exchange in May 2026 (James, 2026b). Moreover, the announcement of Nexperia Europe's collaboration with Polar Semiconductor at its Minnesota foundry to secure US-based MOSFET capacity (Nexperia B.V., 2026d) and the parallel completion of Nexperia China's independent MOSFET and logic IC supply chains in June 2026 (TrendForce, 2026) contribute to the increasing complexity of Nexperia's situation. The parallel development of a western supply chain through collaboration with Polar Semiconductor to secure US-based MOSFET capacity and the development of a Chinese supply chain for the same products appear to indicate an emerging structural bifurcation of Nexperia into separate European and Chinese entities. These decisions further intensify an already complex system, which is already subject to multiple internal and regulatory restrictions (U.S. Department of Commerce, 2023; Bureau of Industry and Security, 2024a; Government of the Netherlands, 2025a), and

appear to make Nexperia's situation unique across multiple dimensions, especially when making allocation decisions under conditions of scarce semiconductor capacity.

This may become even more critical during supply disruptions and changing customer requirements, as allocation decisions may need to balance customer supply commitments, the pressure of constrained supply, operational issues, and regulatory constraints simultaneously. Given this structural complexity, developing an allocation decision framework for Nexperia appears operationally necessary.

All four challenges discussed in this chapter do not seem to affect all product categories with the same intensity. Demand signal distortion becomes more significant in high-demand product categories, the shortage of which is most likely to affect customer production, as panic situations are likely to trigger demand signal distortion. Multi-tier programme opacity becomes more significant in safety-critical applications such as chassis, braking, and electronic power steering (EPS) as downstream exposure of these components is hardest to trace due to multiple Tier-1 and sub-supplier involvement. Qualification lock-in and design-in become strategically important for SiC and GaN power components as the initial design-in can secure the business for a longer period of seven to eight years, while qualification itself takes approximately one to three years. Declaration of force majeure by Nexperia (October-November 2025) due to Dutch government intervention (30 September 2025) and export controls imposed by China on Nexperia's operations in Dongguan (4 October 2025) heavily disrupted all product lines and affected almost all European automotive OEMs (NL Times, 2025; S&P Global Mobility, 2025a). This incident appears to illustrate the research problem addressed in this thesis and shows an operational managerial challenge and measurable industry consequences. Due to the changing automotive market landscape, the same automotive customer accounts, for instance, a major German OEM, can operate simultaneously under declining ICE programmes and growing EV programmes, but Nexperia's allocation based on account-level information cannot reflect the granular programme-level dynamics and differentiation that determine which programmes are using Nexperia's SiC and GaN products and which are integrating its legacy silicon discrete products, which may become critical for the future strategic positioning of the OEM. As a result, account-level allocation strategies may misdirect semiconductor allocations towards ICE programmes instead of future EV programmes, and with declining ICE-related demand, the demand from the account may eventually reduce significantly. This analysis becomes even more critical when competitors secure semiconductor positions in future EV applications.

SiC and GaN product categories appear to be at the intersection of the risks and opportunities created by the changing market landscape. As indicated by Yole Group

(2024), SiC market growth continues to increase and is currently at a critical growth stage due to increasing adoption of high-voltage platforms, which makes the situation even more strategically significant. Diverting the constrained SiC and GaN capacity towards conventional applications may permanently damage the firm's position in new markets due to missed opportunities. The challenges, risks, and opportunities being analysed in this chapter identify six different dimensions, namely regulatory eligibility, contractual severity, qualification lock-in, strategic importance, geopolitical risk, and margin contribution, which form the basis for the assumptions and decision logic developed in the following chapter.

Failure to qualify during the design-in phase for new vehicle programmes may lead to significantly lower future revenue and weaker future strategic positioning in the changing automotive semiconductor market. The USD 200 million investment in Hamburg (Nexperia B.V., 2024a) and the strategic partnership with KOSTAL (Nexperia B.V., 2025b) signify Nexperia's commitment to EVs and SiC/GaN technologies. However, these initiatives are likely to deliver their full benefits only if the future EV design-ins are secured and customer relationships are strengthened today by prioritising vehicle programmes focusing on new technologies and giving comparatively lower priority to traditional ICE-related programmes.

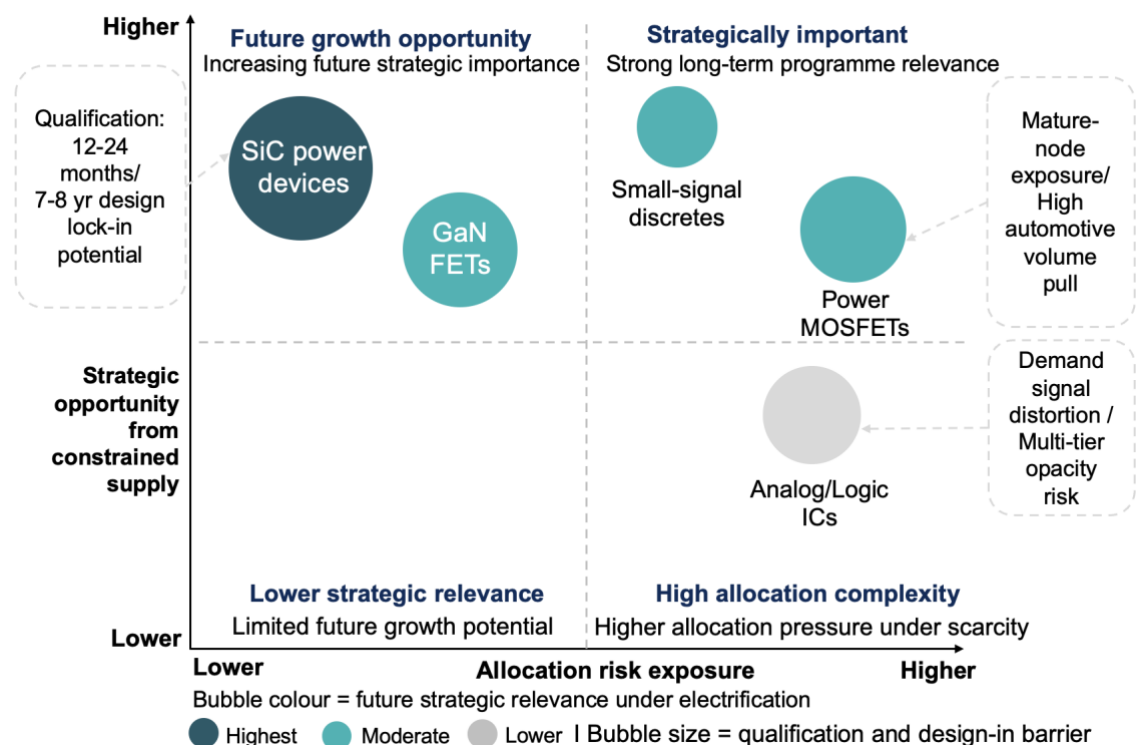


Figure 10: Nexperia product portfolio: Allocation risk exposure vs. strategic opportunity from constrained supply

Source: Own illustration based on Yole Group (2024), Nexperia B.V. (2025a, 2025b), KPMG (2026)

Figure 10 maps different product categories offered by Nexperia with respect to the degree of risk exposure during constrained supply conditions and the strategic opportunities arising from this constrained supply. This map gives a bird's-eye view of how different product categories of Nexperia require different levels of managerial attention. X-axis represents risk exposure during constrained supply conditions and Y-axis represents the degree of strategic opportunity arising from constrained supply. The size of the bubble represents the qualification and design-in barriers for that product category. Different colours represent the strategic importance that a product category of Nexperia is expected to gain as EV adoption increases. The darker the colour, the more strategically important that particular product category is expected to become.

The map is divided into four quadrants based on the position of both the X-axis and Y-axis. SiC power devices and GaN FETs are placed in the first quadrant, named "Future growth opportunity". This quadrant has been named "Future growth opportunity" because these product categories, if allocated correctly during scarcity, can create design-in opportunities and future growth potential for Nexperia. These categories also have low allocation risks as most opportunities are still at the qualification and design-in stage and Nexperia can still influence future programme participation before a particular supplier is fixed. The comparatively larger bubble size in this quadrant also indicates that they have higher entry barriers in terms of qualification standards and procedures. In turn, this also shows that once qualified for design-in, the business may become more secure for Nexperia for the entire lifecycle of the programme due to high entry barriers.

The second quadrant of the map is named "Strategically important" as product categories mapped in this quadrant bring higher strategic opportunity during scarcity along with higher allocation risk as well. The higher strategic opportunity is caused by their design-in importance and their relevance as EV and SDV demand increases. Associated allocation risks are also higher as these are the products that already have strong programme relevance and high demand. Additionally, allocation mistakes in these categories affect large volume customers as these are widely used semiconductor components across multiple automotive programmes. Because of this, any mistake is more visible and becomes commercially more significant.

Product categories placed in the third quadrant show a "higher allocation complexity" due to higher allocation risks. These higher risks are caused by demand signal distortion, limited visibility across multiple downward supply-chain tiers, and difficulty in accurately identifying the actual demand from customers. As a result, allocation decisions for this product category are more difficult. However, because this category has analog and logic

ICs, the strategic opportunity associated with it remains lower than the other product categories in the portfolio.

The fourth quadrant indicates both lower allocation risks and lower strategic opportunities, therefore named “Lower strategic relevance”, where no major Nexperia product categories currently appear to fall.

With Nexperia’s automotive-heavy semiconductor product portfolio, new investments in SiC and GaN technologies in Hamburg (Nexperia B.V., 2024a), regulatory restrictions imposed on Nexperia by Chinese, Dutch, U.S., and European governments simultaneously, and 80% of end-product capacity concentrated in mainland China (Renner *et al.*, 2025; S&P Global Mobility, 2025a), the degree of freedom to make semiconductor allocation decisions while also balancing Nexperia’s commercial and strategic interests becomes increasingly limited for Nexperia. This appears to be far more complex than commercially-oriented prioritisation logic alone due to multiple critical factors operating simultaneously. Production disruptions at a large number of European automotive OEMs and global supply chain disruptions resulting from these factors intensified in 2025 and led to the 2025 crisis (European Automobile Manufacturers’ Association, 2026), which led Nexperia to declare force majeure. This, in turn, contributed to Honda revising its full-year operating profit guidance downward (WardsAuto, 2025). Multiple critical factors such as regulatory exposure imposed by multiple jurisdictions on Nexperia, its Chinese ownership, concentration of back-end manufacturing capacity in China, the strategic KOSTAL partnership in SiC technologies, and 9.7% market-share position interact simultaneously (Nexperia B.V., 2024a, 2025b, 2025d; S&P Global Mobility, 2025a). A structured, multi-criteria customer prioritisation framework capable of balancing profitability, the importance of strategic programmes, geopolitical risk, and supply resilience appears operationally necessary for Nexperia.

Chapter 4 presents the detailed framework logic that will support Nexperia’s semiconductor allocation under these conflicting boundary conditions.

4. Development of a customer prioritisation framework

4.1. Basic assumptions and decision logic

The structure of being headquartered in the Netherlands but fully owned by a Shanghai-listed Chinese company Wingtech (S&P Global Mobility, 2025a) adds to the complexity. This became problematic as the parent company Wingtech was placed on the U.S. BIS Entity List in December 2024 (Bureau of Industry and Security, 2024a). As the U.S. Entity List restricts a company's access to certain U.S. technologies, products, and suppliers (Bureau of Industry and Security, 2024a) and the BIS Affiliates Rule of 29 September 2025 automatically applies this rule to subsidiaries having more than 50% ownership, this rule also became applicable to Nexperia as Nexperia is fully owned by Wingtech, although the U.S. subsequently suspended the Affiliates Rule for one year, from 10 November 2025 to 9 November 2026 (Brattli, Meldahl and Hermann, 2025; Nexperia B.V., 2025f), Invocation of the 1952 Goods Availability Act by the Dutch government to take over important governance and decision-making powers at Nexperia (Government of the Netherlands, 2025a) further intensified the structural problems. Following the action taken by the Dutch government, the subsequent response of the Chinese government on 4 October 2025 to block Nexperia's chips from leaving China (Nexperia B.V., 2025f) appears to have made the complicated structural problems at Nexperia even more severe. García-Herrero (2025) argues that this chain reaction of ownership and export control exposed the fact that Chinese foreign direct investment (FDI) into Europe is 'inherently conditional and reversible, based on geopolitics', and suggests that Nexperia's ownership-related risk is structural rather than cyclical.

This risk has also been identified independently: Moody's (2026) highlights how geopolitical tensions can interrupt semiconductor supply chains and can create bottlenecks across downstream industries. Chatham House (2025) highlights that the export ban imposed by China on Nexperia created a risk of disruption for the European automotive industry (Messmer, Katja and Cole, 2025). Mercator Institute for China Studies (MERICS) identified the Nexperia case as one of the major China-related risks for Europe in 2026 (Legarda *et al.*, 2025).

These structural complexities, along with three constraints identified in Chapters 2 and 3, shape the basis of framework assumptions. The framework therefore operationalises customer prioritisation through programme-level allocation decisions while maintaining customer prioritisation as the strategic objective. First, as around 80% of Nexperia's total end-product capacity is located in mainland China (S&P Global Mobility, 2025a), the supply chain becomes highly geographically concentrated, especially when that location is China due to geopolitical issues. Nexperia's Dongguan facility, with an annual capacity

of 90 billion units and being Nexperia's largest assembly site (Feng, 2025), plays an important role in Nexperia's global supply chain. Blocking Nexperia from exporting from China impacted the circular supply chain structure as European wafers were sent to China for assembly, testing, and packaging, then re-exported to European customers (S&P Global Mobility, 2025a). This increasing supply-chain complexity due to regulations and geopolitics appears to create an immediate chokepoint in the global supply chain for Nexperia, especially for European automakers. Temporary workforce suspension measures imposed by Bosch at Salzgitter and Ansbach facilities (TrendForce, 2025c), and recent warnings issued by Volvo and Volkswagen of possible factory closures (Tan, 2026) suggest that route concentration is a first-order allocation variable at Nexperia, rather than a background condition. Second, with a large portfolio of AEC-Q100/Q101 automotive products used in safety-critical vehicle systems, including airbag safing MOSFETs (Nexperia B.V., 2025e) and qualified AEC-Q101 for use in linear mode in airbag applications (Nexperia B.V., 2019), replacing an approved component is difficult and time-consuming due to lock-in, once the design-in phase is already over and Nexperia's product has already been integrated (Automotive Electronics Council, 2023). This is because a minimum of six months is required for laboratory testing alone and 12-24 months, including OEM vehicle-level validation (Automotive Electronics Council, 2023). Third, as Nexperia's USD 2.06 billion revenue in FY2024 and 9.7% market share (Nexperia B.V., 2025d) include a variety of products from small-signal transistors, which are priced at less than one US cent per unit, to 650V GaN FETs and 1200V SiC MOSFETs from the \$200 million Hamburg wide-bandgap investment (Nexperia B.V., 2024a), allocating scarce capacity to higher-margin and strategically important products may create greater long-term value for Nexperia both commercially and strategically. To achieve Nexperia's \$10 billion revenue target by 2030 as indicated by a Nexperia executive during an interview with EEPower (Bodo's Power Systems, 2022), prioritisation of not only customers but also strategic products is considered important.

Based on the above realities especially applicable in the case of Nexperia, mainly due to geopolitical and product-related factors, an allocation decision based only on high volume, customer relationships, or first-come, first-served principles may cause long-term negative financial and strategic impacts. A volume-based allocation decision at Nexperia could prioritise the highest-volume commodity programmes, which can also be replaced easily by other suppliers. On the other hand, low-volume but strategically important products such as BUK9M-series airbag safing MOSFETs, which are specifically designed and qualified to AEC-Q101 for use in airbag applications (Nexperia B.V., 2022), may receive lower priority during allocation decisions. These highly qualified products, on the other hand, demand extensive automotive validation and approval

processes before they can be used (International Automotive Task Force, 2016). This means that in case these products are not received by the customer, they may not be able to source them from another supplier due to time and qualification constraints (International Automotive Task Force, 2016). The October-November 2025 Nexperia crisis appears to illustrate the fragility of relationship-based allocation as it favours high-volume customers, which generally hold greater bargaining power. The ability of large customers to negotiate and purchase Hamburg wafers and directly contract Dongguan for packaging (Sterling, 2025a) showcased their bargaining power and highlighted the issue that mid-sized and small-sized customers cannot compete with these big customers, and due to the absence of bargaining power compared to larger customers, they may not receive the products. As relationship strength alone does not necessarily direct scarce capacity toward the programmes that create the greatest long-term value for Nexperia, a structured framework based on objective criteria rather than bargaining power or procurement influence is required. On the other hand, prioritisation based on demand-signal urgency may direct scarce capacity to whoever signals the most urgent demand rather than to strategically important OEM and Tier-1 customers. As a result, responding to demand urgency may not serve Nexperia's strategic objectives.

Therefore, the allocation decision framework developed in this chapter is based on the following assumptions:

Assumption 1: The strategic importance of a demand is assessed at the product-programme level and not at the account level as described in Section 1.3. Chapter 2 showed that the uneven redistribution of semiconductor demand across product families can result from electrification. A clear product-level programme separation would make the analysis more precise rather than general account-level noise. Section 3.1 showed that once qualification lock-in is done, it makes it harder for customers to replace that part in the vehicle programme. This analysis from Section 3.1 emphasises product-level details. Moreover, a single OEM account can operate simultaneously on declining ICE programmes and growing EV programmes with different replacement difficulties and future values associated with each part. An account-level consideration at Nexperia may overlook these important details and the resulting assessment may not reflect the true practical situation. At Nexperia, the same OEM can purchase automotive semiconductors with high qualification lock-in requirements such as safety-related semiconductors, and the same OEM can also purchase standard semiconductor products with lower switching barriers. Because account-level prioritisation aggregates these different product categories together, it does not capture this distinction, which is why programme-level allocation is used as the operational tool for customer prioritisation.

Assumption 2: Except for regulatory/export control, the allocation decision for the remaining products balances contractual obligation severity, qualification lock-in, strategic importance, geopolitical and operational risks, gross margin contribution, and demand stability simultaneously. As shown by Chapters 2 and 3, prioritisation error at Nexperia can result in multiple issues at once: economic misallocation can result when volume-based accounts are given priority instead of strategic-value-based allocation. It can also result in loss of strategic position in future value pools due to concentrating only on volume-based parameters in making decisions. Chapter 3 also showed the risk of relationship erosion when allocation decisions are made based on a single factor, especially during scarce supply. These arguments from previous chapters suggest that no single criterion is intended to produce an optimal allocation outcome when applied in isolation while making allocation decisions at Nexperia, and therefore the criteria when analysed in combination may produce more balanced allocation outcomes.

Assumption 3: Allocation decisions at Nexperia must function under limited information availability and a compressed time window because the 2025 crisis at Nexperia escalated rapidly and decisions had to be made before full visibility could be established. As discussed at the start of this chapter, the potential application of the BIS Affiliates Rule extending the restrictions imposed on the parent company Wingtech to its subsidiaries and as Nexperia is fully owned by Wingtech (Bureau of Industry and Security, 2024a; Brattli, Meldahl and Hermann, 2025; Tan, 2026), if the U.S. Department of Commerce decides again to put the BIS Affiliates Rule into effect, Nexperia would automatically be considered affected due to the BIS Affiliates Rule. This was followed by the Dutch government taking over important governance and decision-making powers at Nexperia on 30 September 2025 (Government of the Netherlands, 2025a), China blocking exports of Nexperia's products (Nexperia B.V., 2025f), and the issue of a notification to customers by Nexperia on 10 October 2025 (Nexperia B.V., 2025f). These disruptions occurred within a period of eleven days and may have changed the overall decision environment significantly. Therefore, an allocation decision framework that requires full information before producing a decision may not fit Nexperia's operational reality.

Assumption 4: The allocation decision framework covers contractually binding orders only. Nexperia's allocation problem concerns existing and already committed supplies and does not cover future orders or potential orders (TrendForce, 2025b). The 2025 crisis appears to illustrate this: some customers with direct programme relationships and binding commitments appear to have been able to negotiate with Nexperia and secure Hamburg wafer and Dongguan packaging capacity through direct negotiations and pre-

agreed commitments (TrendForce, 2025b). This is valid for programme lifecycles, which usually range from 7-8 years (Center for Automotive Research, 2023).

Assumption 5: The geopolitical and operational risk weight remains constant across all allocation time horizons considered in the framework development. This is because these restrictions can block supplies in any period and are not dependent on the time horizon moving from a period of crisis (0-3 months) to recovery (3-12 months). As discussed at the start of this chapter, for Nexperia, Chinese back-end concentration (S&P Global Mobility, 2025a), U.S. Entity List restrictions via Wingtech (Bureau of Industry and Security, 2024a; Tan, 2026), Dutch Goods Availability Act exposure (Government of the Netherlands, 2025a), and Chinese export controls (Nexperia B.V., 2025f) exist regardless of the supply cycle. Developments in November 2025 did not eliminate the possibility that similar ownership-related restrictions could reappear in the future (Tan, 2026). Furthermore, García-Herrero (2025) argues that Chinese FDI into Europe has proven to be ‘inherently conditional and reversible, based on geopolitics,’ confirming that Nexperia’s ownership-related risk is structural rather than cyclical or time-dependent.

Assumption 6: The weight assigned to demand stability remains constant throughout the allocation time horizon, but demand stability itself is not assumed to be identical across different customers. Section 3.1 showed that demand can be inflated because of overordering during shortage periods and the bullwhip effect (Lee, Padmanabhan and Whang, 1997). During the 2025 crisis, ACEA warned that manufacturers were relying on reserve stocks (European Automobile Manufacturers’ Association, 2025), indicating that demand forecasts may have become distorted. This also highlights the importance of evaluating demand signal accuracy. Demand stability is defined as C6 in this thesis for framework development, and C6 therefore measures actual forecast accuracy and order consistency across customers rather than assuming that demand stability is identical across all customers.

The above assumptions lead to the development of a two-stage allocation architecture for the Nexperia customer prioritisation framework. Stage 1 applies Gate G1 as a mandatory regulatory and export-control screen, and restrictions applicable to Nexperia are evaluated before any commercial scoring is performed. G1 is taken as binary: if the destination of supply, end-user, routing of supply, payment structure, or ownership-related compliance violates Dutch, EU, U.S., or China regulations, the order is excluded from consideration before any parameters defined in Stage 2 are applied. Considering Nexperia’s current situation, this gate becomes particularly important in the framework development. The argument given by the European Council on Foreign Relations (ECFR), which is a pan-European foreign policy think tank, suggests that the intervention

of the Dutch government in Nexperia may have been influenced by U.S. export-control pressure, similar to the earlier case of U.S. pressure on ASML Holding N.V. (ASML), which is a Dutch semiconductor lithography equipment manufacturer (Quarles van Ufford, 2025). This argument given by ECFR indicates another uncertainty in the decision framework at Nexperia, namely that regulatory risks for Nexperia may not only arise from Nexperia's own actions but can be influenced by the political coordination between governments.

Stage 2 then operates on the programmes that remain eligible after Gate G1. It evaluates and scores six different criteria, namely C1 to C6 as defined later in this chapter, covering different aspects of allocation decision-making. The weights assigned to these criteria change with the time horizon because the objective of the allocation decision itself changes depending on the time horizon. During the immediate response phase, the priority is to prevent irreversible contractual and technical damage. During the recovery phase, however, the allocation objective begins to shape future customer access and programme positioning. In the future growth phase, allocation must also protect future business and design-in opportunities, deepen customer relationships, and make an effort to monetise Nexperia's Hamburg SiC/GaN investment (Nexperia B.V., 2024a) and the KOSTAL and Polar Semiconductor partnerships (Nexperia B.V., 2025b).

The weights were not assigned arbitrarily to criteria C1-C6. Chapters 2 and 3 identified different risks and factors affecting the semiconductor allocation decision at Nexperia. These risks and factors do not hold the same importance for Nexperia. Some risks can create immediate contractual or technical damage, while others can affect future growth opportunities, customer relationships, and design-in opportunities for Nexperia. These weights therefore reflect this relative importance and are based on the findings from Chapters 2 and 3 and managerial judgement supported by the prioritisation logic of the Analytic Hierarchy Process (AHP) (Saaty, 1980).

C1 (Contractual obligation severity) receives the highest weight of 35% in the immediate response time horizon because evidence from the 2025 crisis suggests that non-fulfilment of committed vehicle programmes was immediately transmitted to downstream operations of Tier-1s, OEMs, and other downstream tiers as shown in Section 3.2. Other disruptions observed in this series show a 21% downward revision of Honda's operating profit, including approximately 150 billion yen attributed to the Nexperia disruption (WardsAuto, 2025), temporary workforce suspensions of 300 to 400 workers at Bosch Salzgitter and approximately 650 at Bosch Ansbach locations (TrendForce, 2025c), and the production cuts at Nissan, Kyushu (TrendForce, 2025c), which collectively reflect the immediate disruption caused by non-fulfilment of contractual obligations by Nexperia. Furthermore, the finding by the UK Parliamentary Office of Science and Technology that

semiconductor shortage alone caused the automotive industry to lose more than USD 200 billion in revenue (Parliamentary Office of Science and Technology, 2024) further highlights the importance of contractual obligations. Non-supply to contractually committed vehicle programmes therefore carries the highest economic consequence during crisis periods and to reflect this in the decision framework, the highest weight has been assigned to C1.

C2 (Qualification lock-in) receives 30% weight, which is the second-highest allocation in the semiconductor allocation decision framework being developed in this thesis. This is because automotive semiconductor qualification creates strong switching barriers for customers due to stringent qualification and testing requirements. AEC-Q101 qualification and OEM validation typically require 12-24 months (Automotive Electronics Council, 2023). This highlights the fact that once the supply is not fulfilled by Nexperia, customers may face disruption as qualification of a replacement would typically take 12-24 months. Despite the availability of alternative suppliers during the 2025 crisis, many customers were unable to change the supplier due to high qualification barriers (European Automobile Manufacturers' Association, 2025), which also emphasises the importance of C2. Furthermore, Chapter 3 also shows how this sparked a chain reaction and the crisis was observed across the entire supply chain. Based on these implications, C2 receives the second-highest weight for this criterion.

Criterion C1 protects existing contractual commitments, whereas C2 protects programmes where Nexperia cannot easily be replaced once the component has been qualified into the vehicle, representing switching barriers and reflecting the technical dependency of customers on Nexperia created by the qualification process.

C3 (Strategic importance) holds the third-highest weight in the allocation decision framework and C3 has been defined in this thesis to protect the future business opportunities of Nexperia and protect Nexperia's long-term market position and growth potential by prioritising strategic customers, strategic vehicle programmes, and technologies. The weight assigned to C3 rises from 15% to 25% and then to 30% as we move further along the time horizon away from the immediate response period (0-3 month) towards the future growth period (>1 year) because C3 allocation also shapes Nexperia's future position in EV, SiC, and GaN markets. The increasing weight of C3 represents the shift in the purpose of the allocation decision as we move from harm prevention toward capturing growth identified in Chapters 2 and 3. Section 1.3 showed that batteries and EV powertrain-related components are expected to grow by roughly 13% annually, while the demand for ICE powertrain components will continue to decline between 2025 and 2035 (Kilian *et al.*, 2026). Nexperia's partnership with KOSTAL for SiC MOSFET qualification in EV onboard charger programmes (Nexperia B.V., 2025b)

also reflects the same. This parameter may become important for monetising the USD 200 million investment in Hamburg to expand SiC and GaN manufacturing capacity (Nexperia B.V., 2024a).

While Gate G1 defined earlier in this chapter evaluates current regulatory eligibility, C4 (Geopolitical and operational risk) evaluates the exposure of eligible vehicle programmes to future geopolitical and operational disruptions. C4 is assigned the fourth-highest weight in the allocation decision framework and it remains constant at 10% across all time horizons. As described in Assumption 5, ownership- and supply-chain-related risks at Nexperia are structural and not cyclical (García-Herrero, 2025). The increasing operational separation of Nexperia Europe and Nexperia China following the 2025 crisis (Reuters, 2025) suggests that these risks at Nexperia can influence allocation decisions irrespective of the time horizon and therefore justify an equal weight across all time horizons.

C5 (Gross margin contribution) is the criterion that protects Nexperia's profitability by prioritising customers, vehicle programmes, and products generating higher profit margins for Nexperia. C5 has been assigned the fifth-highest weight in the decision framework development and receives 5% weight during the immediate-response horizon and increases to 10% in the future-growth horizon. This is based on the conclusion arrived at in Section 3.2 showing the importance of maintaining supply continuity during the immediate-response period and the focus shifts to maximising profitability as the situation stabilises. As revealed by Nexperia executives, Nexperia targets USD 10 billion revenue by 2030 (Bodo's Power Systems, 2022). C5 takes this long-term growth ambition into account. Furthermore, recent research conducted by the Peterson Institute for International Economics suggests that geopolitical disruptions are increasing the importance of strategic capacity allocation and value creation in semiconductor supply chains (Bown, 2026), which supports assigning greater weight to C5.

C6 (Demand stability) evaluates the reliability of customer demand signals by analysing the forecast accuracy and order consistency. C6 receives the lowest weight of 5% in the allocation framework. The lowest weight assigned to C6 is based on the findings of Section 3.1 and Assumption 6 that showed that during disruptions, demand signals inflate due to overordering and the bullwhip effect (Lee, Padmanabhan and Whang, 1997; European Automobile Manufacturers' Association, 2025). Although C6 remains relevant for the decision framework at Nexperia, it serves as a validation check rather than a differentiator with significant influence on the allocation decision. This supports its lowest weight across all time horizons.

Stage	Criteria	Immediate response (0 - 3 months)	Recovery (3 - 12 months)	Future growth (12+ months)
Stage 1	G1: Regulatory and export control eligibility	Screen before shipment	Screen before shipment	Screen before shipment
Stage 2	C1: Contractual obligation severity	35%	30%	25%
Stage 2	C2: Qualification lock-in	30%	25%	20%
Stage 2	C3: Strategic importance	15%	25%	30%
Stage 2	C4: Geopolitical and operational risk	10%	10%	10%
Stage 2	C5: Gross margin contribution	5%	5%	10%
Stage 2	C6: Demand stability	5%	5%	5%
Total stage 2		100%	100%	100%

Table 2: Nexperia customer prioritisation framework: Stage 1 Gates and Stage 2 criteria weights by allocation horizon

Source: Own illustration based on Saaty (1980), Government of the Netherlands (2025a), TrendForce (2025a)

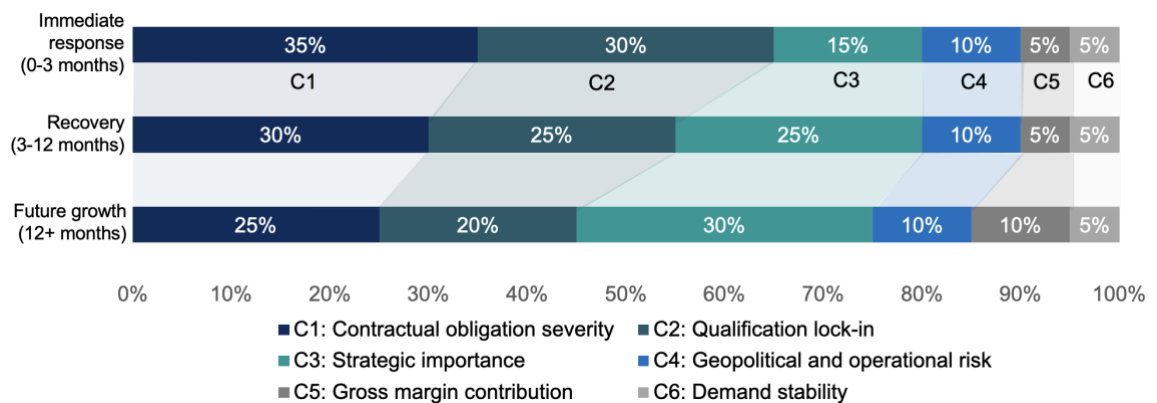


Figure 11: Visualisation of criteria weight shifts across allocation horizons in the Nexperia customer prioritisation framework

Source: Own illustration

Table 2 and Figure 11 show the weight assigned to each criterion (C1-C6) across the three allocation horizons (immediate-response, recovery and future growth), together summing to 100% in each horizon. In addition to the weighted evaluation criteria C1-C6 shown in Figure 11, Table 2 also illustrates Gate G1, which functions as a mandatory regulatory and export-control eligibility screen applied before Stage 2 scoring is performed. Table 2 and Figure 11, which is the graphical representation of Table 2, show that during immediate response (crisis period), C1 (Contractual obligation severity such as liquidation damage clause and others) and C2 (Qualification lock-in) together account for 65% of the total weight during the prioritisation decision at Nexperia. During the future growth period, this combined weight of C1 and C2 reduces to 45% due to the shift in the decision logic from firefighting to strategic growth. With an increase in the time span from the crisis phase to the growth phase, C3 (Strategic Importance) rises to 30%, which reflects that the decision logic shifts from crisis management toward strategic positioning. C4 (Geopolitical and operational risk) remains constant at 10% across all allocation horizons because Nexperia's geopolitical and operational exposure does not change across allocation horizons (Assumption 5). C5 (Gross margin contribution) increases to 10% during the future growth period. C6 (Demand Stability) remains at the lowest weight of 5% throughout the time horizon as it acts as a validation check rather than a differentiator with significant influence on the allocation decision.

4.2. Evaluation criteria for prioritisation

Section 4.1 defined six allocation framework criteria (C1-C6) and the weights assigned to them depending on the time horizon of analysis, along with the mandatory framework screening gate G1. Section 4.2 establishes a scoring system on a uniform 1-3-5 scale that Nexperia's semiconductor allocation committee can apply to each vehicle programme under constrained supply conditions (Assumption 3, Section 4.1).

This framework design is consistent with Assumption 3 which requires the framework to function under limited information availability and a constrained time window because the decision model must be usable by managers in a constrained time window after constraint confirmation. This has been done in order to include the constraints observed during the 2025 Nexperia crisis (European Automobile Manufacturers' Association, 2025; Ozsevim, 2025).

Gate G1: G1 assesses regulatory and export control eligibility at shipment level and produces either a pass or fail outcome (binary output) based on five dimensions, namely destination of supply, end-user, routing of supply, payment structure, and ownership, and is applied as described in Section 4.1. The ruling for mismanagement investigation at Nexperia and upholding the suspension of the CEO on 14 January 2026 and 11

February 2026 by the Amsterdam Enterprise Chamber (Reuters, 2026a), and subsequent appeal by Wingtech to the Dutch Supreme Court (Sterling, 2025b), indicates judicial uncertainty in Nexperia's governance. Furthermore, initiation of the review for the research cooperation between DESY and Nexperia SiC by the German Federal Ministry for Research (BMFTR) (Heide, 2025), indicates that regulatory scrutiny appeared to have expanded beyond trade compliance to potential technology-transfer concerns. Poitiers and Schenk (2026) suggest that in export-control negotiations, the EU's role may be constrained because operational competence remains primarily national.

Such semiconductor-related regulatory boundaries place semiconductor-related decisions within broader international dynamics and suggest that G1 should not rely on a single-jurisdiction compliance assessment.

As defined in Section 3.4, the structural bifurcation of Nexperia into separate European and Chinese entities further complicates the G1 requirements. Domestic sourcing of wafers by Nexperia China and collaboration of Nexperia Europe with Polar Semiconductor in Minnesota (Nexperia B.V., 2026d) and its expansion into Malaysia (van Gerven, 2025) suggest that G1 must also screen the manufacturing location of the orders. A hypothetical European OEM programme supplied from Hamburg-fabricated wafers and assembled through non-Chinese back-end capacity would pass G1 on all five dimensions. However, the same programme when supplied from the Dongguan assembly facility may fail due to China-related restrictions because of the unresolved Wingtech Entity List status and the BIS Affiliates Rule.

A customer programme failing gate G1 on any parameter will be excluded from Stage 2 and supply will not be made regardless of its commercial and strategic value, in order for Nexperia to comply with regulatory and export controls.

Stage 2: After vehicle programmes clear Gate G1, criteria C1-C6 are assigned a score of either 1, 3 or 5. Table 3 shows the customer prioritisation scoring matrix and allocates the score 1, 3 or 5. This framework deliberately uses only three scores (1, 3 or 5) instead of many detailed numbers as decision-making during a crisis is often based on judgement and limited information and not exact calculations. Using 1, 3 and 5 keeps the process simple while still allowing Nexperia to differentiate between low-, medium-, and high-priority programmes.

As shown in Table 3, C1 (Contractual severity) holds the lowest value of 1 if the contract allows flexible delivery and there is no concrete penalty assigned for non-delivery. If non-supply can lead to standard exposure and commercial consequences such as customer complaints, issue escalation, loss of goodwill, or possible normal commercial impact, but the situation is not serious enough to lead to a customer's line stoppage, then C1 is

assigned a value of 3. On the contrary, if non-delivery as per contract can lead to a customer's line stoppage, liquidated damages and other financial penalties of significant magnitude, or the affected programme is close to or during the Start of Production (SOP), such that any supply disruption can directly affect a new vehicle launch, then C1 is assigned the highest value of 5 in the decision framework.

Score	C1 Contractual severity	C2 Qualification lock-in	C3 Strategic importance	C4 Geopolitical risk	C5 Gross margin contribution	C6 Demand stability
5	Formal penalties, line-stop risk, SOP*-critical	No alternative, safety-critical AEC-Q101	Core EV/ADAS/SDV design-in, Hamburg SiC/GaN aligned	Dongguan-dependent, U.S.-sensitive destination	Highest margin per constrained capacity unit	Accurate; MAPE** <10%, verified demand
3	Standard exposure, commercial consequences	Alternative exists, 6-12 months of switching barrier	Mixed current/future relevance	Dual route available	Mid-range margin contribution	Moderate, MAPE** 10–25%
1	No penalty exposure, flexible delivery allowed	Easily substitutable, multiple alternatives	Declining ICE platform, limited relevance	European-only route, low political risk	Commodity pricing, low margin contribution	Volatile, MAPE** >25%, over-ordering pattern

Table 3: Customer prioritisation scoring matrix

Source: Own illustration

(*SOP= Start of Production; **MAPE= Mean Absolute Percentage Error)

As shown in Table 3, C2 (Qualification lock-in) receives the lowest score of 1, if the material supplied by Nexperia can be replaced easily or has multiple alternative suppliers. C2 holds a value of 3 if the alternative option is available in the market but switching cannot be done immediately and requires 6-12 months for the validation process. However, if the product supplied by Nexperia has no alternative or it is AEC-Q101 qualified or for safety-critical applications, C2 holds the highest value of 5.

If the vehicle programme is associated with the declining ICE platform or has limited relevance, C3 (Strategic importance) receives the lowest value of 1. However, if the vehicle programme is associated with the core EV/ADAS/SDV design-in or Hamburg SiC/GaN aligned, then C3 receives the maximum value of 5, showing the relevance of such programmes for Nexperia. For moderate importance, C3 receives a value of 3.

As described in Section 4.1, C4 (Geopolitical risk) evaluates the exposure of eligible vehicle programmes to future geopolitical and operational disruptions, therefore, if any vehicle programme has the highest risk of geopolitical exposure such as Dongguan-dependency or U.S.-sensitive destinations, then C4 receives the highest value of 5. If an alternate route is available, C4 receives the middle value of 3. However, if there is a low or no probability of such geopolitical risks, such as in the case of a European-only route, C4 receives the lowest value of 1 as shown in Table 3.

In case of scarce capacity, if Nexperia earns a comparatively high profit per unit, C5 (Gross margin contribution) receives the highest value of 5 to prioritise such vehicle programmes. However, if the profit earned is comparatively low, C5 receives the lowest value of 1. For the vehicle programmes falling in between the above two zones, C5 receives the middle value of 3 in the allocation decision framework.

C6 (Demand stability) takes a value of 1, 3 or 5 depending on the forecast accuracy compared to the demand signal observed by Nexperia through its order management process (e.g., SAP). To calculate demand accuracy, Mean Absolute Percentage Error (MAPE) is calculated as follows:

$$MAPE = \left(\frac{1}{n}\right) \times \sum_{t=0}^n \frac{|\text{Forecast}^t - \text{Observed demand}|}{\text{Observed demand}} \times 100\%$$

Where:

Forecast	= Demand forecast submitted by the customer
Observed demand	= Demand signal observed by Nexperia through its order management process (e.g., SAP)
n	= Total number of observation periods used in the calculation of observed demand
t	= Individual month for which forecast and observed demand are compared

A lower MAPE (<10%) represents the higher accuracy of demand projected by the customer and therefore C6 receives the highest value of 5. If MAPE is moderate (10-25%), C6 takes the middle value 3 and if MAPE is higher (>25%), C6 takes the lowest value of 1. In other words, if the framework indicates that the demand signal given by the

customer is inflated, it assigns a lower value to C6 and if the decision framework indicates that the demand signal is close to the normal demand of the customer, the decision framework assigns the highest value to C6 to filter demand-signal-related distortion.

4.3. Balancing profitability, strategic relevance, and risk

After defining the criteria C1-C6, assigning weights in Section 4.1, and the scoring scale in Section 4.2, this section explains how allocation priorities are determined.

The portfolio decision rule: For gate G1 qualified customer-programmes, the composite priority score is defined as:

$$\text{Priority score} = (w_1 \times C1) + (w_2 \times C2) + (w_3 \times C3) + (w_4 \times C4) + (w_5 \times C5) + (w_6 \times C6)$$

Where w_1 to w_6 are the weights assigned as given in Table 2 (Section 4.1). The assigned score for criteria C1 to C6 comes from Table 3 (Section 4.2).

As defined in Section 4.1, C1 and C2 receive the highest weights during the immediate response period to prevent irreversible loss and to provide a protective filter. Therefore, the main trade-off exists between C3, C4 and C5. However, the combined weight of C1 and C2 falls from 65% to 45% and C3 rises as the time horizon moves away from the crisis period, showing that the framework starts to prioritise strategy over time. This decision framework model is not a pure revenue maximisation model. It is a constrained optimisation model that aims to optimise three dimensions: profitability (captured through C5), strategic relevance (captured through C3), and risk (captured through C4 and C6). C1 and C2 protect current obligations and qualification lock-in and C6 confirms the demand signal reliability. A decision framework that maximises only margin (C5 dominant) would under-protect highly qualified safety programmes (C2) and also future design-ins already committed (C3). A programme that protects only the strategic relevance (C3 dominant) can give less importance to binding contractual obligations (C1) and can negatively impact business relationships and may attract liquidated damages. Similarly, a decision model trying hard to avoid risks (C4 dominant) would become too defensive and may fail to shape Nexperia's future customer portfolio.

Priority bands for Nexperia: In order to interpret the output given by the priority score equation defined earlier, customer programmes are grouped into four allocation bands as shown in Table 4. Proposed priority bands for Nexperia shown in Table 4 are decision framework design choices and do not disclose current Nexperia practice. These priority band choices are based on the learnings from the 2025 Nexperia crisis as discussed in Chapter 3 and Sections 4.1 and 4.2. Moreover, the underlying logic of directing the scarce capacity based on strategic and operational criticality rather than commercial size is also based on the post-crisis reassessment done by Bruegel Group about the strategic importance of specific semiconductor capabilities and supply-chain dependencies for

European industry (Poitiers and Schenk, 2026). Furthermore, EE Times argues that the 2025 Nexperia crisis increased attention to the strategic importance of mature-node automotive, power and industrial semiconductors, which further strengthens the different priority bands introduced in Table 4 involving criteria C1-C6 and their respective weights, as the priority bands reflect operational criticality rather than commercial priorities alone (Brans, 2026).

Priority band	Priority score	Allocation rule	Managerial meaning
Protect	≥ 4.0	Target 90-95%+ of confirmed demand	Supply first; losing these causes permanent legal, technical, or strategic damage
Support	≥ 3.2 and < 4.0	Target 70-90%	Keep supplying steadily; adjust details through negotiation
Manage	≥ 2.5 and < 3.2	Target 40-60%	Supply partly; use substitutes or negotiate with the customer
Defer/spot	< 2.5	Target 0-40%	Supply only if spare capacity remains; don't use scarce capacity unless escalated

Table 4: Priority bands and allocation rules for constrained supply

Source: Own illustration

Table 4 shows the allocation decision rule by categorising the priority scores into four categories, namely Protect, Support, Manage, and Defer/spot depending on their managerial meanings. Band percentages indicated in the “allocation rule” column in Table 4 are target fulfilment levels, and do not indicate guaranteed supply levels. The customer programme with the higher band percentage will be prioritised over the lower band percentage. If the priority score is ≥ 4.0 , those programmes would be given priority over programmes with lower priority scores as shown in Table 4. For example, programmes in the “Protect” band (priority score ≥ 4.0) target to receive 90–95%+ of confirmed demand, arranged in descending order of priority score. As the priority band decreases down the table, the target supply decreases to 0-40% depending on availability of material.

Trade-off resolution through weight shifts: Changing weights of C1-C6 depending on the time horizon as shown in Table 2 and as described in Section 4.1 creates several trade-offs. Three of these, most relevant for Nexperia, are discussed below:

Trade-off 1: Current revenue versus future positioning: As discussed in Sections 4.1 and 4.2, a high-volume, mature-node, already established vehicle programme (referred to as a legacy programme) scores 5 on C1 due to binding Long Term Supply Agreement (LTSA) and 5 on C2 (safety-critical, deeply qualified) but scores 1 on C3 (declining ICE). Similarly, a lower-volume SiC design-in scores 1 on C1 (early-stage) but 5 on C3 (Hamburg investment, KOSTAL partnership). Therefore, under the immediate response condition (C1: 35%, C3: 15%), the legacy programme is prioritised by the framework. However, during the future growth period (C1: 25%, C3: 30%) the SiC programme is expected to overtake and become prioritised by the framework, shifting the decision framework focus from firefighting towards Nexperia's long-term strategic goals such as monetising the Hamburg SiC initiative (Nexperia B.V., 2024a). This shift in priority due to the change in weights and scores is also consistent with the argument by Poitiers and Schenk (2026) that semiconductor strategy should focus resources on strategic capabilities where long-term influence can be maintained within the global value chain.

Trade-off 2: Geopolitical compliance versus commercial reach: As discussed in Section 4.1, Nexperia's customer access and business opportunities can increase if Nexperia chooses to serve customers involving China supply chains or customers affected by U.S. export-control rules, but this may cause regulatory exposure. This is the tension that is specific to Nexperia due to Nexperia's ownership structure. Calling Nexperia's situation a "breaking point in the chips feud" by Bloomberg (Leonard, Fan and Jacob, 2026), Wingtech's determination to regain control of Nexperia's governance in the Netherlands as reported by Caixin Global (Min, 2026), and the European Sting questioning whether the EU is "abandoning semiconductor sovereignty" (Brown, 2026), highlight this trade-off between geopolitics and commercial reach. This decision framework in Chapter 4 resolves this by fixing the criterion C4 at 10% and placing mandatory compliance in gate G1 as described in Section 4.1, ensuring the decision framework does not ignore geopolitical or regulatory risks associated with Nexperia's operations and its customers despite the commercial attractiveness of customers involving geopolitical exposure.

Trade-off 3: Customer bargaining power versus allocation fairness: A report published by Handelsblatt (2025) showing the direct involvement of automotive manufacturers and suppliers to ensure the semiconductor supply during the 2025 Nexperia crisis (Hofer, Buchenau and Seckel, 2025), and the argument by Arvidsson, Garcia and Govik (2026) showing the power imbalance during the scarce supply condition and the fact that customers with higher bargaining power generally secure the supply, give rise to a fairness issue in the allocation during scarce capacity situations. The higher bargaining power may arise due to higher purchase volumes from Nexperia.

Under such power imbalance conditions and the report published by Handelsblatt (2025), ensuring supplies as per a predefined decision framework can help to reduce the chances of non-optimal allocation and the planned allocation framework may balance all critical parameters in a structured way.

Override policy for the semiconductor allocation decision framework: Even though a semiconductor allocation decision framework is developed in Chapter 4 for Nexperia, bypassing the framework can become unavoidable in practice. However, if this framework is being ignored beyond a certain predefined number of cases, this model becomes symbolic and may not be able to show the optimum results. Stage 1 (G1) eligibility cannot be overridden under any circumstances. Bypassing a Stage 2 allocation band by more than one allocation band (Table 4) requires joint approval by automotive business unit leadership, operations, and legal/compliance, documented in writing, and reviewed. An override rate above 10% would indicate that the decision framework either requires further adjustments or the decision framework is being ignored for political or commercial reasons.

4.4. Governance and implementation aspects

The implementation of the customer prioritisation framework should be carried out by a dedicated allocation committee led by the automotive business unit at Nexperia. This committee should also consist of decision-makers from operations, product management, legal/compliance, finance, and regional sales to give their inputs. Once the framework is implemented at Nexperia, all orders should be processed based on the allocation score and prioritisation band because if all orders are analysed manually, the allocation decision may again become subjective, especially during scarce periods when there may be higher supply pressure on Nexperia. However, some matters need manual intervention. Therefore, the purpose of the dedicated allocation committee is not to analyse every single order individually, but to verify gate G1 eligibility, data accuracy, review the ranking results generated by the framework, and approve justified framework exceptions as and when necessary (as per override policy defined in Section 4.3). In order to ensure the framework is being followed, the allocation committee should meet weekly during the crisis period. However, the committee should meet bi-weekly during normal shortage periods for review. The allocation committee must sit above account managers as supply scarcity can alter bargaining power and influence allocation decisions (Arvidsson, Garcia and Govik, 2026), making the allocation subjective again rather than considering the overall criteria C1-C6 and their importance level defined by their respective weights.

The open displeasure shown by Beijing following the Dutch court's February 2026 ruling restricting Wingtech from regaining control of Nexperia, as reported by Asia Times (Pao, 2026), indicates the regulatory instability that Nexperia is undergoing and indicates that a court ruling in one country can create political and regulatory reactions in another country (China). Furthermore, considering the Nexperia-Wingtech dispute to be an important global legal case, many governments, lawyers, and companies are watching this case (A&O Shearman, 2026). Therefore, Gate G1 in the framework must be owned by the legal compliance function and must be updated dynamically to avoid any further legal and regulatory issues at Nexperia arising due to non-compliance.

In order to produce actionable results by this decision framework, the minimum information required for each allocation decision includes the product programme, customer and end-customer, destination market, manufacturing route, dynamic gate G1 legal eligibility status, contract type and potential penalties, second source availability to customers, forecast accuracy (MAPE as defined in Section 4.3), margin per constrained capacity-hour, and strategic importance of programmes under consideration (EV-centric or ICE-centric). These represent the minimum inputs required to capture regulatory, legal, operational, strategic and commercial considerations simultaneously. These inputs are also necessary in order to address the challenges exposed during the 2025 Nexperia crisis where regulatory restrictions and supply-chain complexity significantly affected automotive production and the broader automotive supply chain (S&P Global Mobility, 2025a; TrendForce, 2025b).

As discussed in Chapter 3, based on observations during the 2025 Nexperia crisis, an eligible supply can suddenly become ineligible due to changing regulatory restrictions for Nexperia and therefore separation of responsibilities must be structured beforehand so that sudden changes in regulatory boundary conditions are accurately reflected in the framework through gate G1.

Function	Responsibility and ownership
Legal/Compliance	To interpret and approve Gate G1 legal eligibility status
Automotive commercial team	To provide C1, C2 and C3 inputs and manage customer relationships, communication and negotiations
Operations	To assess C4 feasibility, routing, and supply capability
Finance	To validate C5 contribution and margin contribution details
Demand planning	To verify C6 forecast accuracy and MAPE calculations
Allocation committee	To review framework outputs, to approve exceptions under override policy, and review final allocation decisions

Table 5: Functional responsibilities and ownerships within the semiconductor allocation decision framework at Nexperia

Source: Own illustration

Table 5 summarises the responsibilities of different functions, namely Legal/Compliance, Automotive commercial team, Operations, Finance, Demand planning, and the dedicated Allocation committee, and their respective ownership of different criteria C1-C6 within Nexperia. Even though these functions within Nexperia are involved and have ownership of different criteria C1-C6, Table 5 shows that reviewing the decision framework output and final allocation decisions still lie with the dedicated allocation committee. Exception or bypass approval can be initiated by the Automotive commercial team, but approval lies with the allocation committee under the override policy defined in Section 4.3. Clear separation of ownership and responsibilities becomes even more necessary for the designed decision framework for Nexperia. As Automotive Manufacturing Solutions warned that Nexperia's standoff could disrupt the global automotive supply chain within weeks (Ozsevim, 2025), clear division of responsibilities could also reduce pressure build-up during the decision-making process at Nexperia during a crisis.

To ensure that the decision framework at Nexperia is producing consistent results as expected during the crisis and later stages, formulating and tracking governance KPIs becomes important.

KPI	Target	Minimum acceptable
Programme-level scoring coverage	100% of constrained automotive volume	95%
Decision turnaround time (crisis)	<24 hours from constraint confirmation to ranked output	<48 hours
Override rate	<10%	<10%
Protect-band fulfilment rate	>95% of protect programmes supplied at target level	>90%

Table 6: Governance KPIs for the allocation decision framework at Nexperia

Source: Own illustration

Table 6 defines four governance KPIs for the developed customer prioritisation framework at Nexperia to ensure consistency and adherence to the allocation decision framework developed for Nexperia in this thesis. However, the target values and the minimum acceptable rates assigned to each KPI represent managerial assumptions and can be adjusted based on the future operational experience of Nexperia. First KPI: Programme-level scoring coverage ensures that all the programmes available with Nexperia must go through the decision framework. The target for this KPI is deliberately set at 100% in order to ensure that all the programmes are being scanned through the developed framework. However, in order to give space for practical difficulties and difficulties in information access, the minimum acceptable level for this KPI is kept at 95%.

Decision turnaround time (crisis) is defined as the second KPI in the framework to take into account the time taken by Nexperia to make allocation decisions once the shortage situation is identified. The target has been set to make a decision within 24 hours. However, in order to include practicality, the minimum acceptable limit is set at 48 hours. Even though an override policy has been introduced in Section 4.3, ensuring that allocation decisions at Nexperia are not ignoring the decision framework, the third KPI, override rate, is introduced. No more than 10% of allocation decisions should bypass the framework to ensure the optimum decision even during the crisis at Nexperia.

To ensure that the programmes that are prioritised highest within the priority bands in line with Table 4 receive supply, the fourth KPI, Protect-band fulfilment rate, is introduced. The target is to ensure that 95% of the programmes falling under the top priority bands receive supply as per supply contracts. In order to bring a practical angle and to give flexibility, the minimum acceptable rate is set to 90% for the fourth KPI.

4.5. Application to Nexperia

Because the internal customer and programme data of Nexperia are not public, four different Nexperia-relevant programme types are selected to implement the allocation framework developed in Chapter 4. The selection of these four distinct programmes, as described below, represents four distinct allocation situations relevant to Nexperia and is also based on the main allocation situations identified in Chapters 3 and 4.

Programme A is a safety-critical programme and has high qualification lock-in. For the application of this thesis, an airbag MOSFET programme (BUK9M-series) in Europe with strong contractual obligations, high qualification lock-in, a moderate margin profile, and a European supply route with limited alternative routing options has been selected.

Programme B is a commercially attractive programme but it is a strategically declining programme. For the application of this thesis, a high-volume power-management and logic device programme supplied to a mature ICE platform of a major European OEM, with strong contractual obligations, attractive margins, a European supply route, stable demand behaviour and forecast accuracy, but low future strategic relevance due to the declining importance of ICE platforms, has been selected.

Programme C has high future strategic value but relatively low current contractual value. For the application of this thesis, a SiC MOSFET design-in has been selected for an EV onboard charger programme, which is also aligned with the KOSTAL partnership and has moderate current contractual severity, high future platform value, a European supply route via Hamburg, high margin potential, and moderate demand predictability.

To increase the complexity of framework application, Programmes B and C are assumed to belong to the same OEM.

Programme D is introduced to test the decision framework under the hardest conditions, representing a high-value programme, deliberately exposed to the China route. For the application, a high-volume, safety-critical ADAS protection-device programme for a major European OEM, with strong contractual obligations, deep qualification lock-in through AEC-Q101 qualification, a mid-range margin, moderate demand predictability, and assembled by Nexperia in Dongguan, China, has been selected.

Stage 1 screening: Programme A passes gate G1 if it is supplied through the European route (Hamburg wafer and assembly by Nexperia, Malaysia). As Programme B is also supplied through the European route, it passes G1 on all five screening dimensions defined in Section 4.1, namely destination, end-user, routing, payment structure, and ownership-related compliance. Programme C passes gate G1 as well on all dimensions as it avoids Dongguan exposure entirely due to the Hamburg-to-Malaysia route. Even though Programme D is strategically important for Nexperia and Nexperia would have

the strongest incentive to supply this programme, Programme D fails gate G1 as it is assembled in Dongguan, China and therefore exposed to Wingtech Entity List restrictions (Bureau of Industry and Security, 2024a), BIS Affiliates Rule exposure, and China litigation risk (Min, 2026).

Stage 2 scoring: For Stage 2 scoring, each of the selected programmes A, B, C and D is assigned a score on a 1-3-5 scale as per the criteria defined in Table 3 (Section 4.2).

Criterion	Programme A (Airbag MOSFET)	Programme B (ICE power/logic)	Programme C (SiC EV design-in)	Programme D (ADAS design-in)
C1 Contractual severity	5	5	1	5
C2 Qualification lock-in	5	3	3	5
C3 Strategic importance	3	1	5	5
C4 Geopolitical risk	3	1	1	5
C5 Margin contribution	3	5	5	3
C6 Demand stability	5	5	3	3

Table 7: Programme-level scoring of Programmes A-D on criteria C1-C6
Source: Own illustration based on Table 3 (Section 4.2)

Table 7 shows the scoring on the 1-3-5 scale as per the criteria defined in Table 3 (Section 4.2). Programme A scores 5 on C1 as supply interruptions related to airbag applications can create significant operational risks for OEMs (European Automobile Manufacturers' Association, 2025). Due to the high AEC-Q101 qualification and requalification barriers, C2 also scores 5 (Automotive Electronics Council, 2021). As airbag MOSFETs remain important to Nexperia but have limited alignment with future SiC/GaN growth, C3 scores 3 (Nexperia B.V., 2024a, 2025b, 2026d). As defined in the introduction of Programme A, C4, C5 and C6 score 3, 3 and 5 respectively.

Programme B scores 5 on C1 as supply interruption of a critical automotive semiconductor can create production risks for OEMs (European Automobile Manufacturers' Association, 2025). Due to the availability of qualified alternative suppliers and moderate switching barriers (Automotive Electronics Council, 2021), C2 scores 3. As Programme B is linked to a mature ICE platform with limited future relevance, C3 scores 1 (Nexperia B.V., 2024a, 2025b; Yole Group, 2024). The European supply route, attractive margins and stable demand behaviour defined earlier result in C4, C5 and C6 scores of 1, 5 and 5 respectively as defined in Programme B introduction.

Programme C scores 1 on C1 because this programme is at the design-in stage, with limited current contractual exposure. As SiC MOSFET qualification creates moderate switching barriers but alternative solutions may still exist (Automotive Electronics Council, 2021), C2 scores 3. As the programme is aligned with EV growth, SiC adoption and the KOSTAL partnership, C3 scores 5 (Nexperia B.V., 2024a, 2025b; Rosina, Boulay and Yang, 2025). Based on the programme characteristics described earlier, C4, C5 and C6 score 1, 5 and 3 respectively.

Although Programme D is excluded at Stage 1, calculating Stage 2 scores helps identify the strategic and commercial opportunities lost due to the eligibility requirements of Gate G1 and is included only for illustrative and analytical purposes. This programme also scores 5 on C1 because interruption of safety-critical ADAS semiconductor supply can create significant operational and production risks for OEMs (European Automobile Manufacturers' Association, 2025). As Programme D remains strategically important for future ADAS platforms, C3 scores 5 (Nexperia B.V., 2024b; Yole Group, 2025b). Due to deep AEC-Q101 qualification lock-in and limited replacement options, C2 scores 5 (Automotive Electronics Council, 2021). The Dongguan assembly route, mid-range margin and moderate demand predictability result in C4, C5 and C6 scores of 5, 3 and 3 respectively.

Composite priority score calculation using the equation introduced in Section 4.3 and Table 2 weights:

Immediate response

Programme A: $(0.35 \times 5) + (0.30 \times 5) + (0.15 \times 3) + (0.10 \times 3) + (0.05 \times 3) + (0.05 \times 5) = 4.40$

Programme B: $(0.35 \times 5) + (0.30 \times 3) + (0.15 \times 1) + (0.10 \times 1) + (0.05 \times 5) + (0.05 \times 5) = 3.40$

Programme C: $(0.35 \times 1) + (0.30 \times 3) + (0.15 \times 5) + (0.10 \times 1) + (0.05 \times 5) + (0.05 \times 3) = 2.50$

Programme D: $(0.35 \times 5) + (0.30 \times 5) + (0.15 \times 5) + (0.10 \times 5) + (0.05 \times 3) + (0.05 \times 3) = 4.80$

Recovery

Programme A: $(0.30 \times 5) + (0.25 \times 5) + (0.25 \times 3) + (0.10 \times 3) + (0.05 \times 3) + (0.05 \times 5) = 4.20$

Programme B: $(0.30 \times 5) + (0.25 \times 3) + (0.25 \times 1) + (0.10 \times 1) + (0.05 \times 5) + (0.05 \times 5) = 3.10$

Programme C: $(0.30 \times 1) + (0.25 \times 3) + (0.25 \times 5) + (0.10 \times 1) + (0.05 \times 5) + (0.05 \times 3) = 2.80$

Programme D: $(0.30 \times 5) + (0.25 \times 5) + (0.25 \times 5) + (0.10 \times 5) + (0.05 \times 3) + (0.05 \times 3) = 4.80$

Future growth

Programme A: $(0.25 \times 5) + (0.20 \times 5) + (0.30 \times 3) + (0.10 \times 3) + (0.10 \times 3) + (0.05 \times 5) = 4.00$

Programme B: $(0.25 \times 5) + (0.20 \times 3) + (0.30 \times 1) + (0.10 \times 1) + (0.10 \times 5) + (0.05 \times 5) = 3.00$

Programme C: $(0.25 \times 1) + (0.20 \times 3) + (0.30 \times 5) + (0.10 \times 1) + (0.10 \times 5) + (0.05 \times 3) = 3.10$

Programme D: $(0.25 \times 5) + (0.20 \times 5) + (0.30 \times 5) + (0.10 \times 5) + (0.10 \times 3) + (0.05 \times 3) = 4.70$

Summarising the composite priority score results for different programmes:

Programme	Immediate response		Recovery		Future growth	
	Score	Priority band	Score	Priority band	Score	Priority band
Programme A	4.4	Protect	4.2	Protect	4.0	Protect
Programme B	3.4	Support	3.1	Manage	3.0	Manage
Programme C	2.5	Manage	2.8	Manage	3.1	Manage
Programme D*	4.8	Protect	4.8	Protect	4.7	Protect

Table 8: Composite programme scores and priority bands by allocation horizon

Source: Own illustration

Table 8 summarises the composite scores and the respective priority bands calculated for Programmes A, B, C and D. Even though Programme D fails Gate G1 during Stage 1 and is therefore excluded from Stage 2 analysis, Programme D is included in Table 8 for illustrative and analytical purposes and demonstrates the strategic and commercial loss caused to Nexperia due to the implementation of Gate G1, which results from the geopolitical regulatory constraints faced by Nexperia. Depending on the composite score calculated for each programme for different time horizons, the priority band has been assigned as defined in Table 4 in Section 4.3.

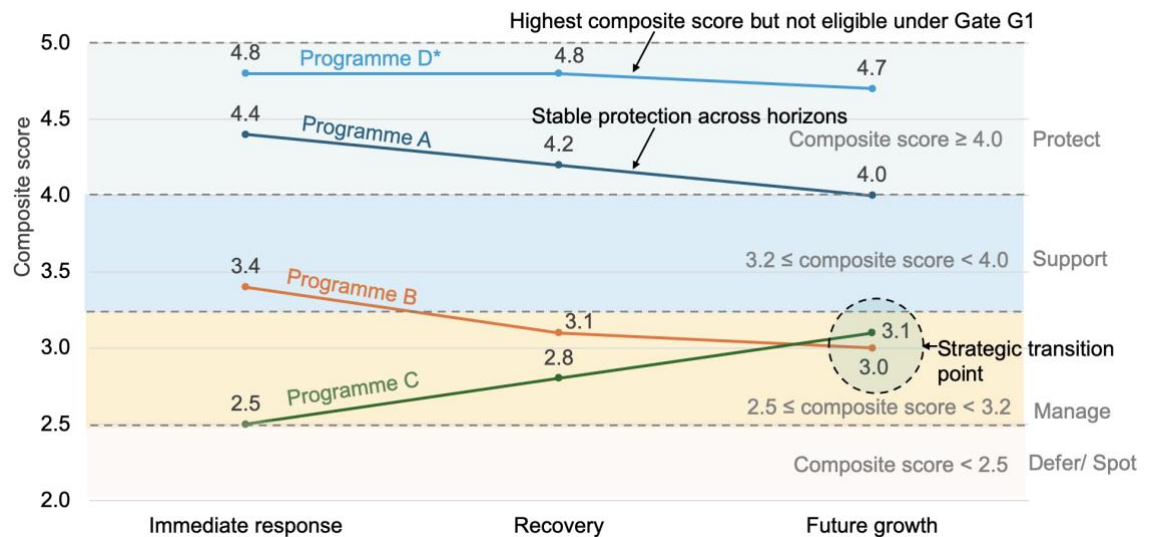


Figure 12: Composite programme scores and priority band evolution across allocation horizons

Source: Own illustration based on Table 4 (Section 4.3) and Table 8

(*Programme D does not qualify under Gate G1 and is included for analytical purposes to demonstrate the strategic and commercial value that Nexperia cannot access due to Gate G1 restrictions)

Figure 12 graphically represents the calculated composite scores summarised in Table 8 and shows how the composite scores change across different time horizons for all four programmes. Figure 12 also shows the strategic transition where Programme C overtakes Programme B in the future growth phase and highlights the framework's reallocation mechanism. Detailed analytical interpretation and its implications for Nexperia are provided in the Analytical interpretation section.

Analytical interpretation: Figure 12 reveals four strategic findings for Nexperia:

First, even though the composite score of Programme A decreases from 4.4 to 4.0, as it moves from the immediate response phase towards the future growth phase, due to the decreasing weights assigned to C1 and C2 as the framework moves away from the immediate response phase, the composite score remains in the "Protect" priority band. This result obtained from the decision framework also reflects the importance of Programme A, which consists of safety-critical airbag-related components with high qualification lock-in. As non-supply of any safety-critical programmes, such as the MOSFET programme (BUK9M-series), can result in contractual consequences and reputational damage for Nexperia, such programmes should always be given priority. This is also achieved by the decision framework. The decision framework also shows that the priority of such programmes does not decay irrespective of the time horizon and remains within the "Protect" priority band, as also shown in Figure 12.

Second, the result obtained for Programme B demonstrates the conflict between the current high-margin programmes and their decreasing strategic importance. As Programme B represents a commercially attractive but strategically less important programme for the future, the framework also captures the same by the increasing C3 weight from 15% (during the immediate response phase) to 30% (during the future growth phase), causing the composite score to decrease from 3.4 (during the immediate response phase) to 3.0 (during the future growth phase), showing the decreasing strategic importance of the programme over time. This is because Programme B is tied to an ICE platform, which has declining strategic importance (captured by a C3 criterion score of 1) for Nexperia. The attractive margin associated with Programme B is no longer sufficient to maintain its ranking. However, as Programme B is associated with strong contractual obligations, the framework assigns a composite score of 3.4 (Support), indicating that Programme B should still receive priority supply during the immediate response phase. The illustration of Programme B and its declining composite score in Figure 12 show that a high-margin vehicle programme cannot always capture the scarce capacity if not strategically aligned with the future of Nexperia.

Third, as Programme C is linked to a vehicle programme with high future strategic value but relatively low current contractual value, the framework output also shows consistency in the result as the composite score increases from 2.5 during the immediate response phase to 3.1 during the future growth phase. Programme C overtakes Programme B (composite score = 3.0) during the future growth phase. This overtake of Programme B by Programme C is the clearest evidence of the framework's core mechanism. Even though the same C3 weight is assigned to both programmes during the future growth phase, Programme C rises while Programme B falls on the composite score scale as also visible in Figure 12. Therefore as the time horizon progresses from immediate response phase towards future growth phase, the framework logic allocates the scarce capacity from profitable legacy programmes towards strategically important future programmes for Nexperia. However, Programme C does not reach the "Support" band (composite score ≥ 3.2 and < 4.0) as C1 criterion score = 1 for Programme C as per the Programme C definition. Thus, the framework encourages Nexperia to secure binding contracts for strategic programmes, which will secure design-in commitments for SiC/GaN programmes.

Even though Stage 2 calculations for Programme D are done only for analytical purposes as it fails Gate G1 on regulatory fronts, its composite scores are the highest among all the considered programmes across all time horizons (composite score = 4.8 during the immediate response phase and composite score = 4.7 during the future growth phase). This high composite score of Programme D, that almost remains constant across all time horizons as also visible in Figure 12, shows the importance of this Programme for Nexperia, but as it fails the hard Gate G1 due to being assembled through Dongguan, which creates regulatory exposure, Nexperia cannot divert scarce capacity to Programme D. This is the framework's central stress test: If regulatory requirements are not fulfilled, no degree of commercial importance or strategic importance can override Gate G1. The calculations done for Programme D in the Stage 2 analysis are therefore not intended for an allocation decision, but to show the strategic and commercial value that remains inaccessible to Nexperia due to Gate G1 restrictions. This analysis also shows that if Programme D is routed through other Nexperia locations such as Malaysia or the Philippines instead of Dongguan, Programme D would pass Gate G1 and the scarce capacity could be considered for supply to Programme D. This restructuring of the supply chain could create both current commercial and future strategic benefits for Nexperia.

In summary, the programmes analysed in Section 4.5 show that the highest-value allocation decision is not always the programme with the highest commercial value to Nexperia. The optimum decision depends on the interaction between contractual

obligations, commercial considerations, strategic goals, risk exposure, regulatory factors and the allocation horizon. Therefore, the framework developed in this thesis enables Nexperia to make semiconductor allocation decisions that are operationally feasible, commercially balanced, and strategically aligned.

4.6. Outlook for Nexperia

Section 4.5 discussed the semiconductor allocation decision framework under the current situation of Nexperia. However, due to changing geopolitical equations, regulations imposed by multiple nations on Nexperia, ownership disputes, and other supply-chain-related constraints, the future operating environment remains uncertain for Nexperia. This chapter tests the decision framework developed in previous chapters under changing external conditions for Nexperia. In order to test the robustness of the semiconductor allocation decision framework under uncertainty, three possible scenarios for Nexperia, namely stabilisation, bifurcation, and renewed regulatory control, are considered.

Stabilisation scenario: If Nexperia reduces its dependence on China-based locations and shifts the back-end work from Dongguan (China) to Malaysia (van Gerven, 2025), Polar Semiconductor and other non-China locations (Nexperia B.V., 2026d), for its European and other non-China customers, the vehicle programmes would now pass the regulatory and export control gate G1, and more programmes would become eligible for Stage 2 analysis. Vehicle programmes similar to Programme D, which could bring high commercial and strategic value to Nexperia, would become eligible and the overall business value of Nexperia would increase. The newly eligible vehicle programmes could now enter Stage 2 evaluation and come under the "Protect" band of the decision framework, as shown in Figure 12. Although geopolitical risks are lower under this scenario, Nexperia should not remove gate G1 from the decision framework as the BIS Affiliates Rule is only a temporary suspension and can be reimposed on Nexperia through Wingtech (Brattli, Meldahl and Hermann, 2025). The new scenario would make more programmes eligible for Stage 2 consideration during allocation decisions.

Bifurcation scenario: As discussed in Section 4.1, Nexperia China and Nexperia Europe may eventually develop separate supply chains to avoid regulatory and export-control exposure (Reuters, 2025; TrendForce, 2026; van Gerven, 2026). If this happens and the Chinese and non-Chinese entities of Nexperia become separate entities, then the semiconductor allocation decision framework developed in this chapter remains applicable. However, Gate G1 would apply differently to both entities. A vehicle programme connected to a Chinese supply chain would not pass Gate G1, but the same programme would pass Gate G1 if it is considered by Nexperia Europe. The scoring

scales and weights defined in Tables 2 and 3 would remain unchanged. However, depending on the circumstances of each entity, different scores may be assigned to criteria C1-C6, resulting in different allocation decisions. Therefore, even under a bifurcation scenario for Nexperia, the decision framework would still perform as long as Gate G1 and criteria C1-C6 are correctly applied.

Renewed-control scenario: As observed during the 2025 Nexperia crisis, the global automotive supply chain was heavily affected within a short period of eleven days (Government of the Netherlands, 2025b, 2025a; Nexperia B.V., 2025f; Tan, 2026), similar scenarios may occur in the future. Any further tightening of the existing restrictions on Nexperia or any upcoming geopolitical or export restrictions imposed by the Netherlands, the U.S., or China on Nexperia, would still be analysed using the decision framework developed in this thesis. The decision framework would absorb any such geopolitical and export restrictions at Gate G1, before the analysis enters Stage 2. However, many affected vehicle programmes may become ineligible for Stage 2, similar to Programme D discussed in Section 4.5, and can impact the commercial and strategic goals of Nexperia. Overall, the semiconductor allocation decision framework remains applicable under such situations, demonstrating the robustness of the framework developed in this thesis.

Because predicting unforeseen circumstances affecting planned operational and strategic goals of Nexperia is difficult, the real strength of the semiconductor allocation decision framework lies in its ability to support allocation decisions even under sudden changes in the external environment. Nexperia cannot predict whether bifurcation or renewed regulatory controls will materialise, but if they do, this decision framework will still generate the composite score and respective priority bands as discussed in earlier chapters. This is because Gate G1 captures regulatory and export-related restrictions imposed or potentially imposed on Nexperia and criteria C1-C6 control other important parameters, fundamentally affecting the allocation decision at Nexperia. The quality of the allocation decision depends on assigning weights and scores that accurately reflect Nexperia's strategic priorities and risk exposure, as shown in the case of the four programmes in Section 4.5, as these values determine how the different parameters interact with each other to arrive at an optimal allocation decision at Nexperia. As the decision logic developed here remains unchanged in principle, this framework would generate results during crisis periods such as the 2025 Nexperia crisis, where decisions must be made within limited time and with limited information.

Section 4.5 analyses Programme D despite its rejection at Gate G1. However, the importance of Programme D extends beyond its exclusion at Gate G1. Programme D shows that the largest losses during periods of scarcity at Nexperia do not necessarily

result from scarce capacity itself. They may result from opportunities lost due to regulatory and export controls imposed on Nexperia due to which programmes such as Programme D were not supplied, despite their high commercial and strategic value. In other words, this decision framework developed in this thesis not only supports semiconductor allocation decisions, but also makes the cost of geopolitical and export controls visible to Nexperia (as shown by Programme D in Figure 12). This insight, which is visible to Nexperia due to this decision framework, would also highlight where future supply-chain diversification could create the greatest strategic benefit for Nexperia.

4.7. Management implications for Nexperia

The decision framework developed for semiconductor allocation was applied to different programmes in Section 4.5 and subsequently, the robustness of the decision framework was tested in Section 4.6. Different findings from these chapters also provide guidance on how Nexperia should react when responding to allocation challenges under supply constraints identified through this thesis. Discussion of the framework development, its application, and testing has led to four managerial implications.

Managerial implication 1: Operationalising customer prioritisation through programme-level allocation: Programme-level allocation should be Nexperia's operational mechanism for applying customer prioritisation, consistent with Section 1.3 and Assumption 1, rather than a separate objective. Because each OEM account contains multiple vehicle programmes with different contractual, strategic and risk profiles, prioritising the right customers in practice requires scoring at the programme level. The following implication therefore shows how programme-level allocation delivers customer prioritisation under constrained supply. Section 4.5 analysed Programme B (ICE-platform-based programme as defined in Section 4.5) and Programme C (SiC MOSFET based programme as defined in Section 4.5) at programme level rather than account level. The composite score for Programme B falls from 3.4 to 3.0 (Figure 12 in Section 4.5), showing the decreasing priority of Programme B, while the composite score for Programme C increases from 2.5 to 3.1 (Figure 12 in Section 4.5), showing the increasing priority of Programme C. If both programmes belonged to the same OEM account, this important difference would never have surfaced and due to the current high volume of Programme B, it would continue to get more of the scarce capacity, reducing the scarce capacity available for Programme C. Nexperia should therefore prioritise customers operationalised through programme-level scoring for all constrained automotive semiconductor volume, because an account-level allocation would quietly starve the rising EV programmes at the time when their strategic importance is increasing and could limit Nexperia's exposure to future growth opportunities.

Managerial implication 2: Commercial value versus regulatory feasibility: The two-stage decision framework developed in this thesis separates the value of a programme from Nexperia's eligibility to serve that programme and demonstrates that the binding constraint can become regulatory rather than commercial or dependent on scarce capacity itself. This is the deeper contribution of Programme D. Revenue, margin contribution and strategic importance already indicate that Programme D is important for Nexperia, but they do not indicate whether Nexperia is able to serve it. By identifying programmes with high commercial and strategic importance that are excluded before allocation begins, the framework treats "feasibility" and "value" as two separate parameters. Programme D is excluded not because capacity is scarce, but because it never reaches the point of being evaluated for capacity allocation. Nexperia should therefore make an effort to shift its focus from optimising allocation among the eligible set of programmes itself towards expanding the eligible set. Despite securing the highest composite score (Figure 12 in Section 4.5), Programme D's ineligibility to enter Stage 2 also suggests to Nexperia which ineligible programmes should be given priority first by rerouting its supply chain to avoid China-related routes by analysing the composite scores of such ineligible programmes, such as Programme D, making the supply-chain redesign a strategic rather than purely operational decision.

Managerial implication 3: Strategic design-ins require commercial anchoring: Section 4.5 shows that for Programme C, which has a high future strategic value but relatively low current contractual value, the composite score increases consistently across the time horizons from 2.5 during the immediate response phase to 3.1 during the future growth phase, but despite the consistent improvement in composite score, Programme C never reaches the "Protect (composite score ≥ 4.0)" or "Support ($3.2 \leq$ composite score < 4.0)" zones as shown in Figure 12. This is because, despite scoring 5 in criterion C3 (strategic importance) during the future growth phase, criterion C1 (contractual severity) still scores 1, reducing the overall composite score of Programme C. This means that, no matter how strategically important a programme may appear to Nexperia, failure to secure contractual commitments places the programme in a situation where the programme may be overlooked or deprioritised despite its strategic importance. This decision framework shows that without contractual commitments, the strategic importance of a programme alone does not place it into a higher allocation band. The managerial lesson for Nexperia is that a strategic design-in alone such as the KOSTAL SiC programme (Nexperia B.V., 2025b) will not protect supply to such programmes during a scarce period, but securing an early design-in and a long-term volume commitment would prioritise the strategic programmes and eventually support Nexperia's future strategic goals.

Managerial implication 4: Consistent allocation governance: Future disruptions may still be difficult to predict for Nexperia due to the changing geopolitical landscape, but the key challenge is not to predict these disruptions but to maintain consistent allocation decisions for scarce capacity. Section 4.6 categorises these disruptions into three scenarios, namely stabilisation, bifurcation, and renewed-control, and shows that despite such changes, the same Gate G1 and C1-C6 logic remains valid and the decision framework is capable of generating reliable results under such disruptions as well. Implementing the decision framework will make allocation decisions consistent under a changing external environment. Nexperia should focus on institutionalising a consistent allocation process through the governance roles, KPIs, and override controls as defined in Section 4.4. When Nexperia implements this, allocation decisions will remain transparent, auditable, and resistant to commercial or political pressure during disruptions, rather than reacting impulsively to a crisis situation, as observed during the 2025 Nexperia crisis.

Managerial implication	Framework evidence	Why it matters for Nexperia	Monitoring measure
Operationalising customer prioritisation through programme-level allocation	Same OEM: Programme B falls (3.4 to 3.0), Programme C rises (2.5 to 3.1) (Figure 12)	Protects future-growth programmes from legacy volume	Programme-level scoring coverage (target 100%, min 95%) (Table 6)
Commercial value versus regulatory feasibility	Programme D scores highest (4.8) but fails G1 (Figure 12)	Identifies high-value but unreachable programmes	High-score programmes blocked by Gate G1
Strategic design-ins require commercial anchoring	C rises (2.5 to 3.1) but remains below support; C3=5, C1=1	Strategic value alone does not guarantee supply protection	Strategic programmes covered by long-term agreements
Consistent allocation governance	Same G1 and C1-C6 logic valid across all scenarios (Section 4.6)	Maintains decision discipline during disruptions	Override rate <10%; crisis decision time <24h

Table 9: Summary of managerial implications derived from the developed customer prioritisation framework

Source: Own illustration

Table 9 summarises the managerial implications derived from the development, application, and testing of the customer prioritisation framework at Nexperia as discussed in Section 4.7. For each implication discussed in Section 4.7, Table 9 shows the framework evidence from which it was derived, its managerial relevance for Nexperia, and a monitoring measure that can be used to track its implementation in practice. Section 4.7 shows that the managerial implications of the semiconductor allocation customer prioritisation framework for Nexperia extend beyond regular support for semiconductor allocation under supply constraints and highlights the importance of programme-level analysis, regulatory feasibility, contractual positioning of strategic design-ins, and governance discipline within the decision framework developed in Chapter 4.

This thesis aimed to convert a recurring customer prioritisation challenge under constrained automotive semiconductor supply into a structured managerial decision framework, by using Nexperia as the central case for the development and application of the semiconductor allocation decision framework under constraints. Chapters 2, 3 and 4 showed that the shortages related to automotive semiconductors are structural and not a temporary problem and are likely to continue in the future. Additionally, Chapters 2, 3 and 4 also established the unique geopolitical situation that Nexperia is facing due to multiple regulatory restrictions from Chinese, Dutch and U.S. authorities. To address these challenges, Chapter 4 developed a two-stage decision framework and stress-tested it against four representative programmes that Nexperia supplies and three likely scenarios that Nexperia may encounter in the future. Building on the analysis presented in chapters 1 to 4, the next chapter explains what the exercise of building, applying, and stress-testing this framework revealed about how semiconductor allocation decisions are actually made, where they create value, where they destroy it, and how their quality can be improved under constraint.

5. Conclusion and next steps

5.1. Answering the research questions

The research questions (RQ1, RQ2, and RQ3) were deliberately structured to build on one another. First, the thesis asks how the allocation decision should be made. Second, it asks which factors affect this allocation decision, and third, it asks how these factors can be combined into a practical framework. This thesis not only answers these research questions but also provides a deeper understanding of the issues behind them.

RQ1: How can semiconductor suppliers (in case of Nexperia) prioritise automotive OEM customers under conditions of limited supply? The direct answer developed throughout Chapter 4 is that customer prioritisation must proceed in two distinct steps. In the first step, a mandatory eligibility gate (Gate G1) filters the programmes exposed to regulatory and export controls. Only the Gate G1-qualified programmes then enter Step 2, where these programmes are assigned weights under six distinct criteria (C1-C6), covering all the major factors affecting the allocation decision at Nexperia. These criteria are set to take into account contractual obligation severity, qualification lock-in, strategic importance, geopolitical and operational risk, gross margin contribution and demand stability of each programme. In order to reflect the practical realities and changing objectives of semiconductor allocation, these criteria C1-C6 are weighted across time horizons (Table 2, Section 4.1). Based on the scores assigned to each programme (Table 3, Section 4.2), the customer prioritisation is based on the composite score (Table 4, Section 4.3). The higher the composite score, the better the chance of securing supply.

The deeper learning, however, is that the research provided a deeper understanding of the issues behind the allocation decision for automotive semiconductors in case of Nexperia. The research revealed that account-level prioritisation alone is insufficient for Nexperia. Analysing individual vehicle programmes leads to more effective allocation decisions. Therefore, these findings demonstrate that customer prioritisation is operationalised through programme-level allocation and analysis, which is also consistent with the conceptual discussion presented in Section 1.3. Because every vehicle programme belongs to a customer, individual programme-level allocation decisions collectively support the broader long-term customer relationship while preserving the programme-specific characteristics that are critical for effective customer relationship under constrained supply.

Because a single OEM account can simultaneously contain a declining internal-combustion programme and a rising electrification one, similar to the relationship between Programme B and Programme C, with very different future prospects, as

discussed in Section 4.5 and illustrated in Figure 12, one programme may not be relevant for the future business of Nexperia while another programme may become significantly more important in the future. As a result, allocation decisions made only at account level may overlook important programme-level differences within the same OEM relationship. The research also reveals a potential hidden cost that Nexperia might have to pay by ignoring the programme-level allocation strategies to prioritise customers. As automotive semiconductors have a long qualification process, and after design-in it is extremely difficult to replace the selected supplier (Automotive Electronics Council, 2021, 2023; Nexperia B.V., 2025e), analysis done at programme level gives Nexperia greater visibility of such programmes securing design-ins and future supply opportunities. If not recognised at an early stage, the loss of a design-in may become the opportunity cost Nexperia pays for diverting scarce capacity to declining ICE programmes within the same OEM account (as illustrated by Programme B and C, Section 4.5). A further finding of this research is specific to Nexperia. Given Nexperia's situation with its front-end-in-Europe, back-end-in-China configuration (S&P Global Mobility, 2025a), allocation decisions are influenced not only by customer demand and programme attractiveness but also by whether a programme can be legally supplied through the available supply route. Therefore, the supply route becomes an important allocation consideration rather than merely a supply-chain detail.

RQ2: Which key factors influence customer prioritisation decisions in such situations? The decision framework developed in this thesis identifies six different factors (C1-C6), namely contractual obligation severity, qualification lock-in, strategic importance, geopolitical and operational risk, gross margin contribution, and demand stability, together with a separate regulatory eligibility gate (Gate G1). These factors are operationalised within the framework through six decision criteria (C1-C6). But the contribution of the research is not the identification of these factors or criteria, but the understanding of how these factors serve different purposes within the same framework and therefore cannot be treated equally. Furthermore, the understanding that these factors hold different levels of importance during the immediate response period (0-3 months), recovery period (3-12 months), and the future growth period (> 12 months) further deepens the understanding of the framework and importance of allocation decisions under constrained capacity for Nexperia. The framework decision to keep the scoring-based model separate from the regulatory eligibility gate shows that the six factors have a different role in the allocation decision than the compliance requirements assessed through Gate G1. The research shows that a programme should not receive supply simply because it is more profitable, strategic or belongs to an important customer if the regulatory restrictions prevent Nexperia from supplying that programme

(Government of the Netherlands, 2025a; Nexperia B.V., 2025f). This reflects a deeper understanding revealed by the research that customer importance can be compared with other commercial factors, but compliance cannot, and should not, be treated the same way. In allocation decisions, financial benefits can be seen immediately and regulatory consequences often appear later (Voit, 2026). The absence of a structured decision framework that considers all these factors with clear separation could result in scarce capacity being allocated to currently profitable customers with stronger bargaining power, ignoring the potential regulatory and compliance risks. By placing Gate G1 before the scoring system begins, the framework shows that regulatory compliance cannot be traded against profitability, customer volume, or strategic importance. The framework further recognises that the six criteria do not serve the same purpose within the allocation decision. C1 and C2 protect Nexperia from contractual penalties and qualification loss, C3 and C5 support future growth and profitability of Nexperia, C4 analyses the level of geopolitical and operational exposure that is associated with a programme once it has passed the eligibility Gate G1, and C6 identifies unreliable demand signals projected by customers. Therefore, the framework treats these criteria as complementary decision factors rather than as equal factors within a single hierarchy. This shows that the framework does not treat all these factors equally, and weighting ensures that these factors interact with each other to optimise allocation decisions for Nexperia. Moreover, the changing weights of different factors across different time horizons deepen the analysis performed by the framework. Therefore, the answer to research question RQ2 is that the importance of different factors discussed above affecting allocation changes across different time horizons. For example, strategic importance (defined as C3 in the framework) carries a weight of only 15% during the immediate shortage phase but it increases to 30% in the future growth phase. As a result, there is no single answer to RQ2 regarding which factor matters the most, as the importance of different factors depends on the time horizon being considered. The contribution of this framework is that it makes these factors explicit and structured instead of leaving them to individual judgement during periods of supply disruption.

RQ3: How can these factors be structured into a practical and applicable decision-making framework? The factors identified in this research are structured through a two-stage allocation framework developed in Chapter 4. Each programme is first analysed through Gate G1, where it is evaluated against regulatory and export controls. Only programmes that clear Gate G1 then enter the next analytical stage of the framework. After Gate G1, the qualified programmes are analysed using weighted criteria (Table 2, Section 4.1), a simplified 1-3-5 scoring scale (Table 3, Section 4.2), four priority bands (Table 4, Section 4.3), and a governance layer consisting of defined ownership and

responsibilities (Table 5, Section 4.4), KPIs (Table 6, Section 4.4), and a capped override mechanism (Section 4.3). When working together, these different factors provide a structured approach for customer prioritisation under scarce capacity. A key finding of this research is that the practical and repetitive use of the decision framework requires simplicity and therefore complexity in the framework is intentionally reduced. As observed during the 2025 Nexperia crisis, when the Dutch government took over important governance and decision-making powers at Nexperia on 30 September 2025 (Government of the Netherlands, 2025a), China blocked the exports of Nexperia's products (Nexperia B.V., 2025f), and the issue of notification to customers by Nexperia on 10 October 2025 (Nexperia B.V., 2025f), it shows that decisions during a crisis are often made under a limited time window, with limited information and limited future visibility, and under uncertainty. Therefore, instead of trying to achieve maximum analytical precision, this framework was designed to remain usable under time pressure and limited information, which is also a characteristic of semiconductor supply disruptions, especially at Nexperia. This shows that the factors used in the framework are structured in a way to prioritise decision speed and operational usability instead of model sophistication. As research finds that the model designed for semiconductor allocation decision at Nexperia must be workable under limited information and a limited time window, the simplicity of different factors and their use in the decision framework becomes a key strength of the framework. They are direct design responses to Nexperia's reality, as the 2025 crisis showed how escalation moved from Dutch intervention to Chinese export block to customer notification in eleven days, leaving no room for continuous optimisation on perfect information and decisions needed an immediate response. A more sophisticated optimisation framework model was deliberately avoided because the effectiveness of such models depends on detailed data availability, which may not be available during supply disruptions. The unique case of Nexperia with a rapidly changing regulatory landscape and operational boundary conditions limits both information quality and decision time available to Nexperia (Government of the Netherlands, 2025a; Nexperia B.V., 2025f). As a result, the developed framework in this thesis prioritises usability under uncertain conditions, and ensures that allocation decisions can still be made consistently even when conditions change faster than data can be collected and analysed in that short duration. The practical use of this framework is very closely linked with its simplified design. Instead of introducing a high level of precision that may quickly become outdated and practically difficult to use during an actual disruption phase, the development of a decision framework that prioritises robustness, speed, and consistent decision-making under supply constraints supports practical decision-making. This also reflects an important

finding of this research that when regulatory, operational, or market conditions can change within a few days, an allocation framework must be designed to remain usable under uncertainty compared to a model that is optimised for maximum theoretical precision.

The answers to the research questions RQ1, RQ2, and RQ3 are connected and built on each other throughout the thesis. The problems that were identified at Nexperia in Chapters 1, 2, and 3 concern how to allocate scarce semiconductor supply when profitability, strategic relevance, and risk take the allocation decision towards different decisions under changing geopolitical regulations imposed by multiple governments and its ownership conflicts. Because of this, Nexperia may be forced to face a persistent and multi-dimensional semiconductor allocation challenge under supply constraints. Based on research presented in Chapters 1, 2, and 3, making customer prioritisation operationalised through programme-level analysis and allocation was highlighted for better visibility of the details that can optimise the semiconductor allocation by balancing these factors optimally. As the effect of any non-optimum allocation can run for the entire lifecycle of the programme, lasting 7-8 years (Center for Automotive Research, 2023), it may affect Nexperia's strategic goals and product placement in the market. In addition, Nexperia's ownership structure and regulatory restrictions add another layer of complexity to the decision-making process. Chapter 4 provides structure to these factors and establishes a relationship among these factors, which also changes as the decision time horizon changes and keeps regulatory restrictions as a mandatory check at the start. Therefore, the answers to RQ1, RQ2 and RQ3 emerge from the logical connection between the problems, the analysis, and the framework developed for Nexperia.

5.2. Contribution of the developed framework

As discussed in Chapter 4, involving Gate G1, multiple criteria or factors C1-C6 scoring system with changing weights as time changes, 1-3-5 scoring system, composite score calculation, their visibility on priority bands and KPIs and governance rules not only contributes to the development of a decision framework but also contributes to making the hidden relationship between these factors and their trade-offs visible and open for discussion and reviewed as and when required during decision making at Nexperia.

The first contribution of the decision framework is the separation of the regulatory and export control Gate G1. It is common for any supplier to check for compliance, and the compliance factor may compete with commercial attractiveness. Because the framework developed in this thesis places this check at the very start before any further commercial or strategic analysis can be done, it helps the decision maker focus on the customer prioritisation for only those programmes that are not exposed to regulatory controls as it

hides any attractive numbers and temptation of such ineligible programmes at the start itself. This is a deeper behavioural and psychological contribution. The fact that Wingtech sits on the U.S. Entity List (Bureau of Industry and Security, 2024a), the concentration of almost 80% of Nexperia's back-end capacity in China (S&P Global Mobility, 2025a), and chances of closure of Dongguan route by either Chinese export controls (Nexperia B.V., 2025f) or the impacts due to BIS Affiliates Rule (Brattli, Meldahl and Hermann, 2025) suggest that the main issue of the 2025 Nexperia crisis may not be the customer prioritisation list, but Nexperia's commitment to customers through these routes, which later made supply legally impossible. This framework ensures that no degree of commercial attractiveness or futuristic goals can override regulatory compliance.

The second contribution of the framework is the shift of the supplier's focus from competing accounts to programme level, which reveals the misallocation that otherwise will remain invisible. The fact that the same OEM can have a high-current-volume legacy programme and a currently lower-volume but strategically rising electrification programme cannot be highlighted in the account-level analysis. Account-level analysis ensures current revenue for the supplier but risks the future design-in opportunities and future strategic place in the semiconductor market. Programmes B and C discussed in Section 4.5 demonstrate the perfect example of this contribution by exposing how the composite score reversed the importance of both programmes in the future growth phase (Figure 12). Therefore, the most significant risk does not lie between the competing accounts to get the scarce semiconductor supplies, but it lies within the same account itself, and if identified by Nexperia at the correct time, could protect and build the future business.

The third contribution is the forced integration of profitability, strategic importance and risk in a single framework structure. Generally, these different departments carry different motivations while making allocation decisions, which may be mostly contradictory. Even during the combined decision, the allocation leans towards the department that holds better negotiating and bargaining powers, making the allocation subjective and non-optimal. All these criteria or factors (C1-C6) assigned with weights, which also represent the expression of strategy, establish an analytical relationship between them that balances profitability, strategic importance and risk across time horizons and contributes to allocation decisions analytically. Even though these criteria C1-C6 are owned and represented by representatives from individual departments within the decision committee (Table 5, Section 4.4) and they can be biased towards their viewpoint, the predefined weight assigned to each criterion reduces any biased allocation decision. This makes allocation decisions structured, auditable, and contributes to the framework organisationally. By formula-based decisions and by removing and replacing department

hierarchy and bargaining power, the framework contributes to allocation decisions both analytically and organisationally and remains auditable and unbiased even under crisis situations.

The fourth contribution of the decision framework developed in this thesis is demonstrated through Programme D, and it is not related to the allocation itself, but is derived from the high-score rejections. Even though Programme D does not clear Gate G1 due to regulatory issues arising from its Dongguan connection, further analysis revealed that it scored the highest composite score of 4.8, which remained almost constant throughout the time horizons (Figure 12, Section 4.5). Even though the commercial value of such regulatory-blocked orders is already known to Nexperia based on order value, composite score generated by the framework combines all six criteria or factors C1-C6 and thus represents a bigger picture for Nexperia (Chapter 4). Therefore, the almost constant score 4.8 (out of 5) shows that Programme D is not only important today but it also holds strong strategic and commercial value for Nexperia. This converts a theoretical and difficult-to-measure restriction "geopolitical risk" into a numeric value that shows overall importance for Nexperia. If more than one programme is rejected at Gate G1, analysis of all the rejected programmes and ranking them within the allocation bands as in Figure 12 would reveal which blocked programme should be rescued first (by changing the supply chain route from China to Malaysia or other non-China routes) so that it passes the Gate G1 and Nexperia can deliver supply. This supply will not only already be optimised due to weights and score allocations as discussed in chapter 4, but it will also be the effort made by Nexperia to get maximum current and strategic benefit by taking the lowest corrective action. This is because rescuing the blocked programmes with the highest composite score already incorporates all the factors C1-C6.

The fifth contribution is the separation of the framework from the decision inputs, demonstrating the robustness of this decision framework and making the output scenario-independent. As demonstrated during the scenario analysis in Section 4.6, the output of the framework remains stable and reliable during the changing scenarios of stabilisation, bifurcation and renewed regulatory control. The decision architecture of the framework never changes because changing scenarios are absorbed by Gate G1 and the 1-3-5 score system (Table 3) allocated to criteria (C1- C6). In case of Nexperia, where the scenarios can change within the period of eleven days as observed during the 2025 Nexperia crisis (Government of the Netherlands, 2025a; Nexperia B.V., 2025f; Tan, 2026), a framework that does not need to be redesigned to incorporate such changes provides resilience and robustness. This importance becomes especially visible during semiconductor crisis periods and demonstrates the stability of the process under instability of the environment.

The sixth contribution is not related to the current allocation itself, but it goes a step further to recognise and value the future business opportunities for Nexperia by managing future business opportunities. The framework values future opportunities today. Programme C in Section 4.5 shows high future strategic importance for Nexperia and this is communicated to the framework by higher scores allocated to C3 on the 1-3-5 score system (Table 3). This highlights the future importance of these programmes and as design-in decisions are taken at the early stages by the customers and once locked-in, the business is almost secured for the entire life cycle of the programmes (Nexperia B.V., 2019; Center for Automotive Research, 2023). But if the design-in is lost, the loss continues for the entire period of programme lifecycle and may further weaken the market position of Nexperia. Moreover, Programme C (Section 4.5) also shows that if the attractive strategic programme is not secured by binding contracts (low C1 on 1-3-5 score), Nexperia can still lose the design-in to its competitor. Programme C suggests that a future opportunity without contractual anchoring is not genuinely secured. By making these variables visible to the decision makers, the framework triggers decision makers at Nexperia to seize the opportunity by securing the design-in and contracts to strengthen its market position and this could also favour Nexperia's competitive position during future design-in opportunities. Therefore, the design-in being highlighted by the framework encourages Nexperia to secure the future potential business.

The seventh contribution is that it limits the bargaining power of different customers and keeps the allocation decision objective and not influenced by the customer pressure during scarce supply periods. The 2025 Nexperia disruptions show how large customers were able to negotiate and purchase Hamburg wafers and directly contract Dongguan for packaging (Sterling, 2025a) This suggests how bargaining power of customers can strongly affect the distribution of scarce capacity available with Nexperia. Exactly due to this reason, criteria C1-C6 do not carry any influence of customer bargaining power so that the final allocation decision suggested by the framework remains objective and remains free of such influences. This framework, however, takes care of a few cases through override policy defined in Section 4.3, allowing only 10% of the total programmes to bypass this policy through a predefined approval process. This ensures that scarce capacity is allocated according to contractual, strategic, operational, and economic criteria rather than the customer's bargaining and negotiation power, and therefore the framework protects both current value and future design-in opportunities.

5.3. Limitations of the framework

The limitations of the framework seem to appear mostly from design choices made during the development of the framework and therefore provide additional insights into

the logic and the trade-offs. The objective of the thesis was to develop a framework to prioritise customers at Nexperia under semiconductors scarcity. In order to have better clarity of the problem and to arrive at a structured and systematic framework, several assumptions were made in Section 4.1. On one side, these assumptions help to analyse the problem better, but at the same time, such assumptions also limit the boundaries under which the developed framework performs and delivers the desired results. Understanding these limitations of the framework becomes important in order to clearly understand where the framework can support managerial decision-making and where it needs further adjustments, additional information, additional organisational capabilities and future research. This chapter lists such limitations associated with the framework developed in this thesis.

The first limitation is the framework's dependence on input data quality. Even though the time-horizon-dependent weights of criteria C1-C6, the 1-3-5 scale and the Gate G1 are fixed in the framework architecture, the inputs that decide the 1-3-5 score allocated to C1-C6 and the inputs for Gate G1 depend on the accuracy of the input data. Chapter 3 also showed that these inputs become distorted during supply disruptions through the bullwhip effect (Lee, Padmanabhan and Whang, 1997), and the visibility across downstream supply chain of the component may become limited (Garcia, Arvidsson and Jonsson, 2024). The MAPE calculation for C6, margin calculation for C5, route eligibility, clear and up-to-date understanding and judgement of regulatory and compliance requirements related to global supply chains for Gate G1, downstream supply chain visibility for C4 make the framework output highly dependent on the quality of data available to operate the framework.

The second limitation is that the 1-3-5 scoring system discussed in Section 4.2 (Table 3) is dependent on human judgement. Although this scoring system makes the framework simple and easy to use, at the same time it also makes the system open to subjective interpretation. As defined in Section 4.4, different functions at Nexperia may have different motivations for a particular supply, the score can be influenced by such motivations and this can lead to changing the score from 3 to 5 for any of the criteria to favour any particular supply. This can be done without much resistance, if not resisted by the other committee members. As this can change the overall composite score, the one score increased for one criterion could change the framework output. The framework can make the allocation decisions standard and transparent but it does not make the decision completely objective due to the involvement of human judgement. The framework defines which criterion is owned by which function to improve accountability, expertise and consistency in scoring but as each function may have different priorities as described in Section 4.4, the human factor still influences the allocation of scores. The

override rate KPI as defined in Section 4.4 and the requirement of the joint approval from all allocation committee members for such override cases put a barrier on the extent to which allocation decisions can become biased and make scoring visible, auditable and challengeable, but it also fails to remove the human judgement completely.

The third limitation is related to the general use of the framework elsewhere apart from Nexperia. With changing times and a changing geopolitical situation globally where each region wants to increase semiconductor independence (Poitiers and Schenk, 2026), the same semiconductor allocation framework can be used, but needs adjustments. The ownership conflicts between the Dutch government and Nexperia's Chinese parent company Wingtech (Sterling, 2025a, 2025b), Wingtech's presence on the U.S. Entity List (Bureau of Industry and Security, 2024a), Nexperia's major concentration of back-end capacity in China and front-end capacity in Europe (S&P Global Mobility, 2025a), the impacts due to BIS Affiliates Rule (Brattli, Meldahl and Hermann, 2025), and chances of closure of Dongguan route by Chinese export controls (Nexperia B.V., 2025f), make the importance given to Gate G1 much higher and thus Nexperia may lose its competitive edge as compared to other semiconductor suppliers. The weights and scores are all calibrated with respect to Nexperia. Thus, the overall framework would still work elsewhere too, but would need adjustments specific to a particular company where the semiconductor allocation decision needs to be done.

The fourth limitation is that the framework is tested using the four programmes A, B, C and D, covering a wide variety of realistic combinations of events, but it does not test the framework on the actual Nexperia internal data. This means that the framework can demonstrate that the decision logic works, but how the framework actually behaves with actual Nexperia data still remains unsolved due to limitations in access to Nexperia internal statistics. The weights, thresholds and scoring system were all developed using the evidence and analysis discussed in Chapters 1, 2, 3 and 4, but their validation against actual Nexperia allocation outcomes remains outside the scope of this research.

5.4. Future research and future framework development

This thesis created a customer prioritisation framework under semiconductor supply constraints using Nexperia as a central case. During the framework development, the thesis found some important questions that are still not fully answered. These questions originate from the unique situation of Nexperia involving ownership conflicts, semiconductor shortages, difficulties in switching suppliers once qualified, and geopolitical challenges. This chapter addresses such questions and suggests what future research on this topic should be targeted to improve overall depth of the framework. Furthermore, the questions raised by this thesis could also be extended to

other similar semiconductor suppliers with minor adjustments to incorporate the unique situation of the supplier into the framework to achieve realistic customer prioritisation.

The impact of ownership, geopolitical and regulatory constraints on customer prioritisation: While most of the research done so far covers how the allocation should be made when the capacity is limited, research conducted in this thesis around Nexperia revealed something much deeper. This comes from an observation made for Programme D described in Section 4.5. Although Programme D scored the highest composite score of 4.8, which remained almost constant across the time horizons, making Programme D the top in the prioritisation list, Programme D should have been the first to receive the scarce capacity from Nexperia. However, due to its connection with Dongguan, China, it became ineligible for the supply at Gate G1 itself. This gives rise to a deeper research opportunity to further develop this framework. The researchers may take the opportunity to study when the ownership structure, geographical distribution of facilities, geopolitical constraints, export controls, sanctions, and regulations imposed by different countries become more important than supply availability, profitability, strategic goals, contract conditions and the overall development of Nexperia. In other words, future research should study how suppliers should value programmes that they may not be legally allowed to supply. This kind of research will widen the understanding of the conditions under which Nexperia will be able to better balance external conditions and its own interests. The importance of this topic may increase further if future research also focuses on export controls and geopolitical restrictions across the semiconductor industry to get a bigger picture at the industry level as there are increasing regionalisation efforts in the semiconductor industry (European Commission, 2023; U.S. Department of Commerce, 2023).

Measuring the future strategic value of customer programmes: The research done in this thesis already shows that the customer prioritisation decision should be decided not only on the basis of the current revenue but, more importantly, the strategic value of the programme as discussed in Chapters 4 by introducing a criterion C3 (Strategic Importance) in the prioritisation framework. This is because the improvement of the composite score of Programme C and its overtaking of Programme B during the future growth period (Figure 12) show that the strategic importance becomes critical for the future of Nexperia. This thesis incorporates the concept using a 1-3-5 score scale. However, it becomes important to also analyse the actual future strategic value of the programmes in monetary terms or how much future value a particular programme actually creates for Nexperia. The future research should therefore focus on the part that can further make the allocation decision much clearer. This is important as once design-in is confirmed, it almost secures semiconductor business for the coming 7 to 8 years

(Center for Automotive Research, 2023). However, if this design-in opportunity is lost, this also indicates the future business loss and subsequently the loss of Nexperia's market position. As value creation is shifting towards electrification and software-defined vehicles (Kilian *et al.*, 2026) and also as strategic positioning within semiconductor value chains is also becoming increasingly important (Poitiers and Schenk, 2026), future research may also focus on the broader semiconductor industry to gain a broader perspective with respect to calculation of strategic value of different programmes.

The current framework developed in this thesis already contains forward-looking elements such as C2 (Qualification lock-in), C3 (Strategic importance), C4 (Geopolitical and operational risk), and C6 (Demand stability) in customer prioritisation, and can be used as a scenario-based and forward-looking framework. A future framework development and future research can move one step ahead and conduct research in a direction where Nexperia can identify potential allocation risks before they occur. The idea may start by connecting the real-time demand and capacity information across the automotive supply chain using standards such as Catena-X (Catena-X Automotive Network e.V., 2024), because the suppliers currently do not have full access to all supply chain levels, which creates delays in decision-making. Another future framework development could focus on simulation model development by creating a digital twin of the allocation environment. This model would allow Nexperia to test potential allocation risks and disruption scenarios before they occur.

Even though the thesis provides the framework for Nexperia that prioritises customers under semiconductor scarcity, the main value of the developed framework comes from its ability to support difficult decisions during periods of scarcity with only limited information and within a limited time window. Another broader observation made during the thesis development shows that the way Nexperia is forced to make supply-chain-related decisions due to ownership structures, supply-chain design and geopolitical exposure, the competitive advantage of Nexperia in the automotive semiconductor market may not depend only on how Nexperia allocates its scarce capacity during scarcity, but may also depend to a greater extent on how Nexperia designs its future supply chains, chooses partnerships, decides its ownership structure and production footprint long before capacity constraints actually require customer prioritisation.

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